Linear Algebra

A Solution Manual for Axler (1997), Lax (2007), and Roman (2008)

Jianfei Shen

School of Economics, The University of New South Wales

Sydney, Australia
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I hear, I forget;
I see, I remember;
I do, I understand.
Old Proverb

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Acronyms

Notation	Description
$U \trianglelefteq V$	U is a subspace of V
$\mathfrak{L}(X)$	The set of operators
$\mathfrak{R}_{\mathrm{T}}$	The range of T
$\mathfrak{N}_{\mathrm{T}}$	The null space of T
$\mathbb{F}$, K	The field on which a vector (linear) space is defined
$V \cong U$	V is isomorphic to U
$[\boldsymbol{x}] = \boldsymbol{x} + Y$	The coset Y in X and $\boldsymbol{x}$ is called a coset representative for $[\boldsymbol{x}]$
X/Y	The quotient space module Y
$\mathfrak{P}_n(\mathbb{F})$	The set of polynomials with degree $\leqslant n$, whose coefficients are in $\mathbb{F}$
$\mathrm{Sym}\,(X)$	The set of all permutations of the set X: the symmetric group on X
$\mathrm{sign}\,(\sigma)$	The signature of a permutation σ

Part I

Linear Algebra Done Right (Axler, 1997)

1

VECTOR SPACES

"As You Should Verify"

Remark 1.1. $P = \{p \in \mathfrak{P}(\mathbb{F}) \colon p(3) = 0\}$ is a subspace of $\mathfrak{P}(\mathbb{F})$.

Proof. The additive identity $0_{\mathfrak{P}(\mathbb{F})}$ is in the set; let $p, q \in P$, then $(p+q)(3) = p(3) + q(3) = 0$; for any $a \in \mathbb{F}$ and $p \in P$, we have $(ap)(3) = a \cdot 0 = 0$. □

Remark 1.2. If $U_1, \ldots, U_m$ are subspaces of V, then the sum $U_1 + \cdots + U_m$ is a subspace of V.

Proof. First, $\mathbf{0} \in U_i$ for all U_i implies that $\mathbf{0} = \mathbf{0} + \cdots + \mathbf{0} \in \sum_{i=1}^m U_i$. Now let $\boldsymbol{u}, \boldsymbol{v} \in \sum_{i=1}^m U_i$. Then $\boldsymbol{u} = \sum_{i=1}^m \boldsymbol{u}_i$ and $\boldsymbol{v} = \sum_{i=1}^m \boldsymbol{v}_i$, where $\boldsymbol{u}_i, \boldsymbol{v}_i \in U_i$, and so $\boldsymbol{u} + \boldsymbol{v} = \sum_{i=1}^m (\boldsymbol{u}_i + \boldsymbol{v}_i) \in \sum_{i=1}^m U_i$ since $\boldsymbol{u}_i + \boldsymbol{v}_i \in U_i$ for all i. Finally, let $\boldsymbol{u} = \sum_{i=1}^m \boldsymbol{u}_i \in \sum_{i=1}^m U_i$ and $a \in \mathbb{F}$. Then $a\boldsymbol{u} = \sum_{i=1}^m (a\boldsymbol{u}_i) \in \sum_{i=1}^m U_i$. □

Exercises

► Exercise 1.3 (1.1). *Suppose a and b are real numbers, not both 0. Find real numbers c and d such that $1/(a+bi) = c + di$.*

Solution. Note that for $\boldsymbol{z} \in \mathbb{C}$ with $\boldsymbol{z} \neq 0$, there exists a unique $\boldsymbol{w} \in \mathbb{C}$ such that $\boldsymbol{z}\boldsymbol{w} = 1$; that is, $\boldsymbol{w} = 1/\boldsymbol{z}$. Let $\boldsymbol{z} = a + bi$ and $\boldsymbol{w} = c + di$. Then

$$(a+bi)(c+di) = (ac - bd) + (ad + bc)i = 1 + 0i$$

yields

$$\begin{cases} ac - bd = 1, \\ ad + bc = 0 \end{cases} \implies \begin{cases} c = a/(a^2+b^2), \\ d = -b/(a^2+b^2). \end{cases}$$ □

► Exercise 1.4 (1.2). *Show that $(-1+\sqrt{3}i)/2$ is a cube root of 1.*

PROOF. We have

$$\begin{aligned}\left(\frac{-1+\sqrt{3}i}{2}\right)^3 &= \left(\frac{-1+\sqrt{3}i}{2}\right)^2 \cdot \left(\frac{-1+\sqrt{3}i}{2}\right)\\ &= \left(-\frac{1}{2}-\frac{\sqrt{3}}{2}i\right)\cdot\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)\\ &= 1.\end{aligned}$$

□

► EXERCISE 1.5 (1.3). *Prove that $-(-\boldsymbol{v}) = \boldsymbol{v}$ for very $\boldsymbol{v} \in V$.*

PROOF. We have $\boldsymbol{v} + (-\boldsymbol{v}) = 0$, so by the uniqueness of additive inverse, the additive inverse of $-\boldsymbol{v}$, i.e., $-(-\boldsymbol{v})$, is $\boldsymbol{v}$. □

► EXERCISE 1.6 (1.4). *Prove that if $a \in \mathbb{F}$, $\boldsymbol{v} \in V$, and $a\boldsymbol{v} = \mathbf{0}$, then $a = 0$ or $\boldsymbol{v} = \mathbf{0}$.*

PROOF. Suppose that $\boldsymbol{v} \neq \mathbf{0}$ and $a \neq 0$. Then $\boldsymbol{v} = 1 \cdot \boldsymbol{v} = (a\boldsymbol{v})/a = \mathbf{0}/a = \mathbf{0}$. A contradiction. □

► EXERCISE 1.7 (1.5). *For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$:*

a. $U = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$;

b. $U = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$;

c. $U = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1x_2x_3 = 0\}$;

d. $U = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$.

SOLUTION. (a) Additive Identity: $\mathbf{0} \in U$; Closed under Addition: Let $\boldsymbol{x}, \boldsymbol{y} \in U$, then $\boldsymbol{x} + \boldsymbol{y} = (x_1 + y_1, x_2 + y_2, x_3 + y_3)$, and $(x_1 + y_1) + 2(x_2 + y_2) + 3(x_3 + y_3) = (x_1 + 2x_2 + 3x_3) + (y_1 + 2y_2 + 3y_3) = 0 + 0 = 0$; that is, $\boldsymbol{x} + \boldsymbol{y} \in U$. Closed under Scalar Multiplication: Pick any $a \in \mathbb{F}$ and $\boldsymbol{x} \in U$. Then $ax_1 + 2 \cdot (ax_2) + 3 \cdot (ax_3) = a \cdot (x_1 + 2x_2 + 3x_3) = 0$, i.e., $a\boldsymbol{x} \in U$. In sum, U is a subspace of $\mathbb{F}^3$, and actually, U is a hyperplane through the $\mathbf{0}$.

(b) U is not a subspace because $\mathbf{0} \notin U$.

(c) Let $\boldsymbol{x} = (1, 1, 0)$ and $\boldsymbol{y} = (0, 0, 1)$. Then $\boldsymbol{x}, \boldsymbol{y} \in U$, but $\boldsymbol{x} + \boldsymbol{y} = (1, 1, 1) \notin U$.

(d) $\mathbf{0} \in U$; Let $\boldsymbol{x}, \boldsymbol{y} \in U$. Then $x_1 + y_1 = 5(x_3 + y_3)$. Let $a \in \mathbb{F}$ and $\boldsymbol{x} \in U$. Then $ax_1 = a \cdot 5x_3$. □

► EXERCISE 1.8 (1.6). *Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-\boldsymbol{u} \in U$ whenever $\boldsymbol{u} \in U$), but U is not a subspace of $\mathbb{R}^2$.*

SOLUTION. Let $U = \mathbb{Z}^2$, which is not closed under scalar multiplication. □

► EXERCISE 1.9 (1.7). *Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbb{R}^2$.*

SOLUTION. Let

$$U = \left\{(x, y) \in \mathbb{R}^2 \colon x = y\right\} \cup \left\{(x, y) \in \mathbb{R}^2 \colon x = -y\right\}.$$

In this case, $(x, x) + (x, -x) = (2x, 0) \notin U$ unless $x = 0$. □

► EXERCISE 1.10 (1.8). *Prove that the intersection of any collection of subspaces of V is a subspace of V.*

PROOF. Let $\{U_i\}$ be a collection of subspaces of V. (i) Every U_i is a subspace, then $\mathbf{0} \in U_i$ for all i and so $\mathbf{0} \in \bigcap U_i$. (ii) Let $\boldsymbol{x}, \boldsymbol{y} \in \bigcap U_i$. Then $\boldsymbol{x}, \boldsymbol{y} \in U_i$ for all i and so $\boldsymbol{x} + \boldsymbol{y} \in U_i$, which implies that $\boldsymbol{x} + \boldsymbol{y} \in \bigcap U_i$. (iii) Let $a \in \mathbb{F}$ and $\boldsymbol{x} \in \bigcap U_i$. Then $a\boldsymbol{x} \in U_i$ for all i implies that $a\boldsymbol{x} \in \bigcap U_i$. □

► EXERCISE 1.11 (1.9). *Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.*

PROOF. Let U and W be two subspaces of V. The "If" part is trivial. So we focus on the "Only if" part. Let $U \cup W$ be a subspace. Suppose $U \not\subseteq W$ and $W \not\subseteq U$. Pick $\boldsymbol{x} \in U \smallsetminus W$ and $\boldsymbol{y} \in W \smallsetminus U$. Then $\boldsymbol{x} + \boldsymbol{y} \notin U$; for otherwise $\boldsymbol{y} = (\boldsymbol{x} + \boldsymbol{y}) - \boldsymbol{x} \in U$; similarly, $\boldsymbol{x} + \boldsymbol{y} \notin W$. But then $\boldsymbol{x} + \boldsymbol{y} \notin U \cup W$, which contradicts the fact that $\boldsymbol{x}, \boldsymbol{y} \in U \cup W$ and $U \cup W$ is a subspace.

A nontrivial vector space V over an infinite field $\mathbb{F}$ is not the union of a finite number of proper subspaces; see Roman (2008, Theorem 1.2). □

► EXERCISE 1.12 (1.10). *Suppose that U is a subspace of V. What is $U + U$?*

SOLUTION. Since $U \subset U$ and $U + U$ is the smallest subspace containing U, we have $U + U \subset U$; on the other hand, $U \subset U + U$ is clear. Hence, $U + U = U$. □

► EXERCISE 1.13 (1.11). *Is the operation of addition on the subspaces of V commutative? Associative?*

SOLUTION. Yes. Let U_1, U_2 and U_3 be subspaces of V.

$$\begin{aligned} U_1 + U_2 &= \{\boldsymbol{u}_1 + \boldsymbol{u}_2 \colon \boldsymbol{u}_1 \in U_1, \boldsymbol{u}_2 \in U_2\} \\ &= \{\boldsymbol{u}_2 + \boldsymbol{u}_1 \colon \boldsymbol{u}_2 \in U_2, \boldsymbol{u}_1 \in U_1\} \\ &= U_2 + U_1. \end{aligned}$$

Similarly for associativity. □

► EXERCISE 1.14 (1.12). *Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?*

SOLUTION. The set $\{\mathbf{0}\}$ is the additive identity: $U + \{\mathbf{0}\} = \{\boldsymbol{u} + \mathbf{0} \colon \boldsymbol{u} \in U\} = U$.

Only the set $\{\mathbf{0}\}$ has additive inverse. Suppose that U is a subspace, and its additive inverse is W, i.e., $U + W = \{\boldsymbol{u} + \boldsymbol{w} \colon \boldsymbol{u} \in U \text{ and } \boldsymbol{w} \in W\} = \{\mathbf{0}\}$. Since $\mathbf{0} \in U$, we have $\mathbf{0} + \boldsymbol{w} = \mathbf{0}$ for all $\boldsymbol{w} \in W$, which means that $W = \{\mathbf{0}\}$. But it is clearly that $U + \{\mathbf{0}\} = \{\mathbf{0}\}$ iff $U = \{\mathbf{0}\}$. □

► EXERCISE 1.15 (1.13). *Prove or give a counterexample: if U_1, U_2, W are subspaces of V such that $U_1 + W = U_2 + W$, then $U_1 = U_2$.*

SOLUTION. Suppose $U_1, U_2 \subseteq W$. Then $U_1 + W = U_2 + W$ for any U_1 and U_2. Hence, the statement is false in general. □

► EXERCISE 1.16 (1.14). *Suppose U is the subspace of $\mathfrak{P}(\mathbb{F})$ consisting of all polynomials p of the form $p(z) = az^2 + bz^5$, where $a, b \in \mathbb{F}$. Find a subspace W of $\mathfrak{P}(\mathbb{F})$ such that $\mathfrak{P}(\mathbb{F}) = U \oplus W$.*

SOLUTION. Let

$$W = \left\{p \in \mathfrak{P}(\mathbb{F}) \colon p(z) = a_0 + a_1 z + a_3 z^3 + a_4 z^4\right\}. \qquad \square$$

► EXERCISE 1.17 (1.15). *Prove or give a counterexample: if U_1, U_2, W are subspaces of V such that $V = U_1 \oplus W$ and $V = U_2 \oplus W$, then $U_1 = U_2$.*

SOLUTION. Let $V = \mathbb{R}^2$, $W = \{(x, 0) \in \mathbb{R}^2 \colon x \in \mathbb{R}\}$, $U_1 = \{(x, x) \in \mathbb{R}^2 \colon x \in \mathbb{R}\}$, and $U_2 = \{(x, -x) \in \mathbb{R}^2 \colon x \in \mathbb{R}\}$. Then

$$\begin{aligned} U_1 + W &= \left\{(x + y, x) \in \mathbb{R}^2 \colon x, y \in \mathbb{R}\right\} = \mathbb{R}^2 = V, \\ U_2 + W &= \left\{(x + y, -x) \in \mathbb{R}^2 \colon x, y \in \mathbb{R}\right\} = \mathbb{R}^2 = V, \\ U_i \cap W &= \{(0, 0)\}, i = 1, 2. \end{aligned}$$

Therefore, $V = U_i \oplus W$ for $i = 1, 2$, but $U_1 \neq U_2$. □

2

FINITE-DIMENSIONAL VECTOR SPACES

"As You Should Verify"

Remark 2.1 (p.22). The span of any list of vectors in V is a subspace of V.

Proof. If $U = ()$, define $\operatorname{span}(U) = \{\mathbf{0}\}$, which is a subspace of V. Now let $U = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ be a list of vectors in V. Then $\operatorname{span}(U) = \{\sum_{i=1}^n a_i \boldsymbol{v}_i : a_i \in \mathbb{F}\}$. (i) $\mathbf{0} = \sum_{i=1}^n 0\boldsymbol{v}_i \in \operatorname{span}(U)$. (ii) Let $\boldsymbol{u} = \sum_{i=1}^n a_i \boldsymbol{v}_i$ and $\boldsymbol{v} = \sum_{i=1}^n b_i \boldsymbol{v}_i$. Then $\boldsymbol{u} + \boldsymbol{v} = \sum_{i=1}^n (a_i + b_i)\boldsymbol{v}_i \in \operatorname{span}(U)$. (iii) For every $\boldsymbol{u} = \sum_{i=1}^n a_i \boldsymbol{v}_i$, we have $a\boldsymbol{u} = \sum_{i=1}^n (aa_i)\boldsymbol{v}_i \in \operatorname{span}(U)$. □

Remark 2.2 (p.23). $\mathfrak{P}_m(\mathbb{F})$ is a subspace of $\mathfrak{P}(\mathbb{F})$.

Proof. (i) $0_{\mathfrak{P}(\mathbb{F})} \in \mathfrak{P}_m(\mathbb{F})$ since its degree is $-\infty < m$ by definition. (ii) Let $p = \sum_{i=0}^{\ell} a_\ell z^\ell$ and $q = \sum_{j=0}^n b_j z^j$, where $\ell, n \leqslant m$ and $a_\ell, b_n \neq 0$. Without loss of generality, suppose $\ell \geqslant n$. Then $p + q = \sum_{i=0}^n (a_i + b_i) z^i + \sum_{j=n+1}^{\ell} a_j z^j \in \mathfrak{P}_m(\mathbb{F})$. (iii) It is easy to see that if $p \in \mathfrak{P}_m(\mathbb{F})$ then $ap \in \mathfrak{P}_m(\mathbb{F})$. □

Exercises

► Exercise 2.3 (2.1). *Prove that if $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ spans V, then so does the list $(\boldsymbol{v}_1 - \boldsymbol{v}_2, \boldsymbol{v}_2 - \boldsymbol{v}_3, \ldots, \boldsymbol{v}_{n-1} - \boldsymbol{v}_n, \boldsymbol{v}_n)$ obtained by subtracting from each vector (except the last one) the following vector.*

Proof. We first show that $\operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n) \subseteq \operatorname{span}(\boldsymbol{v}_1 - \boldsymbol{v}_2, \ldots, \boldsymbol{v}_{n-1} - \boldsymbol{v}_n, \boldsymbol{v}_n)$. Suppose that $V = \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$. Then, for any $\boldsymbol{v} \in V$, there exist $a_1, \ldots, a_n \in \mathbb{F}$ such that

$$\begin{aligned}
\boldsymbol{v} &= a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n \\
&= a_1(\boldsymbol{v}_1 - \boldsymbol{v}_2) + (a_1 + a_2)\boldsymbol{v}_2 + a_3\boldsymbol{v}_3 + \cdots + a_n\boldsymbol{v}_n \\
&= a_1(\boldsymbol{v}_1 - \boldsymbol{v}_2) + (a_1 + a_2)(\boldsymbol{v}_2 - \boldsymbol{v}_3) + (a_1 + a_2 + a_3)\boldsymbol{v}_3 + a_4\boldsymbol{v}_4 + \cdots + a_n\boldsymbol{v}_n \\
&= \sum_{i=1}^{n-1}\left[\left(\sum_{j=1}^{i} a_j\right)(\boldsymbol{v}_i - \boldsymbol{v}_{i+1})\right] + a_n\boldsymbol{v}_n \\
&\in \operatorname{span}(\boldsymbol{v}_1 - \boldsymbol{v}_2, \boldsymbol{v}_2 - \boldsymbol{v}_3, \ldots, \boldsymbol{v}_{n-1} - \boldsymbol{v}_n, \boldsymbol{v}_n).
\end{aligned}$$

For the converse direction, let $\boldsymbol{u} \in \operatorname{span}(\boldsymbol{v}_1 - \boldsymbol{v}_2, \boldsymbol{v}_2 - \boldsymbol{v}_3, \ldots, \boldsymbol{v}_{n-1} - \boldsymbol{v}_n, \boldsymbol{v}_n)$. Then there exist $b_1, \ldots, b_n \in \mathbb{F}$ such that

$$\begin{aligned}
\boldsymbol{u} &= b_1(\boldsymbol{v}_1 - \boldsymbol{v}_2) + b_2(\boldsymbol{v}_2 - \boldsymbol{v}_3) + \cdots + b_{n-1}(\boldsymbol{v}_{n-1} - \boldsymbol{v}_n) + b_n \cdot \boldsymbol{v}_n \\
&= b_1\boldsymbol{v}_1 + (b_2 - b_1)\boldsymbol{v}_2 + (b_3 - b_2)\boldsymbol{v}_3 + \cdots + (b_n - b_{n-1})\boldsymbol{v}_n \\
&\in \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n).
\end{aligned}$$

□

► EXERCISE 2.4 (2.2). *Prove that if $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is linearly independent in V, then so is the list $(\boldsymbol{v}_1 - \boldsymbol{v}_2, \boldsymbol{v}_2 - \boldsymbol{v}_3, \ldots, \boldsymbol{v}_{n-1} - \boldsymbol{v}_n, \boldsymbol{v}_n)$ obtained by subtracting from each vector (except the last one) the following vector.*

PROOF. Let

$$0 = \sum_{i=1}^{n-1} a_i(\boldsymbol{v}_i - \boldsymbol{v}_{i+1}) + a_n\boldsymbol{v}_n = a_1\boldsymbol{v}_1 + (a_2 - a_1)\boldsymbol{v}_2 + \cdots + (a_n - a_{n-1})\boldsymbol{v}_n.$$

Since $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is linear independent, we have $a_1 = a_2 - a_1 = \cdots = a_n - a_{n-1} = 0$, i.e., $a_1 = a_2 = \cdots = a_n = 0$. □

► EXERCISE 2.5 (2.3). *Suppose $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is linearly independent in V and $\boldsymbol{w} \in V$. Prove that if $(\boldsymbol{v}_1 + \boldsymbol{w}, \ldots, \boldsymbol{v}_n + \boldsymbol{w})$ is linearly dependent, then $\boldsymbol{w} \in \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$.*

PROOF. If $(\boldsymbol{v}_1 + \boldsymbol{w}, \ldots, \boldsymbol{v}_n + \boldsymbol{w})$ is linearly dependent, then there exists a list $(a_1, \ldots, a_n) \neq \boldsymbol{0}$ such that

$$\sum_{i=1}^{n} a_i(\boldsymbol{v}_i + \boldsymbol{w}) = \sum_{i=1}^{n} a_i\boldsymbol{v}_i + \left(\sum_{i=1}^{n} a_i\right)\boldsymbol{w} = 0. \tag{2.1}$$

Since $(a_1, \ldots, a_n) \neq \boldsymbol{0}$, we know that $\sum_{i=1}^{n} a_i \neq 0$. It follows from (2.1) that

$$\boldsymbol{w} = \sum_{i=1}^{n}\left(-a_i \Big/ \sum_{j=1}^{n} a_j\right)\boldsymbol{v}_i \in \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n).$$

□

► EXERCISE 2.6 (2.4). *Suppose m is a positive integer. Is the set consisting of $\boldsymbol{0}$ and all polynomials with coefficients in $\mathbb{F}$ and with degree equal to m a subspace of $\mathfrak{P}(\mathbb{F})$?*

SOLUTION. No. Consider p, q with

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m,$$
$$q(z) = b_0 + b_1 z + \cdots - a_m z^m,$$

where $a_m \neq 0$. Then $p(z)+q(z) = (a_0+b_0)+(a_1+b_1)z+\cdots+(a_{m-1}+b_{m-1})z^{m-1}$, whose degree is less than or equal to $m-1$. Hence, this set of polynomials with degree equal to m is not closed under addition. □

► EXERCISE 2.7 (2.5). *Prove that $\mathbb{F}^\infty$ is infinite dimensional.*

PROOF. Suppose that $\mathbb{F}^\infty$ is finite dimensional. Then every linearly independent list of vectors in a finite dimensional vector space can be extended to a basis of the vector space. Consider the following list

$$((1,0,0,0,\ldots),(0,1,0,0,\ldots),(0,0,1,0,\ldots),\ldots,(0,\ldots,1,0,\ldots)),$$

where each vector is in $\mathbb{F}^\infty$, and the length of the above list is n. It is easy to show that this list is linearly independent, but it can not be expanded to a basis of $\mathbb{F}^\infty$. □

► EXERCISE 2.8 (2.6). *Prove that the real vector space consisting of all continuous real-valued functions on the interval* $[0,1]$ *is infinite dimensional.*

PROOF. Consider the following set $\{p(z) \in \mathfrak{P}(\mathbb{F}) \colon z \in [0,1]\}$, which is a subspace, but is infinite dimensional. □

► EXERCISE 2.9 (2.7). *Prove that V is infinite dimensional if and only if there is a sequence $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots$ of vectors in V such that $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is linearly independent for every positive integer n.*

PROOF. Let V be infinite dimensional. Clearly, there exists a nonzero vector $\boldsymbol{v}_1 \in V$; for otherwise, $V = \{\mathbf{0}\}$ and so V is finite dimensional. Since V is infinite dimensional, $\operatorname{span}(\boldsymbol{v}_1) \neq V$; hence there exists $\boldsymbol{v}_2 \in V \smallsetminus \operatorname{span}(\boldsymbol{v}_1)$; similarly, $\operatorname{span}(\boldsymbol{v}_1, \boldsymbol{v}_2) \neq V$; thus we can choose $\boldsymbol{v}_3 \in V \smallsetminus \operatorname{span}(\boldsymbol{v}_1, \boldsymbol{v}_2)$. We thus construct an infinite sequence $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots$

We then use the Induction Principle to prove that for every positive integer n, the list $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is linearly independent. Obviously, $\boldsymbol{v}_1$ is linear independent since $\boldsymbol{v}_1 \neq \mathbf{0}$. Let us assume that $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is linear independent for some positive integer n. We now show that $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n, \boldsymbol{v}_{n+1})$ is linear independent. If not, then there exist $a_1, \ldots, a_n, a_{n+1} \in \mathbb{F}$, not all 0, such that $\sum_{i=1}^{n+1} a_i \boldsymbol{v}_i = \mathbf{0}$. We must have $a_{n+1} \neq 0$: if $a_{n+1} = 0$, then $\sum_{i=1}^{n} a_i \boldsymbol{v}_i = 0$ implies that $a_1 = \cdots = a_n = a_{n+1} = 0$ since $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is linear independent by the induction hypothesis. Hence,

$$\boldsymbol{v}_{n+1} = \sum_{i=1}^{n} (-a_i/a_{n+1}) \boldsymbol{v}_i,$$

i.e. $\boldsymbol{v}_{n+1} \in \operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$, which contradicts the construction of $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_{n+1})$.

Conversely, assume that there exists an infinite sequence $\boldsymbol{v}_1, \boldsymbol{v}_2, \dots$ of vectors in V, and $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is linearly independent for any positive integer n. Suppose V is finite dimensional; that is, there is a spanning list of vectors $(\boldsymbol{u}_1, \dots, \boldsymbol{u}_m)$ of V, and such that the length of every linearly independent list of vectors is less than or equal to m (by Theorem 2.6). A contradiction. □

► EXERCISE 2.10 (2.8). *Let U be the subspace of $\mathbb{R}^5$ defined by*

$$U = \left\{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 \colon x_1 = 3x_2 \text{ and } x_3 = 7x_4\right\}.$$

Find a basis of U.

PROOF. A particular basis of U can be $((3, 1, 0, 0, 0), (0, 0, 7, 1, 0), (0, 0, 0, 0, 1))$. □

► EXERCISE 2.11 (2.9). *Prove or disprove: there exists a basis (p_0, p_1, p_2, p_3) of $\mathfrak{P}_3(\mathbb{F})$ such that none of the polynomials p_0, p_1, p_2, p_3 has degree 2.*

PROOF. Notice that $p_0 = 1$, $p_1 = z$, $p_2' = z^2$, and $p_3 = z^3$ is a standard basis of $\mathfrak{P}_3(\mathbb{F})$, but p_2' has degree 2. So we can let $p_2 = p_2' + p_3 = z^2 + z^3$. Then $\operatorname{span}(p_0, p_1, p_2, p_3) = \mathfrak{P}_3(\mathbb{F})$ and so (p_0, p_1, p_2, p_3) is a basis of $\mathfrak{P}_3(\mathbb{F})$ by Theorem 2.16. □

► EXERCISE 2.12 (2.10). *Suppose that V is finite dimensional, with $\dim V = n$. Prove that there exist one-dimensional subspaces $U_1, \dots, U_n$ of V such that*

$$V = U_1 \oplus \cdots \oplus U_n.$$

PROOF. Let $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ be a basis of V. For each $i = 1, \dots, n$, let $U_i = \operatorname{span}(\boldsymbol{v}_i)$. Then each U_i is a subspace of V and so $U_1 + \cdots + U_n \subseteq V$. Clearly, $\dim V = \sum_{i=1}^n \dim U_i = n$. By Proposition 2.19, it suffices to show that $V \subseteq U_1 + \cdots + U_n$. It follows because for every $\boldsymbol{v} \in V$,

$$\boldsymbol{v} = \sum_{i=1}^n a_i \boldsymbol{v}_i \in U_1 + \cdots + U_n. \qquad \square$$

► EXERCISE 2.13 (2.11). *Suppose that V is finite dimensional and U is a subspace of V such that $\dim U = \dim V$. Prove that $U = V$.*

PROOF. Let $(\boldsymbol{u}_1, \dots, \boldsymbol{u}_n)$ be a basis of U. Since $(\boldsymbol{u}_1, \dots, \boldsymbol{u}_n)$ is linearly independent in V and the length of $(\boldsymbol{u}_1, \dots, \boldsymbol{u}_n)$ is equal to $\dim V$, it is a basis of V. Therefore, $V = \operatorname{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_n) = U$. □

► EXERCISE 2.14 (2.12). *Suppose that $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_j(2) = 0$ for each j. Prove that $(p_0, p_1, \dots, p_m)$ is not linearly independent in $\mathfrak{P}_m(\mathbb{F})$.*

PROOF. $\dim \mathfrak{P}_m(\mathbb{F}) = m+1$ since $(1, z, \dots, z^m)$ is a basis of $\mathfrak{P}_m(\mathbb{F})$. If $(p_0, \dots, p_m)$ is linear independent, then it is a basis of $\mathfrak{P}_m(\mathbb{F})$ by Proposition 2.17. Then $p = \sum_{i=0}^m p_i$ for every $p \in \mathfrak{P}_m(\mathbb{F})$. Take an arbitrary $p \in \mathfrak{P}_m(\mathbb{F})$ with $p(2) \neq 0$ and we get a contradiction. □

► EXERCISE 2.15 (2.13). *Suppose U and W are subspaces of $\mathbb{R}^8$ such that* $\dim U = 3$, $\dim W = 5$, *and $U + W = \mathbb{R}^8$. Prove that $U \cap W = \{\mathbf{0}\}$.*

PROOF. Since $\mathbb{R}^8 = U + W$ and $\dim \mathbb{R}^8 = \dim U + \dim V$, we have $\mathbb{R}^8 = U \oplus V$ by Proposition 2.19; then Proposition 1.9 implies that $U \cap W = \{\mathbf{0}\}$. □

► EXERCISE 2.16 (2.14). *Suppose that U and W are both five-dimensional subspaces of $\mathbb{R}^9$. Prove that $U \cap W \neq \{\mathbf{0}\}$.*

PROOF. If $U \cap W = \{\mathbf{0}\}$, then $\dim U + W = \dim U + \dim W - \dim U \cap W = 5 + 5 - 0 = 10 > 9$; but $U + W \subseteq \mathbb{R}^9$. A contradiction. □

► EXERCISE 2.17 (2.15). *Prove or give a counterexample that*

$$\begin{aligned}\dim U_1 + U_2 + U_3 = {} & \dim U_1 + \dim U_2 + \dim U_3 \\ & - \dim U_1 \cap U_2 - \dim U_1 \cap U_3 - \dim U_2 \cap U_3 \\ & + \dim U_1 \cap U_2 \cap U_3.\end{aligned}$$

SOLUTION. We construct a counterexample to show the proposition is faulse. Let

$$\begin{aligned} U_1 &= \left\{(x, 0) \in \mathbb{R}^2 \colon x \in \mathbb{R}\right\}, \\ U_2 &= \left\{(0, x) \in \mathbb{R}^2 \colon x \in \mathbb{R}\right\}, \\ U_3 &= \left\{(x, x) \in \mathbb{R}^2 \colon x \in \mathbb{R}\right\}. \end{aligned}$$

Then $U_1 \cap U_2 = U_1 \cap U_3 = U_2 \cap U_3 = U_1 \cap U_2 \cap U_3 = \{(0, 0)\}$; hence

$$\dim U_1 \cap U_2 = \dim U_1 \cap U_3 = \dim U_2 \cap U_3 = \dim U_1 \cap U_2 \cap U_3 = 0.$$

But $\dim U_1 + U_2 + U_3 = 2$ since $U_1 + U_2 + U_3 = \mathbb{R}^2$. □

► EXERCISE 2.18 (2.16). *Prove that if V is finite dimensional and $U_1, \dots, U_m$ are subspaces of V, then* $\dim U_1 + \cdots + U_m \leqslant \sum_{i=1}^m \dim U_i$.

PROOF. Let $(\boldsymbol{u}_i^1, \dots, \boldsymbol{u}_i^{n_i})$ be a basis of U_i for each $i = 1, \dots, m$. Then

$$\sum_{i=1}^m \dim U_i = \sum_{i=1}^m n_i.$$

Let

$$(\boldsymbol{u}_1^1, \dots, \boldsymbol{u}_1^{n_1}, \dots, \boldsymbol{u}_m^1, \dots, \boldsymbol{u}_m^{n_m}) = B.$$

Clearly, $U_1 + \cdots + U_m = \operatorname{span}(B)$, and $\dim \operatorname{span}(B) \leqslant \sum_{i=1}^m n_i$ by Theorem 2.10. Therefore, $\dim U_1 + \cdots + U_m \leqslant \sum_{i=1}^m \dim U_i$. □

► EXERCISE 2.19 (2.17). *Suppose V is finite dimensional. Prove that if $U_1, \ldots, U_m$ are subspaces of V such that $V = U_1 \oplus \cdots \oplus U_m$, then* $\dim V = \sum_{i=1}^m \dim U_i$.

PROOF. Let the list $(\boldsymbol{u}_i^1, \ldots, \boldsymbol{u}_i^{n_i})$ be a basis of U_i for all $i = 1, \ldots, m$. Then $\sum_{i=1}^m \dim U_i = \sum_{i=1}^m n_i$. Let

$$(\boldsymbol{u}_1^1, \ldots, \boldsymbol{u}_1^{n_1}, \ldots, \boldsymbol{u}_m^1, \ldots, \boldsymbol{u}_m^{n_m}) = U.$$

Then $\operatorname{span}(U) = V$. We show that $(\boldsymbol{u}_1^1, \ldots, \boldsymbol{u}_1^{n_1}, \ldots, \boldsymbol{u}_m^1, \ldots, \boldsymbol{u}_m^{n_m})$ is linear independent. Let

$$\mathbf{0} = \underbrace{(a_1^1 \boldsymbol{u}_1^1 + \cdots + a_1^{n_1} \boldsymbol{u}_1^{n_1})}_{\boldsymbol{u}_1} + \cdots + \underbrace{(a_m^1 \boldsymbol{u}_m^1 + \cdots + a_m^{n_m} \boldsymbol{u}_m^{n_m})}_{\boldsymbol{u}_m}.$$

Then $\sum_{i=1}^m \boldsymbol{u}_i = \mathbf{0}$ and so $\boldsymbol{u}_i = \mathbf{0}$ for each $i = 1, \ldots, m$ (since $V = \bigoplus_{i=1}^m U_i$). But then $a_1^1 = \cdots = a_m^{n_m} = 0$. Thus, $(\boldsymbol{u}_1^1, \ldots, \boldsymbol{u}_m^{n_m})$ is linear independent and spans V, i.e. it is a basis of V. □

3

LINEAR MAPS

"As You Should Verify"

Remark 3.1 (p. 40). Given a basis $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ of V and any choice of vectors $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n \in W$, we can construct a linear map $\mathrm{T}\colon V \to W$ such that

$$\mathrm{T}(a_1\boldsymbol{v}_1 + \cdots + a_n\boldsymbol{v}_n) = a_1\boldsymbol{w}_1 + \cdots + a_n\boldsymbol{w}_n,$$

where $a_1, \ldots, a_n$ are arbitrary elements of $\mathbb{F}$. Then T is linear.

Proof. Let $\boldsymbol{u}, \boldsymbol{w} \in V$ with $\boldsymbol{u} = \sum_{i=1}^n a_i\boldsymbol{v}_i$ and $\boldsymbol{v} = \sum_{i=1}^n b_i\boldsymbol{v}_i$; let $a \in \mathbb{F}$. Then

$$\begin{aligned}\mathrm{T}(\boldsymbol{u}+\boldsymbol{v}) = \mathrm{T}\left(\sum_{i=1}^n (a_i+b_i)\boldsymbol{v}_i\right) &= \sum_{i=1}^n (a_i+b_i)\boldsymbol{w}_i \\ &= \sum_{i=1}^n a_i\boldsymbol{w}_i + \sum_{i=1}^n b_i\boldsymbol{w}_i \\ &= \mathrm{T}\boldsymbol{u} + \mathrm{T}\boldsymbol{v},\end{aligned}$$

and

$$\mathrm{T}(a\boldsymbol{u}) = \mathrm{T}\left(\sum_{i=1}^n (aa_i)\boldsymbol{v}_i\right) = \sum_{i=1}^n (aa_i)\boldsymbol{w}_i = a\left(\sum_{i=1}^n a_i\boldsymbol{w}_i\right) = a\mathrm{T}\boldsymbol{u}. \qquad \square$$

Remark 3.2 (p. 40-41). Let $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V, W)$. Then $\mathrm{S}+\mathrm{T}, a\mathrm{T} \in \mathfrak{L}(V, W)$.

Proof. As for $\mathrm{S}+\mathrm{T}$, we have $(\mathrm{S}+\mathrm{T})(\boldsymbol{u}+\boldsymbol{v}) = \mathrm{S}(\boldsymbol{u}+\boldsymbol{v}) + \mathrm{T}(\boldsymbol{u}+\boldsymbol{v}) = \mathrm{S}\boldsymbol{u} + \mathrm{S}\boldsymbol{v} + \mathrm{T}\boldsymbol{u} + \mathrm{T}\boldsymbol{v} = (\mathrm{S}+\mathrm{T})(\boldsymbol{u}) + (\mathrm{S}+\mathrm{T})(\boldsymbol{v})$, and $(\mathrm{S}+\mathrm{T})(a\boldsymbol{v}) = \mathrm{S}(a\boldsymbol{v}) + \mathrm{T}(a\boldsymbol{v}) = a(\mathrm{S}+\mathrm{T})(\boldsymbol{v})$.

As for $a\mathrm{T}$, we have $(a\mathrm{T})(\boldsymbol{u}+\boldsymbol{v}) = a\left[\mathrm{T}(\boldsymbol{u}+\boldsymbol{v})\right] = a\left[\mathrm{T}\boldsymbol{u} + \mathrm{T}\boldsymbol{v}\right] = a\mathrm{T}\boldsymbol{u} + a\mathrm{T}\boldsymbol{v} = (a\mathrm{T})\boldsymbol{u} + (a\mathrm{T})\boldsymbol{v}$, and $(a\mathrm{T})(b\boldsymbol{v}) = a\left[\mathrm{T}(b\boldsymbol{v})\right] = ab\mathrm{T}\boldsymbol{v} = b(a\mathrm{T})\boldsymbol{v}$. $\square$

Exercises

► Exercise 3.3 (3.1). *Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if* $\dim V = 1$ *and* $\mathrm{T} \in \mathfrak{L}(V, V)$*, then there exists* $a \in \mathbb{F}$ *such that* $\mathrm{T}\boldsymbol{v} = a\boldsymbol{v}$ *for all* $\boldsymbol{v} \in V$*.*

Proof. Let $\boldsymbol{w} \in V$ be a basis of V. Then $\mathrm{T}\boldsymbol{w} = a\boldsymbol{w}$ for some $a \in \mathbb{F}$. For an arbitrary $\boldsymbol{v} \in V$, there exists $b \in \mathbb{F}$ such that $\boldsymbol{v} = b\boldsymbol{w}$. Then

$$\mathrm{T}\boldsymbol{v} = \mathrm{T}(b\boldsymbol{w}) = b(\mathrm{T}\boldsymbol{w}) = b(a\boldsymbol{w}) = a(b\boldsymbol{w}) = a\boldsymbol{v}. \qquad \square$$

► Exercise 3.4 (3.2). *Give an example of a function* $f\colon \mathbb{R}^2 \to \mathbb{R}$ *such that* $f(a\boldsymbol{v}) = af(\boldsymbol{v})$ *for all* $a \in \mathbb{R}$ *and all* $\boldsymbol{v} \in \mathbb{R}^2$ *but* f *is not linear.*

Proof. For any $\boldsymbol{v} = (v_1, v_2) \in \mathbb{R}^2$, let

$$f(v_1, v_2) = \begin{cases} v_1 & \text{if } v_1 = v_2 \\ 0 & \text{if } v_1 \neq v_2. \end{cases}$$

Now consider $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^2$ with $u_1 \neq u_2$, $v_1 \neq v_2$, but $u_1 + v_1 = u_2 + v_2 > 0$. Notice that

$$f(\boldsymbol{u} + \boldsymbol{v}) = u_1 + v_1 > 0 = f(\boldsymbol{u}) + f(\boldsymbol{v}).$$

Hence, f is not linear. $\square$

► Exercise 3.5 (3.3). *Suppose that* V *is finite dimensional. Prove that any linear map on a subspace of* V *can be extended to a linear map on* V*. In other words, show that if* U *is a subspace of* V *and* $\mathrm{S} \in \mathfrak{L}(U, W)$*, then there exists* $\mathrm{T} \in \mathfrak{L}(V, W)$ *such that* $\mathrm{T}\boldsymbol{u} = \mathrm{S}\boldsymbol{u}$ *for all* $\boldsymbol{u} \in U$*.*

Proof. Let $(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m)$ be a basis of U, and extend it to a basis of V:

$$(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_n).$$

Choose n vectors $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$ from W. Define a map $\mathrm{T}\colon V \to W$ by letting

$$\mathrm{T}\left(\sum_{i=1}^{m} a_i \boldsymbol{u}_i + \sum_{j=1}^{n} a_j \boldsymbol{v}_j\right) = \sum_{i=1}^{m} a_i \mathrm{S}\boldsymbol{u}_i + \sum_{j=1}^{n} a_j \boldsymbol{w}_j.$$

It is trivial to see that $\mathrm{S}\boldsymbol{u} = \mathrm{T}\boldsymbol{u}$ for all $\boldsymbol{u} \in U$. So we only show that T is a linear map. Let $\boldsymbol{u}, \boldsymbol{v} \in V$ with $\boldsymbol{u} = \sum_{i=1}^{m} a_i \boldsymbol{u}_i + \sum_{j=1}^{n} a_j \boldsymbol{v}_j$ and $\boldsymbol{v} = \sum_{i=1}^{m} b_i \boldsymbol{u}_i + \sum_{j=1}^{n} b_j \boldsymbol{v}_j$; let $a \in \mathbb{F}$. Then

$$\begin{aligned}
\mathrm{T}(\boldsymbol{u}+\boldsymbol{v}) &= \mathrm{T}\left(\sum_{i=1}^{m}(a_i+b_i)\boldsymbol{u}_i + \sum_{j=1}^{n}\left(a_j+b_j\right)\boldsymbol{v}_j\right) \\
&= \sum_{i=1}^{m}(a_i+b_i)\mathrm{S}\boldsymbol{u}_i + \sum_{j=1}^{n}(a_j+b_j)\boldsymbol{w}_j \\
&= \left[\sum_{i=1}^{m}a_i\mathrm{S}\boldsymbol{u}_i + \sum_{j=1}^{n}a_j\boldsymbol{w}_j\right] + \left[\sum_{i=1}^{m}b_i\mathrm{S}\boldsymbol{u}_i + \sum_{j=1}^{n}b_j\boldsymbol{w}_j\right] \\
&= \mathrm{T}\boldsymbol{u} + \mathrm{T}\boldsymbol{v},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{T}a\boldsymbol{u} = \mathrm{T}\left(a\left(\sum_{i=1}^{m}a_i\boldsymbol{u}_i + \sum_{j=1}^{n}a_j\boldsymbol{v}_j\right)\right) &= \mathrm{T}\left(\sum_{i=1}^{m}aa_i\boldsymbol{u}_i + \sum_{j=1}^{n}aa_j\boldsymbol{v}_j\right) \\
&= \sum_{i=1}^{m}aa_i\mathrm{S}\boldsymbol{u}_i + \sum_{j=1}^{n}aa_j\boldsymbol{w}_j \\
&= a\left[\sum_{i=1}^{m}a_i\mathrm{S}\boldsymbol{u}_i + \sum_{j=1}^{n}a_j\boldsymbol{w}_j\right] \\
&= a\mathrm{T}\boldsymbol{u}.
\end{aligned}$$

□

► EXERCISE 3.6 (3.4). *Suppose that* T *is a linear map from* V *to* $\mathbb{F}$. *Prove that if* $\boldsymbol{u} \in V$ *is not in* $\mathcal{N}_{\mathrm{T}}$, *then*

$$V = \mathcal{N}_{\mathrm{T}} \oplus \{a\boldsymbol{u} \colon a \in \mathbb{F}\}.$$

PROOF. Let $\mathrm{T} \in \mathfrak{L}(V, \mathbb{F})$. Since $\boldsymbol{u} \in V \smallsetminus \mathcal{N}_{\mathrm{T}}$, we get $\boldsymbol{u} \neq \boldsymbol{0}$ and $\mathrm{T}\boldsymbol{u} \neq 0$. Thus, $\dim \mathcal{R}_{\mathrm{T}} \geqslant 1$. Since $\dim \mathcal{R}_{\mathrm{T}} \leqslant \dim \mathbb{F} = 1$, we get $\dim \mathcal{R}_{\mathrm{T}} = 1$. It follows from Theorem 3.4 that

$$\dim V = \dim \mathcal{N}_{\mathrm{T}} + 1 = \dim \mathcal{N}_{\mathrm{T}} + \dim \{a\boldsymbol{u} \colon a \in \mathbb{F}\}. \tag{3.1}$$

Let $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m)$ be a basis of $\mathcal{N}_{\mathrm{T}}$. Then $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{u})$ is linear independent since $\boldsymbol{u} \notin \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m) = \mathcal{N}_{\mathrm{T}}$. It follows from (3.1) that $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{u})$ is a basis of V (by Proposition 2.17). Therefore

$$\begin{aligned}
V = \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{u}) &= \left\{\sum_{i=1}^{m}a_i\boldsymbol{v}_i + a\boldsymbol{u} \colon a_1, \ldots, a_m, a \in \mathbb{F}\right\} \\
&= \left\{\sum_{i=1}^{m}a_i\boldsymbol{v}_i \colon a_1, \ldots, a_m \in \mathbb{F}\right\} + \{a\boldsymbol{u} \colon a \in \mathbb{F}\} \\
&= \mathcal{N}_{\mathrm{T}} + \{a\boldsymbol{u} \colon a \in \mathbb{F}\}.
\end{aligned} \tag{3.2}$$

It follows from (3.1) and (3.2) that $V = \mathcal{N}_{\mathrm{T}} \oplus \{a\boldsymbol{u} \colon a \in \mathbb{F}\}$ by Proposition 2.19. □

► EXERCISE 3.7 (3.5). *Suppose that* $\mathrm{T} \in \mathfrak{L}(V, W)$ *is injective and* $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ *is linearly independent in* V. *Prove that* $(\mathrm{T}\boldsymbol{v}_1, \ldots, \mathrm{T}\boldsymbol{v}_n)$ *is linearly independent in* W.

PROOF. Let

$$\mathbf{0} = \sum_{i=1}^{n} a_i \cdot \mathrm{T}\boldsymbol{v}_i = \mathrm{T}\left(\sum_{i=1}^{n} a_i \boldsymbol{v}_i\right).$$

Then $\sum_{i=1}^{n} a_i \boldsymbol{v}_i = \mathbf{0}$ since $\mathcal{N}_{\mathrm{T}} = \{\mathbf{0}\}$. The linear independence of $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ implies that $a_1 = \cdots = a_n = 0$. □

► EXERCISE 3.8 (3.6). *Prove that if* $\mathrm{S}_1, \ldots, \mathrm{S}_n$ *are injective linear maps such that* $\mathrm{S}_1 \cdots \mathrm{S}_n$ *makes sense, then* $\mathrm{S}_1 \cdots \mathrm{S}_n$ *is injective.*

PROOF. We use mathematical induction to prove this claim. It holds for $n = 1$ trivially. Let us suppose that $\mathrm{S}_1 \cdots \mathrm{S}_n$ is injective if $\mathrm{S}_1, \ldots, \mathrm{S}_n$ are. Now assume that $\mathrm{S}_1, \ldots, \mathrm{S}_{n+1}$ are all injective linear maps. Let $\mathrm{T} = \mathrm{S}_1 \cdots \mathrm{S}_{n+1}$. For every $\boldsymbol{v} \in \mathcal{N}_{\mathrm{T}}$ we have

$$\mathbf{0} = \mathrm{T}\boldsymbol{v} = (\mathrm{S}_1 \cdots \mathrm{S}_n)(\mathrm{S}_{n+1}\boldsymbol{v}).$$

But the above display implies that $\mathrm{S}_{n+1}\boldsymbol{v} = \mathbf{0}$ since $(\mathrm{S}_1 \cdots \mathrm{S}_n)$ is injective by the induction hypothesis, which implies further that $\boldsymbol{v} = \mathbf{0}$ since S_{n+1} is injective. This proves that $\mathcal{N}_{\mathrm{T}} = \{\mathbf{0}\}$ and so T is injective. □

► EXERCISE 3.9 (3.7). *Prove that if* $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ *spans* V *and* $\mathrm{T} \in \mathfrak{L}(V, W)$ *is surjective, then* $(\mathrm{T}\boldsymbol{v}_1, \ldots, \mathrm{T}\boldsymbol{v}_n)$ *spans* W.

PROOF. Since T is surjective, for any $\boldsymbol{w} \in W$, there exists $\boldsymbol{v} \in V$ such that $\mathrm{T}\boldsymbol{v} = \boldsymbol{w}$; since $V = \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$, there exists $(a_1, \ldots, a_n) \in \mathbb{F}^n$ such that $\boldsymbol{v} = \sum_{i=1}^{n} a_i \boldsymbol{v}_i$. Hence,

$$\boldsymbol{w} = \mathrm{T}\left(\sum_{i=1}^{n} a_i \boldsymbol{v}_i\right) = \sum_{i=1}^{n} a_i \mathrm{T}\boldsymbol{v}_i,$$

that is, $W = \operatorname{span}(\mathrm{T}\boldsymbol{v}_1, \ldots, \mathrm{T}\boldsymbol{v}_n)$. □

► EXERCISE 3.10 (3.8). *Suppose that* V *is finite dimensional and that* $\mathrm{T} \in \mathfrak{L}(V, W)$. *Prove that there exists a subspace* U *of* V *such that* $U \cap \mathcal{N}_{\mathrm{T}} = \{\mathbf{0}\}$ *and* $\mathcal{R}_{\mathrm{T}} = \{\mathrm{T}\boldsymbol{u} \colon \boldsymbol{u} \in U\}$.

PROOF. Let $(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m)$ be a basis of $\mathcal{N}_{\mathrm{T}}$, which can be extended to a basis $(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ of V. Let $U = \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$. Then $U \cap \mathcal{N}_{\mathrm{T}} = \{\mathbf{0}\}$ (see the proof of Proposition 2.13).

To see $\mathcal{R}_{\mathrm{T}} = \{\mathrm{T}\boldsymbol{u} : \boldsymbol{u} \in U\}$, take an arbitrary $\boldsymbol{v} \in V$. Then

$$\mathrm{T}\boldsymbol{v} = \mathrm{T}\left(\sum_{i=1}^{m} a_i \boldsymbol{u}_i + \sum_{j=1}^{n} a_j \boldsymbol{v}_j\right) = \mathrm{T}\left(\sum_{j=1}^{n} a_j \boldsymbol{v}_j\right) = \mathrm{T}\boldsymbol{u}$$

for some $\boldsymbol{u} = \sum_{j=1}^{n} a_j \boldsymbol{v}_j \in U$. □

► EXERCISE 3.11 (3.9). *Prove that if* T *is a linear map from* $\mathbb{F}^4$ *to* $\mathbb{F}^2$ *such that*

$$\mathcal{N}_{\mathrm{T}} = \left\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \colon x_1 = 5x_2 \text{ and } x_3 = 7x_4\right\},$$

then T *is surjective.*

PROOF. Let

$$\boldsymbol{v}_1 = \begin{pmatrix} 5 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 0 \\ 7 \\ 1 \end{pmatrix}.$$

It is easy to see that $(\boldsymbol{v}_1, \boldsymbol{v}_2)$ is a basis of $\mathcal{N}_{\mathrm{T}}$; that is, $\dim \mathcal{N}_{\mathrm{T}} = 2$. Then

$$\dim \mathcal{R}_{\mathrm{T}} = \dim \mathbb{F}^4 - \dim \mathcal{N}_{\mathrm{T}} = 4 - 2 = 2 = \dim \mathbb{F}^2,$$

and so T is surjective. □

► EXERCISE 3.12 (3.10). *Prove that there does not exist a linear map from* $\mathbb{F}^5$ *to* $\mathbb{F}^2$ *whose null space equals*

$$\left\{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{F}^5 \colon x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\right\}.$$

PROOF. It is easy to see that the following two vectors consist of a basis of $\mathcal{N}_{\mathrm{T}}$ if $\mathrm{T} \in \mathcal{L}(\mathbb{F}^5, \mathbb{F}^2)$:

$$\boldsymbol{v}_1 = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then, $\dim \mathcal{N}_{\mathrm{T}} = 2$ and so $\dim \mathcal{R}_{\mathrm{T}} = 5 - 2 = 3 > \dim \mathbb{F}^2$, which is impossible. □

► EXERCISE 3.13 (3.11). *Prove that if there exists a linear map on* V *whose null space and range are both finite dimensional, then* V *is finite dimensional.*

PROOF. If $\dim \mathcal{N}_{\mathrm{T}} < \infty$ and $\dim \mathcal{R}_{\mathrm{T}} < \infty$, then $\dim V = \dim \mathcal{N}_{\mathrm{T}} + \dim \mathcal{R}_{\mathrm{T}} < \infty$. □

► EXERCISE 3.14 (3.12). *Suppose that* V *and* W *are both finite dimensional. Prove that there exists a surjective linear map from* V *onto* W *if and only if* $\dim W \leqslant \dim V$.

PROOF. If there exists a surjective linear map $\mathrm{T} \in \mathcal{L}(V, W)$, then $\dim W = \dim \mathcal{R}_{\mathrm{T}} = \dim V - \dim \mathcal{N}_{\mathrm{T}} \geqslant \dim V$.

Now let $\dim W \leqslant \dim V$. Let $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ be a basis of V, and let $(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_m)$ be a basis of W, with $m \leqslant n$. Define $\mathrm{T} \in \mathcal{L}(V, W)$ by letting

$$\mathrm{T}\left(\sum_{i=1}^{m} a_i \boldsymbol{v}_i + \sum_{j=m+1}^{n} a_j \boldsymbol{v}_j\right) = \sum_{i=1}^{m} a_i \boldsymbol{w}_i .$$

Then for every $\boldsymbol{w} = \sum_{i=1}^{m} a_i \boldsymbol{w}_i \in W$, there exists $\boldsymbol{v} = \sum_{j=1}^{n} a_j \boldsymbol{v}_j$ such that $\mathrm{T}\boldsymbol{v} = \boldsymbol{w}$, i.e. T is surjective. □

► EXERCISE 3.15 (3.13). *Suppose that V and W are finite dimensional and that U is a subspace of V. Prove that there exists $\mathrm{T} \in \mathfrak{L}(V, W)$ such that $\mathcal{N}_{\mathrm{T}} = U$ if and only if $\dim U \geqslant \dim V - \dim W$.*

PROOF. For every $\mathrm{T} \in \mathfrak{L}(V, W)$, if $\mathcal{N}_{\mathrm{T}} = U$, then $\dim U = \dim V - \dim \mathcal{R}_{\mathrm{T}} \geqslant \dim V - \dim W$.

Now let $\dim U \geqslant \dim V - \dim W$. Let $(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m)$ be a basis of U, which can be extended to a basis $(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ of V. Let $(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_p)$ be a basis of W. Then $m \geqslant (m+n) - p$ implies that $n \leqslant p$. Define $\mathrm{T} \in \mathfrak{L}(V, W)$ by letting

$$\mathrm{T}\left(\sum_{i=1}^{m} a_i \boldsymbol{u}_i + \sum_{j=1}^{n} a_j \boldsymbol{v}_j\right) = \sum_{j=1}^{n} a_j \boldsymbol{w}_j .$$

Then $\mathcal{N}_{\mathrm{T}} = U$. □

► EXERCISE 3.16 (3.14). *Suppose that W is finite dimensional and $\mathrm{T} \in \mathfrak{L}(V, W)$. Prove that T is injective if and only if there exists $\mathrm{S} \in \mathfrak{L}(W, V)$ such that ST is the identity map on V.*

PROOF. Suppose first that $\mathrm{ST} = \mathrm{Id}_V$. Then for any $\boldsymbol{u}, \boldsymbol{v} \in V$ with $\boldsymbol{u} \neq \boldsymbol{v}$, we have $\boldsymbol{u} = (\mathrm{ST})\boldsymbol{u} \neq (\mathrm{ST})\boldsymbol{v} = \boldsymbol{v}$; that is, $\mathrm{S}(\mathrm{T}\boldsymbol{u}) \neq \mathrm{S}(\mathrm{T}\boldsymbol{v})$, and so $\mathrm{T}\boldsymbol{u} \neq \mathrm{T}\boldsymbol{v}$.

For the inverse direction, let T be injective. Then $\dim V \leqslant \dim W$ by Corollary 3.5. Also, $\dim W < +\infty$. Let $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ be a basis of V. It follows from Exercise 3.7 that $(\mathrm{T}\boldsymbol{v}_1, \ldots, \mathrm{T}\boldsymbol{v}_n)$ is linearly independent, and so can be extended to a basis $(\mathrm{T}\boldsymbol{v}_1, \ldots, \mathrm{T}\boldsymbol{v}_n, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_m)$ of W. Define $\mathrm{S} \in \mathfrak{L}(W, V)$ by letting

$$\mathrm{S}\,(\mathrm{T}\boldsymbol{v}_i) = (\mathrm{ST})\,\boldsymbol{v}_i = \boldsymbol{v}_i, \quad \text{and} \quad \mathrm{S}\,(\boldsymbol{w}_i) = \boldsymbol{0}_V .$$ □

► EXERCISE 3.17 (3.15). *Suppose that V is finite dimensional and $\mathrm{T} \in \mathfrak{L}(V, W)$. Prove that T is surjective if and only if there exists $\mathrm{S} \in \mathfrak{L}(W, V)$ such that TS is the identity map on W.*

PROOF. If $\mathrm{TS} = \mathrm{Id}_W$, then for any $\boldsymbol{w} \in W$, we have $\mathrm{T}\,(\mathrm{S}\boldsymbol{w}) = \mathrm{Id}_W\,(\boldsymbol{w}) = \boldsymbol{w}$, that is, there exists $\mathrm{S}\boldsymbol{w} \in V$ such that $\mathrm{T}\,(\mathrm{S}\boldsymbol{w}) = \boldsymbol{w}$, and so T is surjective.

If T is surjective, then $\dim W \leqslant \dim V$. Let $(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_m)$ be a basis of W, and let $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ be a basis of V with $n \geqslant m$. Define $\mathrm{S} \in \mathfrak{L}(W, V)$ by letting

$$\mathrm{S}\boldsymbol{w}_i = \boldsymbol{v}_i, \quad \text{with} \quad \mathrm{T}\boldsymbol{v}_i = \boldsymbol{w}_i .$$ □

► EXERCISE 3.18 (3.16[1]). *Suppose that U and V are finite-dimensional vector spaces and that* $\mathrm{S} \in \mathfrak{L}(V, W)$, $\mathrm{T} \in \mathfrak{L}(U, V)$. *Prove that*

$$\dim \mathcal{N}_{\mathrm{ST}} \leqslant \dim \mathcal{N}_{\mathrm{S}} + \dim \mathcal{N}_{\mathrm{T}}.$$

PROOF. We have $W \overset{\mathrm{S}}{\leftarrow} V \overset{\mathrm{T}}{\leftarrow} U$. Since

$$\mathcal{R}_{\mathrm{ST}} = (\mathrm{ST})\,[U] = \mathrm{S}\big[\mathrm{T}\,[U]\big] = \mathrm{S}\,[\mathcal{R}_{\mathrm{T}}]\,,$$

we have

$$\dim \mathcal{R}_{\mathrm{ST}} = \dim \mathrm{S}\,[\mathcal{R}_{\mathrm{T}}].$$

Let N be the complement of $\mathcal{R}_{\mathrm{T}}$ so that $V = \mathcal{R}_{\mathrm{T}} \oplus N$; then

$$\dim V = \dim \mathcal{R}_{\mathrm{T}} + \dim N, \tag{3.3}$$

and

$$\mathcal{R}_{\mathrm{S}} = \mathrm{S}\,[V] = \mathrm{S}\,[\mathcal{R}_{\mathrm{T}}] + \mathrm{S}\,[N]\,.$$

It follows from Theorem 2.18 that

$$\begin{aligned}
\dim \mathcal{R}_{\mathrm{S}} &= \dim \mathrm{S}\,[\mathcal{R}_{\mathrm{T}}] + \dim \mathrm{S}\,[N] - \dim \mathrm{S}\,[\mathcal{R}_{\mathrm{T}}] \cap \mathrm{S}\,[N] \\
&\leqslant \dim \mathrm{S}\,[\mathcal{R}_{\mathrm{T}}] + \dim \mathrm{S}\,[N] \\
&\leqslant \dim \mathrm{S}\,[\mathcal{R}_{\mathrm{T}}] + \dim N \\
&= \dim \mathcal{R}_{\mathrm{ST}} + \dim N,
\end{aligned}$$

and hence that

$$\begin{aligned}
\dim V - \dim \mathcal{N}_{\mathrm{S}} &= \dim \mathcal{R}_{\mathrm{S}} \\
&\leqslant \dim \mathcal{R}_{\mathrm{ST}} + \dim N \\
&= \dim \mathcal{R}_{\mathrm{ST}} + \dim V - \dim \mathcal{R}_{\mathrm{T}},
\end{aligned} \tag{3.4}$$

where the last equality is from (3.3). Hence, (3.4) becomes

$$\dim \mathcal{R}_{\mathrm{T}} - \dim \mathcal{N}_{\mathrm{S}} \leqslant \dim \mathcal{R}_{\mathrm{ST}},$$

or equivalently,

$$\dim U - \dim \mathcal{N}_{\mathrm{T}} - \dim \mathcal{N}_{\mathrm{S}} \leqslant \dim U - \dim \mathcal{N}_{\mathrm{ST}};$$

that is,

$$\dim \mathcal{N}_{\mathrm{ST}} \leqslant \dim \mathcal{N}_{\mathrm{S}} + \dim \mathcal{N}_{\mathrm{T}}. \qquad \square$$

► EXERCISE 3.19 (3.17). *Prove that the distributive property holds for matrix addition and matrix multiplication.*

PROOF. Let $\mathbf{A} = [a_{ij}] \in \mathrm{Mat}\,(m, n, \mathbb{F})$, $\mathbf{B} = [b_{ij}] \in \mathrm{Mat}\,\big(n, p, \mathbb{F}\big)$, and $\mathbf{C} = [c_{ij}] \in \mathrm{Mat}\,\big(n, p, \mathbb{F}\big)$. Then $\mathbf{B} + \mathbf{C} = [b_{ij} + c_{ij}] \in \mathrm{Mat}\,\big(n, p, \mathbb{F}\big)$. It is evident that $\mathbf{AB}$ and

[1] See Halmos (1995, Problem 95, p.270).

$\mathbf{AC}$ are $m \times p$ matrices. Further,

$$\begin{aligned}
\mathbf{A}\,(\mathbf{B}+\mathbf{C}) &= \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11}+c_{11} & \cdots & b_{1p}+c_{1p} \\ \vdots & \ddots & \vdots \\ b_{n1}+c_{n1} & \cdots & b_{np}+c_{np} \end{pmatrix} \\
&= \begin{pmatrix} \sum_{i=1}^n a_{1i}b_{i1} + \sum_{i=1}^n a_{1i}c_{i1} & \cdots & \sum_{i=1}^n a_{1i}b_{ip} + \sum_{i=1}^n a_{1i}c_{ip} \\ \vdots & \ddots & \vdots \\ \sum_{i=1}^n a_{mi}b_{i1} + \sum_{i=1}^n a_{mi}c_{i1} & \cdots & \sum_{i=1}^n a_{mi}b_{ip} + \sum_{i=1}^n a_{mi}c_{ip} \end{pmatrix} \\
&= \mathbf{AB} + \mathbf{AC}.
\end{aligned}$$

□

► EXERCISE 3.20 (3.18). *Prove that matrix multiplication is associative.*

PROOF. Similar to Exercise 3.19. □

► EXERCISE 3.21 (3.19). *Suppose* $\mathrm{T} \in \mathfrak{L}(\mathbb{F}^n, \mathbb{F}^m)$ *and that*

$$\mathcal{M}(\mathrm{T}) = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix},$$

where we are using the standard bases. Prove that

$$\mathrm{T}(x_1, \ldots, x_n) = \left(\sum_{i=1}^n a_{1i}x_i, \ldots, \sum_{i=1}^n a_{mi}x_i \right)$$

for every $(x_1, \ldots, x_n) \in \mathbb{F}^n$.

PROOF. We need to prove that $\mathrm{T}\boldsymbol{x} = \mathcal{M}(\mathrm{T}) \cdot \boldsymbol{x}$ for any $\boldsymbol{x} \in \mathbb{F}^n$. Let $(\boldsymbol{e}_1^n, \ldots, \boldsymbol{e}_n^n)$ be the standard basis for $\mathbb{F}^n$, and let $(\boldsymbol{e}_1^m, \ldots, \boldsymbol{e}_m^m)$ be the standard basis for $\mathbb{F}^m$. Then

$$\begin{aligned}
\mathrm{T}(x_1, \ldots, x_n) = \mathrm{T}\left(\sum_{i=1}^n x_i \boldsymbol{e}_i^n \right) = \sum_{i=1}^n x_i \mathrm{T}\boldsymbol{e}_i^n &= \sum_{i=1}^n x_i \sum_{j=1}^m a_{ji}\boldsymbol{e}_j^m \\
&= \left(\sum_{i=1}^n a_{1i}x_i, \ldots, \sum_{i=1}^n a_{mi}x_i \right).
\end{aligned}$$

□

► EXERCISE 3.22 (3.20). *Suppose* $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ *is a basis of* V. *Prove that the function* $\mathrm{T}\colon V \to \mathrm{Mat}\,(n, 1, \mathbb{F})$ *defined by* $\mathrm{T}\boldsymbol{v} = \mathcal{M}(\boldsymbol{v})$ *is an invertible linear map of* V *onto* $\mathrm{Mat}\,(n, 1, \mathbb{F})$*; here* $\mathcal{M}(\boldsymbol{v})$ *is the matrix of* $\boldsymbol{v} \in V$ *with respect to the basis* $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$.

PROOF. For every $\boldsymbol{v} = \sum_{i=1}^n b_i \boldsymbol{v}_i \in V$, we have

$$\mathcal{M}(\boldsymbol{v}) = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

Since $a\boldsymbol{v} = \sum_{i=1}^{n} (ab_i)\, \boldsymbol{v}_i$ for any $a \in \mathbb{F}$, we have $\mathcal{M}(a\boldsymbol{v}) = a\mathcal{M}(\boldsymbol{v})$. Further, for any $\boldsymbol{u} = \sum_{i=1}^{n} a_i \boldsymbol{v}_i \in V$, and any $\boldsymbol{v} = \sum_{i=1}^{n} b_i \boldsymbol{v}_i \in V$, we have $\boldsymbol{u} + \boldsymbol{v} = \sum_{i=1}^{n} (a_i + b_i)\boldsymbol{v}_i$; hence, $\mathcal{M}(\boldsymbol{u} + \boldsymbol{v}) = \mathcal{M}(\boldsymbol{u}) + \mathcal{M}(\boldsymbol{v})$. Therefore, T is a liner map.

We now show that T is invertible by proving T is bijective. (i) If $\mathrm{T}\boldsymbol{v} = (0, \ldots, 0)'$, then $\boldsymbol{v} = \sum_{i=1}^{n} 0\boldsymbol{v}_i = \boldsymbol{0}_V$; that is, $\mathcal{N}_{\mathrm{T}} = \{\boldsymbol{0}_T\}$. Hence, T is injective. (ii) Take any $M = (a_1, \ldots, a_n)' \in \mathrm{Mat}(n, 1, \mathbb{F})$. Let $\boldsymbol{v} = \sum_{i=1}^{n} a_i \boldsymbol{v}_i$. Then $\mathrm{T}\boldsymbol{v} = M$; that is, T is surjective. □

► EXERCISE 3.23 (3.21). *Prove that every linear map from* $\mathrm{Mat}(n, 1, \mathbb{F})$ *to* $\mathrm{Mat}(m, 1, \mathbb{F})$ *is given by a matrix multiplication. In other words, prove that if*

$$\mathrm{T} \in \mathfrak{L}(\mathrm{Mat}(n, 1, \mathbb{F}), \mathrm{Mat}(m, 1, \mathbb{F})),$$

then there exists an $m \times n$ *matrix* $\mathbf{A}$ *such that* $\mathrm{T}\mathbf{B} = \mathbf{A}\mathbf{B}$ *for every* $\mathbf{B} \in \mathrm{Mat}(n, 1, \mathbb{F})$.

PROOF. A basis of $\mathrm{Mat}(m, n, \mathbb{F})$ consists of those $m \times n$ matrices that have 0 in all entries except for a 1 in one entry. Therefore, a basis for $\mathrm{Mat}(n, 1, \mathbb{F})$ consists of the standard basis of $\mathbb{F}^n$, $(\boldsymbol{e}_1^n, \ldots, \boldsymbol{e}_n^n)$, where, for example, $\boldsymbol{e}_1^n = (1, 0, \ldots, 0)'$. For any $\mathrm{T} \in \mathfrak{L}(\mathrm{Mat}(n, 1, \mathbb{F}), \mathrm{Mat}(m, 1, \mathbb{F}))$, let

$$\underset{(m\times n)}{\mathbf{A}} := \begin{pmatrix} \mathrm{T}\boldsymbol{e}_1^n & \cdots & \mathrm{T}\boldsymbol{e}_n^n \end{pmatrix}.$$

Then for any $\mathbf{B} = \sum_{i=1}^{n} a_i \boldsymbol{e}_i^n \in \mathrm{Mat}(n, 1, \mathbb{F})$, we have

$$\mathrm{T}\mathbf{B} = \mathrm{T}\left(\sum_{i=1}^{n} a_i \boldsymbol{e}_i^n\right) = \sum_{i=1}^{n} a_i \mathrm{T}\boldsymbol{e}_i^n = \mathbf{A}\mathbf{B}. \qquad \square$$

► EXERCISE 3.24 (3.22). *Suppose that* V *is finite dimensional and* $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V)$. *Prove that* ST *is invertible if and only if both* S *and* T *are invertible.*

PROOF. First assume that both S and T are invertible. Then $(\mathrm{ST})\left(\mathrm{T}^{-1}\mathrm{S}^{-1}\right) = \mathrm{SIdS}^{-1} = \mathrm{Id}$ and $\left(\mathrm{T}^{-1}\mathrm{S}^{-1}\right)(\mathrm{ST}) = \mathrm{Id}$. Hence, ST is invertible and $(\mathrm{ST})^{-1} = \mathrm{T}^{-1}\mathrm{S}^{-1}$.

Now suppose that ST is invertible, so it is injective. Take any $\boldsymbol{u}, \boldsymbol{v} \in V$ with $\boldsymbol{u} \neq \boldsymbol{v}$; then $(\mathrm{ST})\,\boldsymbol{u} \neq (\mathrm{ST})\,\boldsymbol{v}$; that is,

$$\boldsymbol{u} \neq \boldsymbol{v} \Longrightarrow \mathrm{S}(\mathrm{T}\boldsymbol{u}) \neq \mathrm{S}(\mathrm{T}\boldsymbol{v}). \tag{3.5}$$

But then $\mathrm{T}\boldsymbol{u} \neq \mathrm{T}\boldsymbol{v}$, which implies that T is invertible by Theorem 3.21. Finally, for any $\boldsymbol{u}, \boldsymbol{v} \in V$ with $\boldsymbol{u} \neq \boldsymbol{v}$, there exist $\boldsymbol{u}', \boldsymbol{v}' \in V$ with $\boldsymbol{u}' \neq \boldsymbol{v}'$ such that $\boldsymbol{u} = \mathrm{T}\boldsymbol{u}'$ and $\boldsymbol{v} = \mathrm{T}\boldsymbol{v}'$. Hence, by (3.5), $\boldsymbol{u} \neq \boldsymbol{v}$ implies that

$$\mathrm{S}\boldsymbol{u} = \mathrm{S}(\mathrm{T}\boldsymbol{u}') \neq \mathrm{S}(\mathrm{T}\boldsymbol{v}') = \mathrm{S}\boldsymbol{v};$$

that is, S is injective, too. Applying Theorem 3.21 once again, we know that S is invertible. □

► EXERCISE 3.25 (3.23). *Suppose that V is finite dimensional and* $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V)$. *Prove that* $\mathrm{ST} = \mathrm{Id}$ *if and only if* $\mathrm{TS} = \mathrm{Id}$.

PROOF. We only prove the *only if* part; the *if* part can be proved similarly. If $\mathrm{ST} = \mathrm{Id}$, then ST is bijective and so invertible. Then by Exercise 3.24, both S and T are invertible. Therefore,

$$\mathrm{ST} = \mathrm{Id} \iff \mathrm{S}^{-1}\mathrm{ST} = \mathrm{S}^{-1}\mathrm{Id} \iff \mathrm{T} = \mathrm{S}^{-1} \iff \mathrm{TS} = \mathrm{S}^{-1}\mathrm{S} = \mathrm{Id}. \qquad \square$$

► EXERCISE 3.26 (3.24). *Suppose that V is finite dimensional and* $\mathrm{T} \in \mathfrak{L}(V)$. *Prove that* T *is a scalar multiple of the identity if and only if* $\mathrm{ST} = \mathrm{TS}$ *for every* $\mathrm{S} \in \mathfrak{L}(V)$.

PROOF. If $\mathrm{T} = a\mathrm{Id}$ for some $a \in \mathbb{F}$, then for any $\mathrm{S} \in \mathfrak{L}(V)$, we have

$$\mathrm{ST} = a\mathrm{SId} = a\mathrm{S} = a\mathrm{IdS} = \mathrm{TS}.$$

For the converse direction, assume that $\mathrm{ST} = \mathrm{TS}$ for all $\mathrm{S} \in \mathfrak{L}(V)$. □

► EXERCISE 3.27 (3.25). *Prove that if V is finite dimensional with* $\dim V > 1$, *then the set of noninvertible operators on V is not a subspace of* $\mathfrak{L}(V)$.

PROOF. Since every finite-dimensional vector space is isomorphic to some $\mathbb{F}^n$, we just focus on $\mathbb{F}^n$. For simplicity, consider $\mathbb{F}^2$. Let $\mathrm{S}, \mathrm{T} \in \mathbb{F}^2$ with

$$\mathrm{S}(a, b) = (a, 0) \quad \text{and} \quad \mathrm{T}(a, b) = (0, b).$$

Obviously, both S and T are noninvertible since they are not injective; however, $\mathrm{S} + \mathrm{T} = \mathrm{Id}$ is invertible. □

► EXERCISE 3.28 (3.26). *Suppose n is a positive integer and $a_{ij} \in \mathbb{F}$ for $i, j = 1, \ldots, n$. Prove that the following are equivalent:*

a. *The trivial solution $x_1 = \cdots = x_n = 0$ is the only solution to the homogeneous system of equations*

$$\begin{aligned} \sum_{k=1}^{n} a_{1k} x_k &= 0 \\ &\vdots \\ \sum_{k=1}^{n} a_{nk} x_k &= 0. \end{aligned}$$

b. *For every $c_1, \ldots, c_n \in \mathbb{F}$, there exists a solution to the system of equations*

$$\sum_{k=1}^{n} a_{1k} x_k = c_1$$
$$\vdots$$
$$\sum_{k=1}^{n} a_{nk} x_k = c_n.$$

PROOF. Let

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}.$$

If we let $\mathrm{T}\boldsymbol{x} = A\boldsymbol{x}$, then by Exercise 3.22, $\mathrm{T} \in \mathfrak{L}(\mathbb{F}^n, \mathbb{F}^n)$. (a) implies that $\mathcal{N}_{\mathrm{T}} = \{\mathbf{0}\}$; hence

$$\dim \mathcal{R}_{\mathrm{T}} = n - 0 = n.$$

Since $\mathcal{R}_{\mathrm{T}}$ is a subspace of $\mathbb{F}^n$, we have $\mathcal{R}_{\mathrm{T}} = \mathbb{F}^n$, that is, T is surjective: for any $(c_1, \ldots, c_n)$, there is a unique solution $(x_1, \ldots, x_n)$. □

4

POLYNOMIALS

► EXERCISE 4.1 (4.1). *Suppose m and n are positive integers with $m \leqslant n$. Prove that there exists a polynomial $p \in \mathfrak{P}_n(\mathbb{F})$ with exactly m distinct roots.*

PROOF. Let
$$p(z) = \prod_{i=1}^{m} (z - \lambda_i)^{m_i},$$
where $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ are distinct and $\sum_{i=1}^{m} m_i = n$. □

► EXERCISE 4.2 (4.2). *Suppose that $z_1, \ldots, z_{m+1}$ are distinct elements of $\mathbb{F}$ and that $w_1, \ldots, w_{m+1} \in \mathbb{F}$. Prove that there exists a unique polynomial $p \in \mathfrak{P}_m(\mathbb{F})$ such that $p(z_j) = w_j$ for $j = 1, \ldots, m+1$.*

PROOF. Let $p_i(x) = \prod_{j \neq i} (x - z_j)$. Then $\deg p_i = m$ and $p_i(z_j) \neq 0$ if and only if $i = j$. Define
$$p(x) = \sum_{i=1}^{m+1} \frac{w_i}{p_i(z_i)} p_i(x).$$
Then $\deg p = m$ and
$$\begin{aligned} p(z_j) &= \frac{w_1}{p_1(z_1)} p_1(z_j) + \cdots + \frac{w_j}{p_j(z_j)} p_j(z_j) + \cdots + \frac{w_{m+1}}{p_{m+1}(z_{m+1})} p_{m+1}(z_j) \\ &= w_j. \end{aligned}$$
□

► EXERCISE 4.3 (4.3). *Prove that if $p, q \in \mathfrak{P}(\mathbb{F})$, with $p \neq 0$, then there exist unique polynomials $s, r \in \mathfrak{P}(\mathbb{F})$ such that $q = sp + r$ and $\deg r < \deg p$.*

PROOF. Assume that there also exist $s', r' \in \mathfrak{P}(\mathbb{F})$ such that $q = s'p + r'$ and $\deg r' < \deg p$. Then
$$(s - s')p + (r - r') = 0.$$

If $s \neq s'$, then $\deg (s - s')p + \deg (r - r') = \deg (s - s') + \deg p + \deg (r - r') \geqslant 0$; but $\deg 0 = -\infty$. Hence, $s = s'$ and so $r = r'$. □

► EXERCISE 4.4 (4.4). *Suppose $p \in \mathfrak{P}(\mathbb{C})$ has degree m. Prove that p has m distinct roots if and only if p and its derivative p' have no roots in common.*

PROOF. If λ is a root of p, then we can write p as $p(z) = (z - \lambda)q(z)$. Then

$$p'(z) = q(z) + (z - \lambda)q'(z).$$

So λ is also a root for p' if and only if λ is a root of q; that is, λ is a multiple root. A contradiction. □

► EXERCISE 4.5 (4.5). *Prove that every polynomial with odd degree and real coefficients has a real root.*

PROOF. If $p \in \mathfrak{P}(\mathbb{R})$ with $\deg p$ is odd, then $p(-\infty) < 0$ and $p(+\infty) > 0$. Then there exists $x^* \in \mathbb{R}$ such that $p(x^*) = 0$. □

5

EIGENVALUES AND EIGENVECTORS

"As You Should Verify"

Remark 5.1 (p.80). Fix an operator $\mathrm{T} \in \mathfrak{L}(V)$, then the function from $\mathfrak{P}(\mathbb{F})$ to $\mathfrak{L}(V)$ given by $p \mapsto p(\mathrm{T})$ is linear.

Proof. Let the mapping be $A\colon \mathfrak{P}(\mathbb{F}) \to \mathfrak{L}(V)$ with $A(p) = p(\mathrm{T})$. For any $p, q \in \mathfrak{P}(\mathbb{F})$, we have $A(p+q) = (p+q)(\mathrm{T}) = p(\mathrm{T}) + q(\mathrm{T}) = A(p) + A(q)$. For any $a \in \mathbb{F}$, we have $A(ap) = (ap)(\mathrm{T}) = ap(\mathrm{T}) = aA(p)$. □

Exercises

► Exercise 5.2 (5.1). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$. *Prove that if* $U_1, \ldots, U_m$ *are subspaces of* V *invariant under* T, *then* $U_1 + \cdots + U_m$ *is invariant under* T.

Proof. Take an arbitrary $\boldsymbol{u} \in U_1 + \cdots + U_m$; then $\boldsymbol{u} = \boldsymbol{u}_1 + \cdots + \boldsymbol{u}_m$, where $\boldsymbol{u}_i \in U_i$ for every $i = 1, \ldots, m$. Therefore, $\mathrm{T}\boldsymbol{u} = \mathrm{T}\boldsymbol{u}_1 + \cdots + \mathrm{T}\boldsymbol{u}_m \in U_1 + \cdots + U_m$ since $\mathrm{T}\boldsymbol{u}_i \in U_i$. □

► Exercise 5.3 (5.2). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$. *Prove that the intersection of any collection of subspaces of* V *invariant under* T *is invariant under* T.

Proof. Let the collection $\{U_i \trianglelefteq V : i \in I\}$ of subspaces of V invariant under T, where I is an index set. Let $U = \bigcap_{i \in I} U_i$. Then $\boldsymbol{u} \in U_i$ for every $i \in I$ if $\boldsymbol{u} \in U$, and so $\mathrm{T}\boldsymbol{u} \in U_i$ for every $i \in I$. Then $\mathrm{T}\boldsymbol{u} \in U$; that is, U is invariant under T. □

► Exercise 5.4 (5.3). *Prove or give a counterexample: if* U *is a subspace of* V *that is invariant under every operator on* V, *then* $U = \{\boldsymbol{0}\}$ *or* $U = V$.

Proof. Assume that $U \neq \{\boldsymbol{0}\}$ and $U \neq V$. Let $(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m)$ be a basis of U, which then can be extended to a basis $(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ of V, where $n \geqslant 1$ since $U \neq V$. Define an operator $\mathrm{T} \in \mathfrak{L}(V)$ by letting $\mathrm{T}(a_1\boldsymbol{u}_1 + \cdots + a_m\boldsymbol{u}_m +$

$b_1\boldsymbol{v}_1 + \cdots + b_n\boldsymbol{v}_n) = (a_1 + \cdots + a_m + b_1 + \cdots + b_n)\boldsymbol{v}_1$. Then U fails be invariant clearly. □

► EXERCISE 5.5 (5.4). *Suppose that* $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V)$ *are such that* $\mathrm{ST} = \mathrm{TS}$. *Prove that* $\mathcal{N}_{\mathrm{T}-\lambda\mathrm{Id}}$ *is invariant under* S *for every* $\lambda \in \mathbb{F}$.

PROOF. If $\boldsymbol{u} \in \mathcal{N}_{\mathrm{T}-\lambda\mathrm{Id}}$, then $(\mathrm{T} - \lambda\mathrm{Id})(\boldsymbol{u}) = \mathrm{T}\boldsymbol{u} - \lambda\boldsymbol{u} = \mathbf{0}$; hence

$$\begin{aligned} \mathrm{S}(\mathrm{T}\boldsymbol{u} - \lambda\boldsymbol{u}) = \mathrm{S}\mathbf{0} &\iff \mathrm{ST}\boldsymbol{u} - \lambda\mathrm{S}\boldsymbol{u} = \mathbf{0} \\ &\iff \mathrm{TS}\boldsymbol{u} - \lambda\mathrm{S}\boldsymbol{u} = \mathbf{0} \\ &\iff (\mathrm{T} - \lambda\mathrm{Id})(\mathrm{S}\boldsymbol{u}) = \mathbf{0}; \end{aligned}$$

that is, $\mathrm{S}\boldsymbol{u} \in \mathcal{N}_{\mathrm{T}-\lambda\mathrm{Id}}$. □

► EXERCISE 5.6 (5.5). *Define* $\mathrm{T} \in \mathfrak{L}(\mathbb{F}^2)$ *by* $\mathrm{T}(w, z) = (z, w)$. *Find all eigenvalues and eigenvectors of* T.

PROOF. $\mathrm{T}\boldsymbol{u} = \lambda\boldsymbol{u}$ implies that $(z, w) = (\lambda w, \lambda z)$. Hence, $\lambda_1 = 1$, $\lambda_2 = -1$, and the corresponding eigenvectors are $(1, 1)$ and $(1, -1)$. Since $\dim \mathbb{F}^2 = 2$, they are the all eigenvalues and eigenvectors of T. □

► EXERCISE 5.7 (5.6). *Define* $\mathrm{T} \in \mathfrak{L}(\mathbb{F}^3)$ *by* $\mathrm{T}(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$. *Find all eigenvalues and eigenvectors of* T.

PROOF. If $\lambda \in \mathbb{F}$ is a eigenvalue of T and $(z_1, z_2, z_3) \neq \mathbf{0}$ is a corresponding eigenvector, then $\mathrm{T}(z_1, z_2, z_3) = \lambda(z_1, z_2, z_3)$, that is,

$$\begin{cases} 2z_2 = \lambda z_1 & \text{(i)} \\ 0 = \lambda z_2 & \text{(ii)}, \\ 5z_3 = \lambda z_3 & \text{(iii)}. \end{cases} \tag{5.1}$$

- If $z_2 \neq 0$, then $\lambda = 0$ from (ii); but then $z_2 = 0$ from (i). A contradiction. Hence, $z_2 = 0$ and (5.1) becomes

$$\begin{cases} 0 = \lambda z_1 & \text{(i}'\text{)} \\ 5z_3 = \lambda z_3 & \text{(iii}'\text{)}. \end{cases} \tag{5.2}$$

- If $z_3 \neq 0$, then $\lambda = 5$ from (iii′); then (i′) implies that $z_1 = 0$. Hence, $\lambda = 5$ is an eigenvalue, and the corresponding eigenvector is $(0, 0, 1)$.
- If $z_1 \neq 0$, then $\lambda = 0$ from (i′); then (iii′) implies that $z_3 = 0$. Hence, $\lambda = 0$ is an eigenvalue, and the corresponding eigenvector is $(1, 0, 0)$. □

► EXERCISE 5.8 (5.7). *Suppose* n *is a positive integer and* $\mathrm{T} \in \mathfrak{L}(\mathbb{F}^n)$ *is defined by*

$$\mathrm{T}(x_1, \ldots, x_n) = \left(\sum_{i=1}^{n} x_i, \ldots, \sum_{i=1}^{n} x_i \right);$$

in other words, T *is the operator whose matrix (with respect to the standard basis) consists of all* 1*'s. Find all eigenvalues and eigenvectors of* T.

PROOF. If $\lambda \in \mathbb{F}$ is an eigenvalue of T and $(x_1, \dots, x_n) \neq \mathbf{0}$ is a corresponding eigenvector, then $\sum_{i=1}^n x_i \neq 0$ and

$$\begin{pmatrix} \sum_{i=1}^n x_i \\ \vdots \\ \sum_{i=1}^n x_i \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}.$$

Hence, $\lambda \neq 0$, $x_i \neq 0$ for all $i = 1, \dots, n$, and $\lambda x_1 = \cdots = \lambda x_n$ implies that $x_1 = \cdots = x_n$, and so the unique eigenvalue of T is $(\sum_{i=1}^n x_i)/x_i = n$. Then an eigenvector to n is $(1, \dots, 1)$. □

► EXERCISE 5.9 (5.8). *Find all eigenvalues and eigenvectors of the backward shift operator* $\mathrm{T} \in \mathfrak{L}(\mathbb{F}^\infty)$ *defined by* $\mathrm{T}(z_1, z_2, z_3, \dots) = (z_2, z_3, \dots)$.

PROOF. For any $\lambda \in \mathbb{F}$ with $\lambda \neq 0$, we have $\mathrm{T}(\lambda, \lambda^2, \lambda^3, \dots) = (\lambda^2, \lambda^3, \dots) = \lambda \cdot (\lambda, \lambda^2, \dots)$; hence, every $\lambda \neq 0$ is an eigenvalue of T. We now show that $\lambda = 0$ is also an eigenvalue: let $\boldsymbol{z} = (z_1, 0, \dots)$ with $z_1 \neq 0$. Then $\mathrm{T}\boldsymbol{z} = (0, 0, \dots) = 0 \cdot \boldsymbol{z}$. □

► EXERCISE 5.10 (5.9). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$ *and* $\dim \mathfrak{R}_\mathrm{T} = k$. *Prove that* T *has at most* $k + 1$ *distinct eigenvalues.*

PROOF. Suppose that T has more than or equal to $k+2$ distinct eigenvalues. We take the first $k + 2$ eigenvalues: $\lambda_1, \dots, \lambda_{k+2}$. Then there are $k + 2$ corresponding nonzero eigenvectors, $\boldsymbol{u}_1, \dots, \boldsymbol{u}_{k+2}$, satisfying $\mathrm{T}\boldsymbol{u}_1 = \lambda_1 \boldsymbol{u}_1, \dots, \mathrm{T}\boldsymbol{u}_{k+2} = \lambda_{k+2}\boldsymbol{u}_{k+2}$. Since the $k + 2$ eigenvectors are linearly independent, the list $(\lambda_1 \boldsymbol{u}_1, \dots, \lambda_{k+2}\boldsymbol{u}_{k+2})$ is linearly independent, too (there are $n + 1$ vectors if one λ is zero). Obviously, the above list is in $\mathfrak{R}_\mathrm{T}$, which means that $\dim \mathfrak{R}_\mathrm{T} \geqslant k + 1$. A contradiction. □

► EXERCISE 5.11 (5.10). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$ *is invertible and* $\lambda \in \mathbb{F} \smallsetminus \{0\}$. *Prove that* λ *is an eigenvalue of* T *if and only if* $1/\lambda$ *is an eigenvalue of* T^{-1}.

PROOF. If $\lambda \neq 0$ be an eigenvalue of T, then there exists a nonzero $\boldsymbol{u} \in V$ such that $\mathrm{T}\boldsymbol{u} = \lambda \boldsymbol{u}$. Therefore,

$$\mathrm{T}^{-1}(\mathrm{T}\boldsymbol{u}) = \mathrm{T}^{-1}(\lambda \boldsymbol{u}) \iff \boldsymbol{u} = \lambda \mathrm{T}^{-1}\boldsymbol{u} \iff \mathrm{T}^{-1}\boldsymbol{u} = \boldsymbol{u}/\lambda;$$

that is, $1/\lambda$ is an eigenvalue of T^{-1}. The other direction can be proved with the same way. □

► EXERCISE 5.12 (5.11). *Suppose* $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V)$. *Prove that* ST *and* TS *have the same eigenvalues.*

PROOF. Let λ be an eigenvalue of ST, and $\boldsymbol{u} \neq \mathbf{0}$ be the corresponding eigenvector. Then $(\mathrm{ST})\boldsymbol{u} = \lambda \boldsymbol{u}$. Therefore,

$$\mathrm{T}(\mathrm{ST}\boldsymbol{u}) = \mathrm{T}(\lambda\boldsymbol{u}) \iff (\mathrm{TS})(\mathrm{T}\boldsymbol{u}) = \lambda(\mathrm{T}\boldsymbol{u}).$$

Hence, if $\mathrm{T}\boldsymbol{u} \neq \boldsymbol{0}$, then λ is an eigenvalue of TS, and the corresponding eigenvector is $\mathrm{T}\boldsymbol{u}$; if $\mathrm{T}\boldsymbol{u} = \boldsymbol{0}$, then $(\mathrm{ST})\boldsymbol{u} = \mathrm{S}(\mathrm{T}\boldsymbol{u}) = \boldsymbol{0}$ implies that $\lambda = 0$ (since $\boldsymbol{u} \neq \boldsymbol{0}$). In this case, T is not injective, and so TS is not injective (by Exercise 3.24). But this means that there exists $\boldsymbol{v} \neq \boldsymbol{0}$ such that $(\mathrm{TS})\boldsymbol{v} = \boldsymbol{0} = 0\boldsymbol{v}$; that is, 0 is an eigenvalue of TS. The other direction can be proved with the same way. □

► EXERCISE 5.13 (5.12). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$ *is such that every vector in* V *is an eigenvector of* T. *Prove that* T *is a scalar multiple of the identity operator.*

PROOF. Let $B = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ be a basis of V and take arbitrary $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ from B. Then there are λ_i and λ_j such that $\mathrm{T}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ and $\mathrm{T}\boldsymbol{v}_j = \lambda_j \boldsymbol{v}_j$. Since $\boldsymbol{v}_i + \boldsymbol{v}_j$ is also an eigenvector, there is $\lambda \in \mathbb{F}$ such that $\mathrm{T}(\boldsymbol{v}_i + \boldsymbol{v}_j) = \lambda(\boldsymbol{v}_i + \boldsymbol{v}_j)$. Therefore,

$$\lambda_i \boldsymbol{v}_i + \lambda_j \boldsymbol{v}_j = \lambda \boldsymbol{v}_i + \lambda \boldsymbol{v}_j;$$

that is, $(\lambda_i - \lambda)\,\boldsymbol{v}_i + \big(\lambda_j - \lambda\big)\,\boldsymbol{v}_j = \boldsymbol{0}$. Since $(\boldsymbol{v}_i, \boldsymbol{v}_j)$ is linearly independent, we have $\lambda_i = \lambda_j = \lambda$. Hence, for any $\boldsymbol{v} = \sum_{i=1}^n a_i \boldsymbol{v}_i \in V$, we have

$$\mathrm{T}\boldsymbol{v} = \mathrm{T}\left(\sum_{i=1}^n a_i \boldsymbol{v}_i\right) = \sum_{i=1}^n a_i \lambda \boldsymbol{v}_i = \lambda\left(\sum_{i=1}^n a_i \boldsymbol{v}_i\right) = \lambda \boldsymbol{v},$$

i.e., $\mathrm{T} = \lambda \mathrm{Id}$. □

► EXERCISE 5.14 (5.13). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$ *is such that every subspace of* V *with dimension* $\dim V - 1$ *is invariant under* T. *Prove that* T *is a scalar multiple of the identity operator.*

PROOF. Let $\dim V = n$ and $(\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n)$ be a basis of V. We first show that there exists $\lambda_1 \in \mathbb{F}$ such that $\mathrm{T}\boldsymbol{v}_1 = \lambda_1 \boldsymbol{v}_1$.

Let $V_1 = \{a\boldsymbol{v}_1 \colon a \in \mathbb{F}\}$ and $U_1 = \operatorname{span}(\boldsymbol{v}_2, \ldots, \boldsymbol{v}_n)$. Then for any $\boldsymbol{v} = \sum_{i=1}^n a_i \boldsymbol{v}_i \in V$, we have

$$\begin{aligned}
\mathrm{T}\boldsymbol{v} = \mathrm{T}\left(a_1\boldsymbol{v}_1 + \sum_{i=2}^n a_i \boldsymbol{v}_i\right) &= a_1 \mathrm{T}\boldsymbol{v}_1 + \mathrm{T}\left(\sum_{i=2}^n a_i \boldsymbol{v}_i\right) \\
&= a_1\left(\sum_{j=1}^n b_j \boldsymbol{v}_j\right) + \mathrm{T}\left(\sum_{i=2}^n a_i \boldsymbol{v}_i\right) \\
&= (a_1 b_1)\,\boldsymbol{v}_1 + \left[\sum_{i=2}^n (a_1 b_i)\,\boldsymbol{v}_i + \mathrm{T}\left(\sum_{i=2}^n a_i \boldsymbol{v}_i\right)\right] \\
&\in V_1 + U_1,
\end{aligned}$$

where $\mathrm{T}\left(\sum_{i=2}^n a_i \boldsymbol{v}_i\right) \in U_1$ since U_1 is invariant under T.

Since $V = V_1 + U_1$ and $\dim V = \dim V_1 + \dim U_1$, we have $V = V_1 \oplus U_1$ by Proposition 2.19, which implies that $V_1 \cap U_1 = \{\mathbf{0}\}$ by Proposition 1.9. If $\mathrm{T}\boldsymbol{v}_1 \notin V_1$, then $\mathrm{T}\boldsymbol{v}_1 \neq \mathbf{0}$ and $\mathrm{T}\boldsymbol{v}_1 \in U_1$; hence, there exist $c_2, \dots, c_n \in \mathbb{F}$ not all zero such that

$$\mathrm{T}\boldsymbol{v}_1 = \sum_{i=2}^{n} c_i \boldsymbol{v}_i.$$

Without loss of generality, we suppose that $c_n \neq 0$.

Let $V_n = \{a\boldsymbol{v}_n : a \in \mathbb{F}\}$ and $U_n = \operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_{n-1})$. Similarly, $V = V_n \oplus U_n$ and $V_n \cap U_n = \{\mathbf{0}\}$. Since U_n is invariant under T, we have $\mathrm{T}\boldsymbol{v}_1 \in U_n$, that is, $\mathrm{T}\boldsymbol{v}_1 = \sum_{j=1}^{n-1} d_j \boldsymbol{v}_j$, but which means that $c_n = 0$ A contradiction. We thus proved that $\mathrm{T}\boldsymbol{v}_1 \in V_1$, i.e., there is $\lambda_1 \in \mathbb{F}$ such that $\mathrm{T}\boldsymbol{v}_1 = \lambda_1 \boldsymbol{v}_1$. But this way can be applied to every $\boldsymbol{v}_i$. Therefore, every $\boldsymbol{v}_i$ is an eigenvector of T. By Exercise 5.13, T is a scalar multiple of the identity operator. □

► EXERCISE 5.15 (5.14). *Suppose* $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V)$ *and* S *is invertible. Prove that if* $p \in \mathfrak{P}(\mathbb{F})$ *is a polynomial, then* $p(\mathrm{STS}^{-1}) = \mathrm{S}p(\mathrm{T})\mathrm{S}^{-1}$.

PROOF. Let $p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m$. Then

$$p(\mathrm{STS}^{-1}) = a_0\mathrm{Id} + a_1 \cdot (\mathrm{STS}^{-1}) + a_2 \cdot (\mathrm{STS}^{-1})^2 + \cdots + a_m \cdot (\mathrm{STS}^{-1})^m.$$

We also have

$$\begin{aligned}(\mathrm{STS}^{-1})^n &= (\mathrm{STS}^{-1}) \cdot (\mathrm{STS}^{-1}) \cdot (\mathrm{STS}^{-1})^{n-2} \\ &= (\mathrm{ST}^2\mathrm{S}^{-1}) \cdot (\mathrm{STS}^{-1})^{n-2} \\ &= \cdots \\ &= \mathrm{ST}^n\mathrm{S}^{-1}.\end{aligned}$$

Therefore,

$$\begin{aligned}\mathrm{S}p(\mathrm{T})\mathrm{S}^{-1} &= \mathrm{S}(a_0\mathrm{Id} + a_1\mathrm{T} + a_2\mathrm{T}^2 + \cdots + a_m\mathrm{T}^m)\mathrm{S}^{-1} \\ &= a_0\mathrm{Id} + a_1 \cdot (\mathrm{STS}^{-1}) + a_2 \cdot (\mathrm{ST}^2\mathrm{S}^{-1}) + \cdots + a_m \cdot (\mathrm{ST}^m\mathrm{S}^{-1}) \\ &= p(\mathrm{STS}^{-1}).\end{aligned}$$

□

► EXERCISE 5.16 (5.15). *Suppose* $\mathbb{F} = \mathbb{C}$, $\mathrm{T} \in \mathfrak{L}(V)$, $p \in \mathfrak{P}(\mathbb{C})$, *and* $a \in \mathbb{C}$. *Prove that* a *is an eigenvalue of* $p(\mathrm{T})$ *if and only if* $a = p(\lambda)$ *for some eigenvalue* λ *of* T.

PROOF. If λ is an eigenvalue of T, then there exists $\boldsymbol{v} \neq \mathbf{0}$ such that $\mathrm{T}\boldsymbol{v} = \lambda\boldsymbol{v}$. Thus,

$$\begin{aligned}[p(\mathrm{T})](\boldsymbol{v}) &= (a_0\mathrm{Id} + a_1\mathrm{T} + a_2\mathrm{T}^2 + \cdots + a_m\mathrm{T}^m)\boldsymbol{v} \\ &= a_0\boldsymbol{v} + a_1\mathrm{T}\boldsymbol{v} + a_2\mathrm{T}^2\boldsymbol{v} + \cdots a_m\mathrm{T}^m\boldsymbol{v} \\ &= a_0\boldsymbol{v} + a_1\lambda\boldsymbol{v} + a_2\mathrm{T}(\lambda\boldsymbol{v}) + \cdots + a_m\mathrm{T}^{m-1}(\lambda\boldsymbol{v}) \\ &= a_0\boldsymbol{v} + (a_1\lambda)\boldsymbol{v} + (a_2\lambda^2)\boldsymbol{v} + \cdots + (a_m\lambda^m)\boldsymbol{v} \\ &= p(\lambda)\boldsymbol{v};\end{aligned}$$

that is, $p(\lambda)$ is an eigenvalue of $p(\mathrm{T})$.

Conversely, let $a \in \mathbb{C}$ be an eigenvalue of $p(\mathrm{T}) = a_0\mathrm{Id} + a_1\mathrm{T} + \cdots + a_m\mathrm{T}^m$, and $\boldsymbol{v}$ be the corresponding eigenvector. Then $p(\mathrm{T})(\boldsymbol{v}) = a\boldsymbol{v}$; that is,

$$\left[(a_0 - a)\mathrm{Id} + a_1\mathrm{T} + \cdots + a_m\mathrm{T}^m\right]\boldsymbol{v} = \mathbf{0}.$$

It follows from Corollary 4.8 that the above display can be rewritten as follows:

$$\left[c(\mathrm{T} - \lambda_1\mathrm{Id})\cdots(\mathrm{T} - \lambda_m\mathrm{Id})\right]\boldsymbol{v} = \mathbf{0}, \tag{5.3}$$

where $c, \lambda_1, \ldots, \lambda_m \in \mathbb{C}$ and $c \neq 0$. Hence, for some $i = 1, \ldots, m$, we have $(\mathrm{T} - \lambda_i\mathrm{Id})\boldsymbol{v} = \mathbf{0}$; that is, λ_i is an eigenvalue of T. □

► EXERCISE 5.17 (5.16). *Show that the result in the previous exercise does not hold if $\mathbb{C}$ is replaced with $\mathbb{R}$.*

PROOF. Let $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^2)$ defined by $\mathrm{T}(w, z) = (-z, w)$. Then T has no eigenvalue (see p. 78). But $\mathrm{T}^2(w, z) = \mathrm{T}(-z, w) = (-w, -z)$ has an eigenvalue: let $(-w, -z) = \lambda(w, z)$; then

$$\begin{cases} -w = \lambda w \\ -z = \lambda z. \end{cases}$$

Hence, $\lambda = -1$. □

► EXERCISE 5.18 (5.17). *Suppose V is a complex vector space and $\mathrm{T} \in \mathfrak{L}(V)$. Prove that T has an invariant subspace of dimension j for each $j = 1, \ldots, \dim V$.*

PROOF. Suppose that $\dim V = n$. Let $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ be a basis of V with respect to which T has an upper-triangular matrix (by Theorem 5.13)

$$\mathcal{M}\left(\mathrm{T}, (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)\right) = \begin{pmatrix} \lambda_1 & & & * \\ & \lambda_2 & & \\ & & \ddots & \\ \mathbf{0} & & & \lambda_n \end{pmatrix}.$$

Then it follows from Proposition 5.12 that the claim holds. □

► EXERCISE 5.19 (5.18). *Give an example of an operator whose matrix with respect to some basis contains only 0's on the diagonal, but the operator is invertible.*

PROOF. Let $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^2)$. Take the standard basis $\big((0, 1), (1, 0)\big)$ of $\mathbb{R}^2$, with respect to which T has the following matrix

$$\mathcal{M}(\mathrm{T}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $\mathrm{T}(x, y) = \mathcal{M}(\mathrm{T}) \cdot (x, y)' = \big(y, x\big)$ is invertible. □

► EXERCISE 5.20 (5.19). *Give an example of an operator whose matrix with respect to some basis contains only nonzero numbers on the diagonal, but the operator is not invertible.*

PROOF. Consider the standard basis $\big((1,0),(0,1)\big)$ of $\mathbb{R}^2$. Let $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^2)$ be defined as $\mathrm{T}(x,y) = (x,0)$. Then T is not injective and so is not invertible. Its matrix is

$$\mathcal{M}(\mathrm{T}) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \qquad \square$$

► EXERCISE 5.21 (5.20). *Suppose that* $\mathrm{T} \in \mathfrak{L}(V)$ *has* $\dim V$ *distinct eigenvalues and that* $\mathrm{S} \in \mathfrak{L}(V)$ *has the same eigenvectors as* T *(not necessarily with the same eigenvalues). Prove that* $\mathrm{ST} = \mathrm{TS}$.

PROOF. Let $\dim V = n$. Let $\lambda_1, \dots, \lambda_n$ be n distinct eigenvalues of T and $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ be n eigenvector corresponding to the eigenvalues. Then $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is independent and so is a basis of V. Further, the matrix of T with respect to $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is given by

$$\mathcal{M}\big(\mathrm{T}, (\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)\big) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Since S has the same eigenvectors as T, so for any $\boldsymbol{v}_i$, there is some $\widehat{\lambda}_i$ such that $\mathrm{S}\boldsymbol{v}_i = \widehat{\lambda}_i \boldsymbol{v}_i$. For every $\boldsymbol{v} = \sum_{i=1}^n a_i \boldsymbol{v}_i \in V$ we have

$$\begin{aligned}
(\mathrm{ST})(\boldsymbol{v}) = \mathrm{S}\left[\mathrm{T}\left(\sum_{i=1}^n a_i \boldsymbol{v}_i\right)\right] = \mathrm{S}\left(\sum_{i=1}^n a_i \mathrm{T}\boldsymbol{v}_i\right) &= \mathrm{S}\left(\sum_{i=1}^n a_i \lambda_i \boldsymbol{v}_i\right) \\
&= \sum_{i=1}^n (a_i \lambda_i)\, \mathrm{S}\boldsymbol{v}_i \\
&= \sum_{i=1}^n \left(a_i \lambda_i \widehat{\lambda}_i\right) \boldsymbol{v}_i,
\end{aligned}$$

and

$$\begin{aligned}
(\mathrm{TS})(\boldsymbol{v}) = \mathrm{T}\left[\mathrm{S}\left(\sum_{i=1}^n a_i \boldsymbol{v}_i\right)\right] = \mathrm{T}\left(\sum_{i=1}^n a_i \mathrm{S}\boldsymbol{v}_i\right) &= \mathrm{T}\left(\sum_{i=1}^n \left(a_i \widehat{\lambda}_i\right) \boldsymbol{v}_i\right) \\
&= \sum_{i=1}^n \left(a_i \widehat{\lambda}_i\right) \mathrm{T}\boldsymbol{v}_i \\
&= \sum_{i=1}^n \left(a_i \lambda_i \widehat{\lambda}_i\right) \boldsymbol{v}_i.
\end{aligned}$$

Hence, ST = TS. □

► EXERCISE 5.22 (5.21). *Suppose* $\mathbf{P} \in \mathfrak{L}(V)$ *and* $\mathbf{P}^2 = \mathbf{P}$. *Prove that* $V = \mathcal{N}_{\mathbf{P}} \oplus \mathcal{R}_{\mathbf{P}}$.

PROOF. By Theorem 3.4, $\dim V = \dim \mathcal{N}_{\mathbf{P}} + \dim \mathcal{R}_{\mathbf{P}}$, so it suffices to show that $V = \mathcal{N}_{\mathbf{P}} + \mathcal{R}_{\mathbf{P}}$ by Proposition 2.19. Take an arbitrary $\boldsymbol{v} \in V$. Since $\mathbf{P}^2 = \mathbf{P}$, we have

$$\mathbf{P}^2\boldsymbol{v} = \mathbf{P}\boldsymbol{v} \iff \mathbf{P}(\mathbf{P}\boldsymbol{v} - \boldsymbol{v}) = \mathbf{0} \iff \mathbf{P}\boldsymbol{v} - \boldsymbol{v} \in \mathcal{N}_{\mathbf{P}};$$

that is, there exists $\boldsymbol{u} \in \mathcal{N}_{\mathbf{P}}$ such that $\mathbf{P}\boldsymbol{v} - \boldsymbol{v} = \boldsymbol{u}$. Therefore,

$$\boldsymbol{v} = -\boldsymbol{u} + \mathbf{P}\boldsymbol{v} \in \mathcal{N}_{\mathbf{P}} + \mathcal{R}_{\mathbf{P}}.$$ □

► EXERCISE 5.23 (5.22). *Suppose* $V = U \oplus W$, *where* U *and* W *are nonzero subspaces of* V. *Find all eigenvalues and eigenvectors of* $\mathbf{P}_{U,W}$.

PROOF. We first show that $\lambda = 0$ is an eigenvalue of $\mathbf{P}_{U,W}$. Since $W \neq \{\mathbf{0}\}$, we can take $\boldsymbol{w} \in W$ with $\boldsymbol{w} \neq \mathbf{0}$. Obviously, $\boldsymbol{w} \in V$ and $\boldsymbol{w}$ can be written as $\boldsymbol{w} = \mathbf{0} + \boldsymbol{w}$ uniquely. Then

$$\mathbf{P}_{U,W}(\boldsymbol{w}) = \mathbf{0} = 0\boldsymbol{w};$$

that is, 0 is an eigenvalue of $\mathbf{P}_{U,W}$ and any $\boldsymbol{w} \in W$ with $\boldsymbol{w} \neq \mathbf{0}$ is an eigenvector corresponding to 0.

Now let us check whether there is eigenvalue $\lambda \neq 0$. If there is an eigenvalue $\lambda \neq 0$ under $\mathbf{P}_{U,W}$, then there exists $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w} \neq \mathbf{0}$, where $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in W$, such that $\mathbf{P}_{U,W}(\boldsymbol{v}) = \lambda\boldsymbol{v}$, but which means that

$$\boldsymbol{u} = \lambda(\boldsymbol{u} + \boldsymbol{w}).$$

Then $\boldsymbol{w} = (1-\lambda)\boldsymbol{u}/\lambda \in U$ since $\lambda \neq 0$, and which implies that $\boldsymbol{w} = \mathbf{0}$ since $V = U \oplus W$ forces $U \cap W = \{\mathbf{0}\}$. Therefore, $\boldsymbol{v} = \boldsymbol{u} \neq \mathbf{0}$ and

$$\mathbf{P}_{U,W}(\boldsymbol{v})\mathbf{P}_{U,W}(\boldsymbol{u}) = \boldsymbol{u} = 1 \cdot \boldsymbol{u},$$

that is, $\lambda = 1$ is the unique nonzero eigenvalue of $\mathbf{P}_{U,W}$. □

► EXERCISE 5.24 (5.23). *Give an example of an operator* $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^4)$ *such that* T *has no (real) eigenvalues.*

PROOF. Our example is based on (5.4). Let $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^4)$ be defined by

$$\mathrm{T}(x_1, x_2, x_3, x_4) = (-x_2, x_1, -x_4, x_3).$$

Suppose that λ is a (real) eigenvalue of T; then

$$\begin{cases} \lambda x_1 = -x_2 \\ \lambda x_2 = x_1 \\ \lambda x_3 = -x_4 \\ \lambda x_4 = x_3. \end{cases}$$

If $\lambda = 0$, then $(x_1, x_2, x_3, x_4) = \mathbf{0}$. So $\lambda \neq 0$. It is evident that

$$x_1 \neq 0 \iff x_2 \neq 0, \quad \text{and} \quad x_3 \neq 0 \iff x_4 \neq 0.$$

Suppose that $x_1 \neq 0$. Then from the first two equations we have

$$\lambda^2 x_2 = -x_2 \Longrightarrow \lambda^2 = -1,$$

which has no solution in $\mathbb{R}$. Hence, $x_1 = x_2 = 0$ when $\lambda \neq 0$. Similarly, we can show that $x_3 = x_4 = 0$ if $\lambda \neq 0$. □

► EXERCISE 5.25 (5.24). *Suppose V is a real vector space and $\mathrm{T} \in \mathfrak{L}(V)$ has no eigenvalues. Proves that every subspace of V invariant under T has even dimension.*

PROOF. If U is invariant under T and $\dim U$ is odd, then $\mathrm{T}\restriction_U \in \mathfrak{L}(U)$ has an eigenvalue. But this implies that T has an eigenvalue. A contradiction. □

6

INNER-PRODUCT SPACES

"As You Should Verify"

Remark 6.1 (p. 113). The orthogonal projection $\mathbf{P}_U$ has the following properties:

a. $\mathbf{P}_U \in \mathfrak{L}(V)$;

b. $\mathcal{R}_{\mathbf{P}_U} = U$;

c. $\mathcal{N}_{\mathbf{P}_U} = U^\perp$;

d. $\boldsymbol{v} - \mathbf{P}_U \boldsymbol{v} \in U^\perp$ for every $\boldsymbol{v} \in V$;

e. $\mathbf{P}_U^2 = \mathbf{P}_U$;

f. $\|\mathbf{P}_U \boldsymbol{v}\| \leqslant \|\boldsymbol{v}\|$ for every $\boldsymbol{v} \in V$;

g. $\mathbf{P}_U \boldsymbol{v} = \sum_{i=1}^m \langle \boldsymbol{v}, \boldsymbol{e}_i \rangle \boldsymbol{e}_i$ for every $\boldsymbol{v} \in V$, where $(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_m)$ is a basis of U.

Proof. (a) For any $\boldsymbol{v}, \boldsymbol{v}' \in V$, we have

$$\begin{aligned}\mathbf{P}_U(\boldsymbol{v} + \boldsymbol{v}') = \mathbf{P}_U\left[(\boldsymbol{u} + \boldsymbol{w}) + (\boldsymbol{u}' + \boldsymbol{w}')\right] &= \mathbf{P}_U\left[(\boldsymbol{u} + \boldsymbol{u}') + (\boldsymbol{w} + \boldsymbol{w}')\right] \\ &= \boldsymbol{u} + \boldsymbol{u}' \\ &= \mathbf{P}_U \boldsymbol{v} + \mathbf{P}_U \boldsymbol{v}',\end{aligned}$$

where $\boldsymbol{u}, \boldsymbol{u}' \in U$ and $\boldsymbol{w}, \boldsymbol{w}' \in U^\perp$. Also it is true that $\mathbf{P}_U(a\boldsymbol{v}) = a\mathbf{P}_U \boldsymbol{v}$. Therefore, $\mathbf{P}_U \in \mathfrak{L}(V)$.

(b) Write every $\boldsymbol{v} \in V$ as $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$, where $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in U^\perp$. Since $\mathbf{P}_U \boldsymbol{v} = \boldsymbol{u}$, we have one direction that $\mathcal{R}_{\mathbf{P}_U} \subseteq U$. For the other direction, notice that $U = \mathbf{P}_U[U] \subseteq \mathcal{R}_{\mathbf{P}_U}$.

(c) If $\boldsymbol{v} \in \mathcal{N}_{\mathbf{P}_U}$, then $\mathbf{0} = \mathbf{P}_U \boldsymbol{v} = \boldsymbol{u}$; that is, $\boldsymbol{v} = \mathbf{0} + \boldsymbol{w}$ with $\boldsymbol{w} \in U^\perp$. This proves that $\mathcal{N}_{\mathbf{P}_U} \subseteq U^\perp$. The other inclusion direction is clear.

(d) For every $\boldsymbol{v} \in V$, we have $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$, where $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in U^\perp$. Hence, $\boldsymbol{v} - \mathbf{P}_U \boldsymbol{v} = (\boldsymbol{u} + \boldsymbol{w}) - \boldsymbol{u} = \boldsymbol{w} \in U^\perp$.

(e) For every $\boldsymbol{v} \in V$, we have $\mathbf{P}_U^2 \boldsymbol{v} = \mathbf{P}_U(\mathbf{P}_U \boldsymbol{v}) = \mathbf{P}_U \boldsymbol{u} = \boldsymbol{u} = \mathbf{P}_U \boldsymbol{v}$.

(f) We can write every $\boldsymbol{v} \in V$ as $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$ with $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in U^{\perp}$; therefore, $\|\boldsymbol{v}\| = \|\boldsymbol{u} + \boldsymbol{w}\| \overset{*}{=} \|\boldsymbol{u}\| + \|\boldsymbol{w}\| \geqslant \|\boldsymbol{u}\| = \|\mathbf{P}_U \boldsymbol{v}\|$, where $(*)$ holds since $U \perp U^{\perp}$.

(g) It follows from Axler (1997, 6.31, p.112). □

REMARK 6.2 (p. 119-120). Verify that the function $\mathrm{T} \mapsto \mathrm{T}^*$ has the following properties:

a. $(\mathrm{S} + \mathrm{T})^* = \mathrm{S}^* + \mathrm{T}^*$ for all $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V, W)$;

b. $(a\mathrm{T})^* = \overline{a}\,\mathrm{T}^*$ for all $a \in \mathbb{F}$ and $\mathrm{T} \in \mathfrak{L}(V, W)$;

c. $(\mathrm{T}^*)^* = \mathrm{T}$ for all $\mathrm{T} \in \mathfrak{L}(V, W)$;

d. $\mathrm{Id}^* = \mathrm{Id}$, where Id is the identity operator on V;

e. $(\mathrm{ST})^* = \mathrm{T}^*\mathrm{S}^*$ for all $\mathrm{T} \in \mathfrak{L}(V, W)$ and $\mathrm{S} \in \mathfrak{L}(W, U)$.

PROOF. (a) $\langle (\mathrm{S} + \mathrm{T})\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \mathrm{S}\boldsymbol{v}, \boldsymbol{w} \rangle + \langle \mathrm{T}\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, \mathrm{S}^*\boldsymbol{w} \rangle + \langle \boldsymbol{v}, \mathrm{T}^*\boldsymbol{w} \rangle = \langle \boldsymbol{v}, (\mathrm{S}^* + \mathrm{T}^*)\boldsymbol{w} \rangle$.

(b) $\langle (a\mathrm{T})\boldsymbol{v}, \boldsymbol{w} \rangle = a\langle \mathrm{T}\boldsymbol{v}, \boldsymbol{w} \rangle = a\langle \boldsymbol{v}, \mathrm{T}^*\boldsymbol{w} \rangle = \langle \boldsymbol{v}, (\overline{a}\,\mathrm{T}^*)(\boldsymbol{w}) \rangle$.

(c) $\langle \mathrm{T}^*\boldsymbol{w}, \boldsymbol{v} \rangle = \overline{\langle \boldsymbol{v}, \mathrm{T}^*\boldsymbol{w} \rangle} = \overline{\langle \mathrm{T}\boldsymbol{v}, \boldsymbol{w} \rangle} = \langle \boldsymbol{w}, \mathrm{T}\boldsymbol{v} \rangle$.

(d) $\langle \mathrm{Id}\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, \mathrm{Id}\boldsymbol{w} \rangle$.

(e) $\langle (\mathrm{ST})\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \mathrm{S}(\mathrm{T}\boldsymbol{v}), \boldsymbol{w} \rangle = \langle \mathrm{T}\boldsymbol{v}, \mathrm{S}^*\boldsymbol{w} \rangle = \langle \boldsymbol{v}, (\mathrm{T}^*\mathrm{S}^*)\boldsymbol{w} \rangle$. □

EXERCISES

► EXERCISE 6.3 (6.1). *Prove that if $\boldsymbol{x}$, $\boldsymbol{y}$ are nonzero vectors in $\mathbb{R}^2$, then $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \|\boldsymbol{x}\|\,\|\boldsymbol{y}\| \cos\theta$, where θ is the angle between $\boldsymbol{x}$ and $\boldsymbol{y}$.*

PROOF. Using notation as in Figure 6.1, the law of cosines states that

$$\|\boldsymbol{x} - \boldsymbol{y}\|^2 = \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 - 2\|\boldsymbol{x}\|\,\|\boldsymbol{y}\| \cos\theta. \tag{6.1}$$

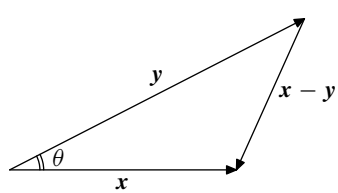

FIGURE 6.1. The law of cosines

After inserting $\|\boldsymbol{x} - \boldsymbol{y}\|^2 = \langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle = \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 - 2\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ into (6.1), we get the conclusion. □

► EXERCISE 6.4 (6.2). *Suppose $\boldsymbol{u}, \boldsymbol{v} \in V$. Prove that $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 0$ if and only if $\|\boldsymbol{u}\| \leqslant \|\boldsymbol{u} + a\boldsymbol{v}\|$ for all $a \in \mathbb{F}$.*

PROOF. If $\langle \boldsymbol{u}, \boldsymbol{v}\rangle = 0$, then $\langle \boldsymbol{u}, a\boldsymbol{v}\rangle = 0$ and so

$$\|\boldsymbol{u} + a\boldsymbol{v}\|^2 = \langle \boldsymbol{u} + a\boldsymbol{v}, \boldsymbol{u} + a\boldsymbol{v}\rangle = \|\boldsymbol{u}\|^2 + \|a\boldsymbol{v}\|^2 \geqslant \|\boldsymbol{u}\|^2.$$

Now suppose that $\|\boldsymbol{u}\| \leqslant \|\boldsymbol{u} + a\boldsymbol{v}\|$ for any $a \in \mathbb{F}$. If $\boldsymbol{v} = \boldsymbol{0}$, then $\langle \boldsymbol{u}, \boldsymbol{v}\rangle = 0$ holds trivially. Thus we assume that $\boldsymbol{v} \neq \boldsymbol{0}$. We first have

$$\begin{aligned}
\|\boldsymbol{u} + a\boldsymbol{v}\|^2 &= \langle \boldsymbol{u} + a\boldsymbol{v}, \boldsymbol{u} + a\boldsymbol{v}\rangle \\
&= \langle \boldsymbol{u}, \boldsymbol{u} + a\boldsymbol{v}\rangle + \langle a\boldsymbol{v}, \boldsymbol{u} + a\boldsymbol{v}\rangle \\
&= \|\boldsymbol{u}\|^2 + \overline{a}\langle \boldsymbol{u}, \boldsymbol{v}\rangle + a\overline{\langle \boldsymbol{u}, \boldsymbol{v}\rangle} + \|a\boldsymbol{v}\|^2 \\
&= \|\boldsymbol{u}\|^2 + \|a\boldsymbol{v}\|^2 + \overline{a}\langle \boldsymbol{u}, \boldsymbol{v}\rangle + \overline{\overline{a}\langle \boldsymbol{u}, \boldsymbol{v}\rangle} \\
&= \|\boldsymbol{u}\|^2 + \|a\boldsymbol{v}\|^2 + 2\mathrm{Re}\left(\overline{a}\langle \boldsymbol{u}, \boldsymbol{v}\rangle\right).
\end{aligned}$$

Therefore, $\|\boldsymbol{u}\| \leqslant \|\boldsymbol{u} + a\boldsymbol{v}\|$ for all $a \in \mathbb{F}$ implies that for all $a \in \mathbb{F}$,

$$2\mathrm{Re}\left(\overline{a}\langle \boldsymbol{u}, \boldsymbol{v}\rangle\right) \geqslant -\|a\boldsymbol{v}\|^2 = -|a|^2\,\|\boldsymbol{v}\|^2. \tag{6.2}$$

Take $a = -\alpha\langle \boldsymbol{u}, \boldsymbol{v}\rangle$, with $\alpha > 0$; then (6.2) becomes

$$2\left|\langle \boldsymbol{u}, \boldsymbol{v}\rangle\right|^2 \leqslant \alpha\left|\langle \boldsymbol{u}, \boldsymbol{v}\rangle\right|^2 \|\boldsymbol{v}\|^2. \tag{6.3}$$

Let $\alpha = 1/\|\boldsymbol{v}\|^2$. Then (6.3) becomes

$$2\left|\langle \boldsymbol{u}, \boldsymbol{v}\rangle\right|^2 \leqslant \left|\langle \boldsymbol{u}, \boldsymbol{v}\rangle\right|^2.$$

Hence, $\langle \boldsymbol{u}, \boldsymbol{v}\rangle = 0$. □

► EXERCISE 6.5 (6.3). *Prove that* $\left(\sum_{j=1}^n a_j b_j\right)^2 \leqslant \left(\sum_{j=1}^n j a_j^2\right)\left(\sum_{j=1}^n b_j^2/j\right)$ *for all* $a_j, b_j \in \mathbb{R}$.

PROOF. Since $a_j, b_j \in \mathbb{R}$, we can write any $\boldsymbol{a} = (a_1, \ldots, a_n)$ and $\boldsymbol{b} = (b_1, \ldots, b_n)$ as $\boldsymbol{a} = (a_1', a_2'/\sqrt{2}, \ldots, a_n'/\sqrt{n})$ and $\boldsymbol{b} = (b_1', \sqrt{2}b_2', \ldots, \sqrt{n}b_n')$ for some $\boldsymbol{a}' = (a_1', \ldots, a_n')$ and $\boldsymbol{b}' = (b_1', \ldots, b_n')$. Then

$$\left(\sum_{j=1}^n a_j b_j\right)^2 = \left(\sum_{j=1}^n a_j' b_j'\right)^2 = \left\langle \boldsymbol{a}', \boldsymbol{b}'\right\rangle^2,$$

$$\sum_{j=1}^n j a_j^2 = \sum_{j=1}^n j\,\frac{a_j'^2}{j} = \sum_{j=1}^n a_j'^2 = \|\boldsymbol{a}'\|^2,$$

and

$$\sum_{j=1}^n \frac{b_j^2}{j} = \sum_{j=1}^n \frac{j\,b_j'^2}{j} = \sum_{j=1}^n b_j'^2 = \|\boldsymbol{b}'\|^2.$$

Hence, by the Cauchy-Schwarz Inequality,

$$\left(\sum_{j=1}^{n} a_j b_j\right)^2 = \langle \boldsymbol{a}', \boldsymbol{b}' \rangle^2 \leqslant \|\boldsymbol{a}'\|^2 \, \|\boldsymbol{b}'\|^2 = \left(\sum_{j=1}^{n} j a_j^2\right)\left(\sum_{j=1}^{n} \frac{b_j^2}{j}\right). \qquad \square$$

► EXERCISE 6.6 (6.4). *Suppose $\boldsymbol{u}, \boldsymbol{v} \in V$ are such that $\|\boldsymbol{u}\| = 3$, $\|\boldsymbol{u} + \boldsymbol{v}\| = 4$, and $\|\boldsymbol{u} - \boldsymbol{v}\| = 6$. What number must $\|\boldsymbol{v}\|$ equal?*

SOLUTION. By the parallelogram equality, $\|\boldsymbol{u} + \boldsymbol{v}\|^2 + \|\boldsymbol{u} - \boldsymbol{v}\|^2 = 2(\|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2)$, so we have $\|\boldsymbol{v}\| = \sqrt{17}$. $\square$

► EXERCISE 6.7 (6.5). *Prove or disprove: there is an inner product on $\mathbb{R}^2$ such that the associated norm is given by $\|(x_1, x_2)\| = |x_1| + |x_2|$ for all $(x_1, x_2) \in \mathbb{R}^2$.*

PROOF. There is no such inner product on $\mathbb{R}^2$. For example, let $\boldsymbol{u} = (1, 0)$ and $\boldsymbol{v} = (0, 1)$. Then $\|\boldsymbol{u}\| = \|\boldsymbol{v}\| = 1$ and $\|\boldsymbol{u} + \boldsymbol{v}\| = \|\boldsymbol{u} - \boldsymbol{v}\| = 2$. But then the Parallelogram Equality fails. $\square$

► EXERCISE 6.8 (6.6). *Prove that if V is a real inner-product space, then $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = (\|\boldsymbol{u} + \boldsymbol{v}\|^2 - \|\boldsymbol{u} - \boldsymbol{v}\|^2)/4$ for all $\boldsymbol{u}, \boldsymbol{v} \in V$.*

PROOF. If V is a real inner-product space, then for any $\boldsymbol{u}, \boldsymbol{v} \in V$,

$$\begin{aligned}
\frac{\|\boldsymbol{u} + \boldsymbol{v}\|^2 - \|\boldsymbol{u} - \boldsymbol{v}\|^2}{4} &= \frac{\langle \boldsymbol{u} + \boldsymbol{v}, \boldsymbol{u} + \boldsymbol{v} \rangle - \langle \boldsymbol{u} - \boldsymbol{v}, \boldsymbol{u} - \boldsymbol{v} \rangle}{4} \\
&= \frac{\left(\|\boldsymbol{u}\|^2 + 2\langle \boldsymbol{u}, \boldsymbol{v} \rangle + \|\boldsymbol{v}\|^2\right) - \left(\|\boldsymbol{u}\|^2 - 2\langle \boldsymbol{u}, \boldsymbol{v} \rangle + \|\boldsymbol{v}\|^2\right)}{4} \\
&= \langle \boldsymbol{u}, \boldsymbol{v} \rangle. \qquad \square
\end{aligned}$$

► EXERCISE 6.9 (6.7). *Prove that if V is a complex inner-product space, then*

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \frac{\|\boldsymbol{u} + \boldsymbol{v}\|^2 - \|\boldsymbol{u} - \boldsymbol{v}\|^2 + \|\boldsymbol{u} + i\,\boldsymbol{v}\|^2\, i - \|\boldsymbol{u} - i\,\boldsymbol{v}\|^2\, i}{4}$$

for all $\boldsymbol{u}, \boldsymbol{v} \in V$.

PROOF. If V is a complex inner-product space, then for any $\boldsymbol{u}, \boldsymbol{v} \in V$ we have

$$\begin{aligned}
\|\boldsymbol{u} + \boldsymbol{v}\|^2 = \langle \boldsymbol{u} + \boldsymbol{v}, \boldsymbol{u} + \boldsymbol{v} \rangle &= \langle \boldsymbol{u}, \boldsymbol{u} + \boldsymbol{v} \rangle + \langle \boldsymbol{v}, \boldsymbol{u} + \boldsymbol{v} \rangle \\
&= \|\boldsymbol{u}\|^2 + \langle \boldsymbol{u}, \boldsymbol{v} \rangle + \langle \boldsymbol{v}, \boldsymbol{u} \rangle + \|\boldsymbol{v}\|^2, \\
\|\boldsymbol{u} - \boldsymbol{v}\|^2 = \langle \boldsymbol{u} - \boldsymbol{v}, \boldsymbol{u} - \boldsymbol{v} \rangle &= \langle \boldsymbol{u}, \boldsymbol{u} - \boldsymbol{v} \rangle - \langle \boldsymbol{v}, \boldsymbol{u} - \boldsymbol{v} \rangle \\
&= \|\boldsymbol{u}\|^2 - \langle \boldsymbol{u}, \boldsymbol{v} \rangle - \langle \boldsymbol{v}, \boldsymbol{u} \rangle + \|\boldsymbol{v}\|^2, \\
\|\boldsymbol{u} + i\,\boldsymbol{v}\|^2\, i = \langle \boldsymbol{u} + i\,\boldsymbol{v}, \boldsymbol{u} + i\,\boldsymbol{v} \rangle\, i &= \big(\langle \boldsymbol{u}, \boldsymbol{u} + i\,\boldsymbol{v} \rangle + \langle i\,\boldsymbol{v}, \boldsymbol{u} + i\,\boldsymbol{v} \rangle\big)\, i \\
&= \left(\|\boldsymbol{u}\|^2 + \bar{i}\,\langle \boldsymbol{u}, \boldsymbol{v} \rangle + i\,\langle \boldsymbol{v}, \boldsymbol{u} \rangle + i\bar{i}\,\|\boldsymbol{v}\|^2\right) i \\
&= \|\boldsymbol{u}\|^2 i + \langle \boldsymbol{u}, \boldsymbol{v} \rangle - \langle \boldsymbol{v}, \boldsymbol{u} \rangle + \|\boldsymbol{v}\|^2 i,
\end{aligned}$$

and

$$\begin{aligned}\|\boldsymbol{u}-i\,\boldsymbol{v}\|^2\,i &= \langle \boldsymbol{u}-i\,\boldsymbol{v},\boldsymbol{u}-i\,\boldsymbol{v}\rangle\,i = \big(\langle \boldsymbol{u},\boldsymbol{u}-i\,\boldsymbol{v}\rangle - \langle i\,\boldsymbol{v},\boldsymbol{u}-i\,\boldsymbol{v}\rangle\big)\,i\\ &= \Big(\|\boldsymbol{u}\|^2 - \bar{i}\,\langle \boldsymbol{u},\boldsymbol{v}\rangle - i\,\langle \boldsymbol{v},\boldsymbol{u}\rangle + i\bar{i}\,\|\boldsymbol{v}\|^2\Big)\,i\\ &= \|\boldsymbol{u}\|^2 i - \langle \boldsymbol{u},\boldsymbol{v}\rangle + \langle \boldsymbol{v},\boldsymbol{u}\rangle + \|\boldsymbol{v}\|^2 i.\end{aligned}$$

Hence,

$$\begin{aligned}&\frac{\|\boldsymbol{u}+\boldsymbol{v}\|^2 - \|\boldsymbol{u}-\boldsymbol{v}\|^2 + \|\boldsymbol{u}+i\,\boldsymbol{v}\|^2\,i - \|\boldsymbol{u}-i\,\boldsymbol{v}\|^2\,i}{4}\\ &\quad= \frac{2\langle \boldsymbol{u},\boldsymbol{v}\rangle + 2\,\langle \boldsymbol{v},\boldsymbol{u}\rangle + 2\langle \boldsymbol{u},\boldsymbol{v}\rangle - 2\,\langle \boldsymbol{v},\boldsymbol{u}\rangle}{4}\\ &\quad= \langle \boldsymbol{u},\boldsymbol{v}\rangle.\end{aligned}$$

□

► EXERCISE 6.10 (6.10). *On $\mathfrak{P}_2(\mathbb{R})$, consider the inner product given by*

$$\langle p,q\rangle = \int_0^1 p\,(x)\,q\,(x)\;\mathrm{d}x.$$

Apply the Gram-Schmidt procedure to the basis $(1,x,x^2)$ to produce an orthonormal basis of $\mathfrak{P}_2(\mathbb{R})$.

SOLUTION. It is clear that $e_1 = 1$ since $\|1\|^2 = \int_0^1 (1\times 1)\;\mathrm{d}x = 1$. As for e_2, let

$$e_2 = \frac{x - \langle x,e_1\rangle\,e_1}{\big\|x - \langle x,e_1\rangle\,e_1\big\|}.$$

Since

$$\langle x,e_1\rangle = \int_0^1 x\,\mathrm{d}x = \frac{1}{2};$$

we have

$$e_2 = \frac{x-1/2}{\|x-1/2\|} = \frac{x-1/2}{\sqrt{\displaystyle\int_0^1 \big(x-1/2\big)^2\;\mathrm{d}x}} = \sqrt{3}\,(2x-1)\,.$$

As for e_3,

$$e_3 = \frac{x^2 - \langle x^2,e_1\rangle e_1 - \langle x^2,e_2\rangle e_2}{\big\|x^2 - \langle x^2,e_1\rangle e_1 - \langle x^2,e_2\rangle e_2\big\|}.$$

Since

$$\begin{aligned}\Big\langle x^2,e_1\Big\rangle &= \int_0^1 x^2\,\mathrm{d}x = \frac{1}{3},\\ \Big\langle x^2,e_2\Big\rangle &= \int_0^1 x^2\Big[\sqrt{3}\,(2x-1)\Big]\;\mathrm{d}x = \frac{\sqrt{3}}{6},\end{aligned}$$

and

$$\big\|x^2 - \langle x^2,e_1\rangle e_1 - \langle x^2,e_2\rangle e_2\big\| = \sqrt{\int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2\;\mathrm{d}x} = \frac{\sqrt{1/5}}{6},$$

we know that

$$e_3 = \frac{x^2 - x + 1/6}{\sqrt{1/5}\big/6} = \sqrt{5}\left(6x^2 - 6x + 1\right). \qquad \square$$

► EXERCISE 6.11 (6.11). *What happens if the Gram-Schmidt procedure is applied to a list of vectors that is not linearly independent.*

SOLUTION. If $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is not linearly independent, then

$$\boldsymbol{e}_j = \frac{\boldsymbol{v}_j - \langle \boldsymbol{v}_j, \boldsymbol{e}_j \rangle \boldsymbol{e}_1 - \cdots - \langle \boldsymbol{v}_j, \boldsymbol{e}_{j-1} \rangle \boldsymbol{e}_{j-1}}{\left\| \boldsymbol{v}_j - \langle \boldsymbol{v}_j, \boldsymbol{e}_j \rangle \boldsymbol{e}_1 - \cdots - \langle \boldsymbol{v}_j, \boldsymbol{e}_{j-1} \rangle \boldsymbol{e}_{j-1} \right\|}$$

may not be well defined since if $\boldsymbol{v}_j \in \operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1})$, then

$$\left\| \boldsymbol{v}_j - \langle \boldsymbol{v}_j, \boldsymbol{e}_j \rangle \boldsymbol{e}_1 - \cdots - \langle \boldsymbol{v}_j, \boldsymbol{e}_{j-1} \rangle \boldsymbol{e}_{j-1} \right\| = 0. \qquad \square$$

► EXERCISE 6.12 (6.12). *Suppose V is a real inner-product space and $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_m)$ is a linearly independent list of vectors in V. Prove that there exist exactly 2^m orthonormal lists $(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m)$ of vectors in V such that*

$$\operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_j) = \operatorname{span}(\boldsymbol{e}_1, \dots, \boldsymbol{e}_j)$$

for all $j \in \{1, \dots, m\}$.

PROOF. Given the linearly independent list $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_m)$, we have a corresponding orthonormal list $(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m)$ by the Gram-Schmidt procedure, such that $\operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_j) = \operatorname{span}(\boldsymbol{e}_1, \dots, \boldsymbol{e}_j)$ for all $j \in \{1, \dots, m\}$.

Now, for every $i = 1, \dots, m$, the list $(\boldsymbol{e}_1, \dots, -\boldsymbol{e}_i, \dots, \boldsymbol{e}_m)$ is also an orthonormal list; further,

$$\operatorname{span}(\boldsymbol{e}_1, \dots, \boldsymbol{e}_i) = \operatorname{span}(\boldsymbol{e}_1, \dots, -\boldsymbol{e}_i).$$

The above shows that there are at least 2^m orthonormal lists satisfying the requirement.

On the other hand, if there is an orthonormal list $(\boldsymbol{f}_1, \dots, \boldsymbol{f}_m)$ satisfying

$$\operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_j) = \operatorname{span}(\boldsymbol{f}_1, \dots, \boldsymbol{f}_j)$$

for all $j \in \{1, \dots, m\}$, then $\operatorname{span}(\boldsymbol{v}_1) = \operatorname{span}(\boldsymbol{f}_1)$ implies that

$$\boldsymbol{f}_1 = \pm \frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|} = \pm \boldsymbol{e}_1;$$

Similarly, $\operatorname{span}(\boldsymbol{e}_1, \boldsymbol{e}_2) = \operatorname{span}(\boldsymbol{v}_1, \boldsymbol{v}_2) = \operatorname{span}(\boldsymbol{e}_1, \boldsymbol{f}_2)$ implies that

$$\boldsymbol{f}_2 = a_1 \boldsymbol{e}_1 + a_2 \boldsymbol{e}_2, \quad \text{for some } a_1, a_2 \in \mathbb{R}.$$

Then the orthonormality implies that

$$\langle \boldsymbol{e}_1, a_1 \boldsymbol{e}_1 + a_2 \boldsymbol{e}_2 \rangle = 0 \Longrightarrow a_1 = 0,$$
$$\langle a_2 \boldsymbol{e}_2, a_2 \boldsymbol{e}_2 \rangle = 1 \Longrightarrow a_2 = \pm 1;$$

that is, $\boldsymbol{f}_2 = \pm \boldsymbol{e}_2$. By induction, we can show that $\boldsymbol{f}_i = \pm \boldsymbol{e}_i$ for all $i = 1, \dots, m$, and this completes the proof. □

► EXERCISE 6.13 (6.13). *Suppose* $(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m)$ *is an orthonormal list of vectors in* V. *Let* $\boldsymbol{v} \in V$. *Prove that* $\|\boldsymbol{v}\|^2 = \left|\langle \boldsymbol{v}, \boldsymbol{e}_1 \rangle\right|^2 + \cdots + \left|\langle \boldsymbol{v}, \boldsymbol{e}_m \rangle\right|^2$ *if and only if* $\boldsymbol{v} \in \operatorname{span}(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m)$.

PROOF. It follows from Corollary 6.25 that the list $(e_1, \dots, e_m)$ can be extended to an orthonormal basis $(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m, \boldsymbol{f}_1, \dots, \boldsymbol{f}_n)$ of V. Then by Theorem 6.17, every vector $\boldsymbol{v} \in V$ can be presented uniquely as $\boldsymbol{v} = \sum_{i=1}^m \langle \boldsymbol{v}, \boldsymbol{e}_i \rangle \boldsymbol{e}_i + \sum_{j=1}^n \langle \boldsymbol{v}, \boldsymbol{f}_j \rangle \boldsymbol{f}_j$, and so

$$\begin{aligned}
\|\boldsymbol{v}\|^2 &= \left\| \sum_{i=1}^m \langle \boldsymbol{v}, \boldsymbol{e}_i \rangle \boldsymbol{e}_i + \sum_{j=1}^n \langle \boldsymbol{v}, \boldsymbol{f}_j \rangle \boldsymbol{f}_j \right\|^2 \\
&= \left\langle \sum_{i=1}^m \langle \boldsymbol{v}, \boldsymbol{e}_i \rangle \boldsymbol{e}_i + \sum_{j=1}^n \langle \boldsymbol{v}, \boldsymbol{f}_j \rangle \boldsymbol{f}_j, \sum_{i=1}^m \langle \boldsymbol{v}, \boldsymbol{e}_i \rangle \boldsymbol{e}_i + \sum_{j=1}^n \langle \boldsymbol{v}, \boldsymbol{f}_j \rangle \boldsymbol{f}_j \right\rangle \\
&= \sum_{i=1}^m \left|\langle \boldsymbol{v}, \boldsymbol{e}_i \rangle\right|^2 + \sum_{j=1}^n \left|\langle \boldsymbol{v}, \boldsymbol{f}_j \rangle\right|^2 .
\end{aligned}$$

Hence,

$$\begin{aligned}
\|\boldsymbol{v}\|^2 = \left|\langle \boldsymbol{v}, \boldsymbol{e}_1 \rangle\right|^2 + \cdots + \left|\langle \boldsymbol{v}, \boldsymbol{e}_m \rangle\right|^2 &\iff \langle \boldsymbol{v}, \boldsymbol{f}_j \rangle = 0, \quad \forall\ j = 1, \dots, n \\
&\iff \boldsymbol{v} = \sum_{i=1}^m \langle \boldsymbol{v}, \boldsymbol{e}_i \rangle \boldsymbol{e}_j \\
&\iff \boldsymbol{v} \in \operatorname{span}(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m).
\end{aligned}$$

□

► EXERCISE 6.14 (6.14). *Find an orthonormal basis of* $\mathfrak{P}_2(\mathbb{R})$ *such that the differentiation operator on* $\mathfrak{P}_2(\mathbb{R})$ *has an upper-triangular matrix with respect to this basis.*

SOLUTION. Consider the orthonormal basis $(1, \sqrt{3}(2x-1), \sqrt{5}(6x^2 - 6x + 1)) = (\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$ in Exercise 6.10. Let T be the differentiation operator on $\mathfrak{P}_2(\mathbb{R})$. We have

$$\mathrm{T}\boldsymbol{e}_1 = 0 \in \operatorname{span}(\boldsymbol{e}_1),$$

$$\mathrm{T}\boldsymbol{e}_2 = \left[\sqrt{3}(2x-1)\right]' = 2\sqrt{3} \in \operatorname{span}(\boldsymbol{e}_1, \boldsymbol{e}_2),$$

and

$$\mathrm{T}\boldsymbol{e}_3 = \left[\sqrt{5}(6x^2 - 6x + 1)\right]' = 12\sqrt{5}x - 6\sqrt{5} \in \operatorname{span}(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3).$$

It follows from Proposition 5.12 that T has an upper-triangular matrix. □

► EXERCISE 6.15 (6.15). *Suppose* U *is a subspace of* V. *Prove that* $\dim U^\perp = \dim V - \dim U$.

PROOF. We have $V = U \oplus U^\perp$; hence,

$$\begin{aligned}\dim V &= \dim U + \dim U^{\perp} - \dim U \cap U^{\perp} \\ &= \dim U + \dim U^{\perp};\end{aligned}$$

that is, $\dim U^{\perp} = \dim V - \dim U$. □

► EXERCISE 6.16 (6.16). *Suppose U is a subspace of V. Prove that $U^{\perp} = \{\mathbf{0}\}$ if and only if $U = V$.*

PROOF. If $U^{\perp} = \{\mathbf{0}\}$, then $V = U \oplus U^{\perp} = U \oplus \{\mathbf{0}\} = U$. To see the converse direction, let $U = V$. For any $\boldsymbol{w} \in U^{\perp}$, we have $\langle \boldsymbol{w}, \boldsymbol{w} \rangle = 0$ since $\boldsymbol{w} \in U^{\perp} \subseteq V = U$; then $\boldsymbol{w} = \mathbf{0}$, that is, $U^{\perp} = \{\mathbf{0}\}$. □

► EXERCISE 6.17 (6.17). *Prove that if $\mathbf{P} \in \mathfrak{L}(V)$ is such that $\mathbf{P}^2 = \mathbf{P}$ and every vector in $\mathcal{N}_{\mathbf{P}}$ is orthogonal to every vector in $\mathcal{R}_{\mathbf{P}}$, then $\mathbf{P}$ is an orthogonal projection.*

PROOF. For every $\boldsymbol{w} \in \mathcal{R}_{\mathbf{P}}$, there exists $\boldsymbol{v}_{\boldsymbol{w}} \in V$ such that $\mathbf{P}\boldsymbol{v}_{\boldsymbol{w}} = \boldsymbol{w}$. Hence,

$$\mathbf{P}\boldsymbol{w} = \mathbf{P}(\mathbf{P}\boldsymbol{v}_{\boldsymbol{w}}) = \mathbf{P}^2\boldsymbol{v}_{\boldsymbol{w}} = \mathbf{P}\boldsymbol{v}_{\boldsymbol{w}} = \boldsymbol{w}.$$

By Exercise 5.22, $V = \mathcal{N}_{\mathbf{P}} \oplus \mathcal{R}_{\mathbf{P}}$ if $\mathbf{P}^2 = \mathbf{P}$. Then any $\boldsymbol{v} \in V$ can be uniquely written as $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$ with $\boldsymbol{u} \in \mathcal{N}_{\mathbf{P}}$ and $\boldsymbol{w} \in \mathcal{R}_{\mathbf{P}}$, and

$$\mathbf{P}\boldsymbol{v} = \mathbf{P}(\boldsymbol{u} + \boldsymbol{w}) = \mathbf{P}\boldsymbol{w} = \boldsymbol{w},$$

Hence, $\mathbf{P} = \mathbf{P}_{\mathcal{R}_{\mathbf{P}}}$ when $\mathcal{N}_{\mathbf{P}} \perp \mathcal{R}_{\mathbf{P}}$. □

► EXERCISE 6.18 (6.18). *Prove that if $\mathbf{P} \in \mathfrak{L}(V)$ is such that $\mathbf{P}^2 = \mathbf{P}$ and $\|\mathbf{P}\boldsymbol{v}\| \leqslant \|\boldsymbol{v}\|$ for every $\boldsymbol{v} \in V$, then $\mathbf{P}$ is an orthogonal projection.*

PROOF. It follows from the previous exercise that if $\mathbf{P}^2 = \mathbf{P}$, then $\mathbf{P}\boldsymbol{v} = \boldsymbol{w}$ for every $\boldsymbol{v} \in V$, where $\boldsymbol{v}$ is uniquely written as $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$ with $\boldsymbol{u} \in \mathcal{N}_{\mathbf{P}}$ and $\boldsymbol{w} \in \mathcal{R}_{\mathbf{P}}$.

It now suffices to show that $\mathcal{N}_{\mathbf{P}} \perp \mathcal{R}_{\mathbf{P}}$. Take an arbitrary $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w} \in V$, where $\boldsymbol{u} \in \mathcal{N}_{\mathbf{P}}$ and $\boldsymbol{w} \in \mathcal{R}_{\mathbf{P}}$. Then $\|\mathbf{P}\boldsymbol{v}\| \leqslant \|\boldsymbol{v}\|$ implies that

$$\langle \mathbf{P}\boldsymbol{v}, \mathbf{P}\boldsymbol{v} \rangle = \langle \boldsymbol{w}, \boldsymbol{w} \rangle \leqslant \langle \boldsymbol{u} + \boldsymbol{w}, \boldsymbol{u} + \boldsymbol{w} \rangle \iff -\|\boldsymbol{u}\|^2 \leqslant 2\mathrm{Re}\left(\langle \boldsymbol{u}, \boldsymbol{w} \rangle\right).$$

The above inequality certainly fails for some $\boldsymbol{v}$ if $\langle \boldsymbol{u}, \boldsymbol{w} \rangle \neq 0$ (see Exercise 6.4). Therefore, $\mathcal{N}_{\mathbf{P}} \perp \mathcal{R}_{\mathbf{P}}$ and $\mathbf{P} = \mathbf{P}_{\mathcal{R}_{\mathbf{P}}}$. □

► EXERCISE 6.19 (6.19). *Suppose $\mathrm{T} \in \mathfrak{L}(V)$ and U is a subspace of V. Prove that U is invariant under T if and only if $\mathbf{P}_U \mathrm{T} \mathbf{P}_U = \mathrm{T}\mathbf{P}_U$.*

PROOF. It follows from Theorem 6.29 that $V = U \oplus U^{\perp}$.

Only if: Suppose that U is invariant under T. For any $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$ with $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in U^{\perp}$, we have

$$(\mathbf{P}_U \mathrm{T} \mathbf{P}_U)(\boldsymbol{v}) = (\mathbf{P}_U \mathrm{T})(\boldsymbol{u}) = \mathbf{P}_U(\mathrm{T}\boldsymbol{u}) = \mathrm{T}\boldsymbol{u},$$

where the last equality holds since $\boldsymbol{u} \in U$ and U is invariant under T. We also have

$$(\mathrm{T}\mathbf{P}_U)(\boldsymbol{v}) = \mathrm{T}\boldsymbol{u}.$$

If: Now suppose that $\mathbf{P}_U \mathrm{T} \mathbf{P}_U = \mathrm{T}\mathbf{P}_U$. Take any $\boldsymbol{u} \in U$ and we have

$$\mathrm{T}\boldsymbol{u} = \mathrm{T}\left(\mathbf{P}_U(\boldsymbol{u})\right) = (\mathrm{T}\mathbf{P}_U)(\boldsymbol{u}) = (\mathbf{P}_U \mathrm{T}\mathbf{P}_U)(\boldsymbol{u}) = \mathbf{P}_U(\mathrm{T}\boldsymbol{u}) \in U$$

by the definition of $\mathbf{P}_U$. This proves that U is invariant under T. □

► EXERCISE 6.20 (6.20). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$ *and* U *is a subspace of* V*. Prove that* U *and* $U^{\perp}$ *are both invariant under* T *if and only if* $\mathbf{P}_U \mathrm{T} = \mathrm{T}\mathbf{P}_U$*.*

PROOF. Suppose first that both U and $U^{\perp}$ are both invariant under T. Then for any $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$, where $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in U^{\perp}$, we have

$$(\mathbf{P}_U \mathrm{T})(\boldsymbol{v}) = (\mathbf{P}_U \mathrm{T})(\boldsymbol{u} + \boldsymbol{w}) = \mathbf{P}_U(\mathrm{T}\boldsymbol{u} + \mathrm{T}\boldsymbol{w}) = \mathrm{T}\boldsymbol{u},$$

and $(\mathrm{T}\mathbf{P}_U)(\boldsymbol{v}) = \mathrm{T}\boldsymbol{u}$.

Now suppose $\mathbf{P}_U \mathrm{T} = \mathrm{T}\mathbf{P}_U$. For any $\boldsymbol{u} \in U$, we have $\mathrm{T}\boldsymbol{u} = (\mathrm{T}\mathbf{P}_U)(\boldsymbol{u}) = (\mathbf{P}_U \mathrm{T})(\boldsymbol{u}) = \mathbf{P}_U(\mathrm{T}\boldsymbol{u}) \in U$. Applying the previous argument to $U^{\perp}$ proves that $U^{\perp}$ is invariant. □

► EXERCISE 6.21 (6.21). *In* $\mathbb{R}^4$*, let* $U = \operatorname{span}((1,1,0,0),(1,1,1,2))$*. Find* $\boldsymbol{u} \in U$ *such that* $\|\boldsymbol{u} - (1,2,3,4)\|$ *is as small as possible.*

SOLUTION. We first need to find the orthonormal basis of U. Using the Gram-Schmidt procedure, we have

$$\boldsymbol{e}_1 = \frac{(1,1,0,0)}{\|(1,1,0,0,)\|} = \left(\sqrt{2}/2, \sqrt{2}/2, 0, 0\right),$$

and

$$\boldsymbol{e}_2 = \frac{(1,1,1,2) - \langle (1,1,1,2), \boldsymbol{e}_1 \rangle \boldsymbol{e}_1}{\left\|(1,1,1,2) - \langle (1,1,1,2), \boldsymbol{e}_1 \rangle \boldsymbol{e}_1\right\|} = \left(0, 0, \sqrt{5}/5, 2\sqrt{5}/5\right).$$

Then by 6.35,

$$\mathbf{P}_U(1,2,3,4) = \langle (1,2,3,4), e_1 \rangle e_1 + \langle (1,2,3,4), e_2 \rangle e_2 = \left(3/2, 3/2, 11/5, 22/5\right).$$

REMARK 6.22. We can use *Maple* to obtain the orthonormal basis easily:

```
>with(LinearAlgebra):
>v1:=<1,1,0,0>:
>v2:=<1,1,1,2>:
>GramSchmidt({v1,v2}, normalized)
```

□

► EXERCISE 6.23 (6.22). *Find* $p \in \mathcal{P}_3(\mathbb{R})$ *such that* $p(0) = 0$*,* $p'(0) = 0$*, and* $\int_0^1 |2 + 3x - p(x)|^2 \, \mathrm{d}x$ *is as small as possible.*

PROOF. $p(0) = p'(0) = 0$ implies that $p(x) = ax^2 + bx^3$, where $a, b \in \mathbb{R}$. We want to find $p \in U \equiv \operatorname{span}(x^2, x^3)$ such that distance from $q = 2_3 x$ to U is as small as possible. With the Gram-Schmidt procedure, the orthonomal basis is

$$e_1 = \frac{x^2}{\|x^2\|} = \frac{x^2}{\sqrt{\int_0^1 \left|x^2 \cdot x^2\right| \mathrm{d}x}} = \sqrt{5}x^2,$$

and

$$e_2 = \frac{x^3 - \left(\int_0^1 \left|\sqrt{5}x^5\right| \mathrm{d}x\right)\sqrt{5}x^2}{\left\|x^3 - \left(\int_0^1 \left|\sqrt{5}x^5\right| \mathrm{d}x\right)\sqrt{5}x^2\right\|} = \frac{x^3 - \frac{5}{6}x^2}{\sqrt{7}/42} = 6\sqrt{7}x^3 - 5\sqrt{7}x^2.$$

Hence,

$$\begin{aligned}\mathbf{P}_U(2+3x) = &\left[\int_0^1 (2+3x)\sqrt{5}x^2 \,\mathrm{d}x\right]\sqrt{5}x^2 \\ &+ \left[\int_0^1 (2+3x)\left(6\sqrt{7}x^3 - 5\sqrt{7}x^2\right) \mathrm{d}x\right]\left(6\sqrt{7}x^3 - 5\sqrt{7}x^2\right). \quad \square\end{aligned}$$

► EXERCISE 6.24 (6.24). *Find a polynomial $q \in \mathfrak{P}_2(\mathbb{R})$ such that*

$$p\left(\frac{1}{2}\right) = \int_0^1 p(x)q(x)\,\mathrm{d}x$$

for every $p \in \mathfrak{P}_2(\mathbb{R})$.

SOLUTION. For every $p \in \mathfrak{P}_2(\mathbb{R})$, we define a function $\mathrm{T}\colon \mathfrak{P}_2(\mathbb{R}) \to \mathbb{R}$ by letting $\mathrm{T}p = p(1/2)$. It is clear that $\mathrm{T} \in \mathfrak{L}(\mathfrak{P}_2(\mathbb{R}), \mathbb{R})$.

It follows from Exercise 6.10 that $(e_1, e_2, e_3) = (1, \sqrt{3}(2x-1), \sqrt{5}(6x^2-6x+1))$ is an orthonormal basis of $\mathfrak{P}_2(\mathbb{R})$. Then, by Theorem,

$$\begin{aligned}\mathrm{T}p &= \mathrm{T}\left(\langle p, e_1\rangle e_1 + \langle p, e_2\rangle e_2 + \langle p, e_3\rangle e_3\right) \\ &= \langle p, \mathrm{T}(e_1)e_1 + \mathrm{T}(e_2)e_2 + \mathrm{T}(e_3)e_3\rangle;\end{aligned}$$

hence,

$$\begin{aligned}q(x) &= e_1(1/2)e_1 + e_2(1/2)e_2 + e_3(1/2)e_3 \\ &= 1 + 0 - \frac{\sqrt{5}}{2}\left[\sqrt{5}(6x^2 - 6x + 1)\right] \\ &= -\frac{3}{2} + 15x - 15x^2. \quad \square\end{aligned}$$

► EXERCISE 6.25 (6.25). *Find a polynomial $q \in \mathfrak{P}_2(\mathbb{R})$ such that*

$$\int_0^1 p(x)(\cos \pi x)\,\mathrm{d}x = \int_0^1 p(x)q(x)\,\mathrm{d}x$$

for every $p \in \mathfrak{P}_2(\mathbb{R})$.

SOLUTION. As in the previous exercise, we let $\mathrm{T}\colon p \mapsto \int_0^1 p(x)(\cos \pi x)\,\mathrm{d}x$ for every $p \in \mathfrak{P}_2(\mathbb{R})$. Then $\mathrm{T} \in \mathfrak{L}(\mathfrak{P}_2(\mathbb{R}), \mathbb{R})$. Let

$$q(x) = \mathrm{T}(e_1)e_1 + \mathrm{T}(e_2)e_2 + \mathrm{T}(e_3)e_3 = 12/\pi^2 - 24x/\pi^2. \qquad \square$$

► EXERCISE 6.26 (6.26). *Fix a vector* $\boldsymbol{v} \in V$ *and define* $\mathrm{T} \in \mathfrak{L}(V, \mathbb{F})$ *by* $\mathrm{T}\boldsymbol{u} = \langle \boldsymbol{u}, \boldsymbol{v}\rangle$. *For* $a \in \mathbb{F}$, *find a formula for* $\mathrm{T}^* a$.

PROOF. Take any $\boldsymbol{u} \in V$. We have $\langle \mathrm{T}\boldsymbol{u}, a\rangle = \big\langle \langle \boldsymbol{u}, \boldsymbol{v}\rangle, a\big\rangle = \langle \boldsymbol{u}, \boldsymbol{v}\rangle a = \langle \boldsymbol{u}, a\boldsymbol{v}\rangle$; thus, $\mathrm{T}^* a = a\boldsymbol{v}$. $\square$

► EXERCISE 6.27 (6.27). *Suppose* n *is a positive integer. Define* $\mathrm{T} \in \mathfrak{L}(\mathbb{F}^n)$ *by* $\mathrm{T}(z_1, \ldots, z_n) = (0, z_1, \ldots, z_{n-1})$. *Find a formula for* $\mathrm{T}^*(z_1, \ldots, z_n)$.

SOLUTION. Take the standard basis of $\mathbb{F}^n$, which is also a orgonormal basis of $\mathbb{F}^n$. We then have

$$\begin{aligned}
\mathrm{T}(1, 0, 0, \ldots, 0) &= (0, 1, 0, 0, \ldots, 0)\,,\\
\mathrm{T}(0, 1, 0, \ldots, 0) &= (0, 0, 1, 0, \ldots, 0)\,,\\
&\cdots\\
\mathrm{T}(0, 0, \ldots, 0, 1) &= (0, 0, 0, 0, \ldots, 0)\,.
\end{aligned}$$

Therefore, $\mathcal{M}(\mathrm{T})$ is given by

$$\mathcal{M}(\mathrm{T}) = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0\\ 1 & 0 & \cdots & 0 & 0\\ 0 & 1 & \cdots & 0 & 0\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix},$$

and so

$$\mathcal{M}(\mathrm{T}^*) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0\\ 0 & 0 & 1 & \cdots & 0\\ \vdots & \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & 0 & \cdots & 1\\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Then $\mathrm{T}^*(z_1, \ldots, z_n) = \mathcal{M}(\mathrm{T}^*)\mathcal{M}(z_1, \ldots, z_n) = (z_2, z_3, \ldots, z_{n-1}, z_n, 0)$. $\square$

► EXERCISE 6.28 (6.28). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$ *and* $\lambda \in \mathbb{F}$. *Prove that* λ *is an eigenvalue of* T *if and only if* $\overline{\lambda}$ *is an eigenvalue of* T^*.

PROOF. If $\lambda \in \mathbb{F}$ is an eigenvalue of T, then there exists $\boldsymbol{v} \neq \mathbf{0}$ such that $\mathrm{T}\boldsymbol{v} = \lambda \boldsymbol{v}$. Take $\boldsymbol{w} \in V$ with $\boldsymbol{w} \neq \mathbf{0}$. Then

$$\langle \mathrm{T}\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \lambda \boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, \bar{\lambda} \boldsymbol{w} \rangle = \langle \boldsymbol{v}, \mathrm{T}^* \boldsymbol{w} \rangle,$$

which implies that $\mathrm{T}^* \boldsymbol{w} = \bar{\lambda} \boldsymbol{w}$; that is, $\bar{\lambda}$ is an eigenvalue of T^*. With the same logic, we can show the inverse direction. □

► EXERCISE 6.29 (6.29). *Suppose* $\mathrm{T} \in \mathcal{L}(V)$ *and* U *is a subspace of* V*. Prove that* U *is invariant under* T *if and only if* $U^\perp$ *is invariant under* T^**.*

PROOF. Take any $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in U^\perp$. If U is invariant under T, then $\mathrm{T}\boldsymbol{u} \in U$ and so

$$0 = \langle \mathrm{T}\boldsymbol{u}, \boldsymbol{w} \rangle = \left\langle \boldsymbol{u}, \mathrm{T}^* \boldsymbol{w} \right\rangle;$$

that is, $\mathrm{T}^* \boldsymbol{w} \in U^\perp$. Applying T^* then we obtain the inverse direction. □

► EXERCISE 6.30 (6.30). *Suppose* $\mathrm{T} \in \mathcal{L}(V, W)$*. Prove that*

a. T *is injective if and only if* T^* *is surjective;*

b. T *is surjective if and only if* T^* *is injective.*

PROOF. (a) If T is injective, then $\dim \mathcal{N}_\mathrm{T} = 0$. Then

$$\dim \mathcal{R}_{\mathrm{T}^*} = \dim \mathcal{R}_\mathrm{T} = \dim V - \dim \mathcal{N}_\mathrm{T} = \dim V,$$

i.e., $\mathrm{T} \in \mathcal{L}(W, V)$ is surjective. If T^* is surjective, then $\dim \mathcal{R}_{\mathrm{T}^*} = \dim V$ and so

$$\dim \mathcal{N}_\mathrm{T} = \dim V - \dim \mathcal{R}_\mathrm{T} = \dim V - \dim \mathcal{R}_{\mathrm{T}^*} = 0,$$

that is, $\mathrm{T} \in \mathcal{L}(V, W)$ is injective.

(b) Using the fact that $(\mathrm{T}^*)^* = \mathrm{T}$ and the result in part (a) we get (b) immediately. □

► EXERCISE 6.31 (6.31). *Prove that* $\dim \mathcal{N}_{\mathrm{T}^*} = \dim \mathcal{N}_\mathrm{T} + \dim W - \dim V$ *and* $\dim \mathcal{R}_{\mathrm{T}^*} = \dim \mathcal{R}_\mathrm{T}$ *for every* $\mathrm{T} \in \mathcal{L}(V, W)$*.*

PROOF. It follows from Proposition 6.46 that $\mathcal{N}_{\mathrm{T}^*} = (\mathcal{R}_\mathrm{T})^\perp$. Since $\mathcal{R}_\mathrm{T}$ is a subspace of W, and $W = \mathcal{R}_\mathrm{T} \oplus (\mathcal{R}_\mathrm{T})^\perp$, we thus have

$$\begin{aligned} \dim V &= \dim \mathcal{N}_\mathrm{T} + \dim \mathcal{R}_\mathrm{T} \\ &= \dim \mathcal{N}_\mathrm{T} + \dim W - \dim \mathcal{R}_\mathrm{T}^\perp \\ &= \dim \mathcal{N}_\mathrm{T} + \dim W - \dim \mathcal{N}_{\mathrm{T}^*}, \end{aligned} \tag{6.4}$$

which proves the first claim. As for the second equality, we first have

$$\begin{aligned} \dim \mathcal{R}_\mathrm{T} &= \dim V - \dim \mathcal{N}_\mathrm{T}, \\ \dim \mathcal{R}_{\mathrm{T}^*} &= \dim W - \dim \mathcal{N}_{\mathrm{T}^*}. \end{aligned}$$

Thus, $\dim \mathcal{R}_{\mathrm{T}} - \dim \mathcal{R}_{\mathrm{T}^*} = 0$ by (6.4), that is, $\dim \mathcal{R}_{\mathrm{T}} = \dim \mathcal{R}_{\mathrm{T}^*}$. □

► EXERCISE 6.32 (6.32). *Suppose* $\mathbf{A}$ *is an* $m \times n$ *matrix of real numbers. Prove that the dimension of the span of the columns of* $\mathbf{A}$ *(in* $\mathbb{R}^m$*) equals the dimension of the span of the rows of* $\mathbf{A}$ *(in* $\mathbb{R}^n$*).*

PROOF. Without loss of generality, we can assume that $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^n, \mathbb{R}^m)$ is the linear map induced by $\mathbf{A}$, where $\mathbf{A}$ corresponds to an orthonormal basis of $\mathbb{R}^n$ and an orthonormal basis of $\mathbb{R}^m$; that is, $\mathrm{T}\boldsymbol{x} = \mathbf{A}\boldsymbol{x}$ for all $\boldsymbol{x} \in \mathbb{R}^n$. By Proposition 6.47, we know that for any $\boldsymbol{y} \in \mathbb{R}^m$,

$$\mathrm{T}^*\boldsymbol{y} = \mathbf{A}'\boldsymbol{y},$$

where $\mathbf{A}'$ is the (conjugate) transpose of $\mathbf{A}$. Let

$$\mathbf{A} = \begin{pmatrix} \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_n \end{pmatrix} = \begin{pmatrix} \boldsymbol{b}_1 \\ \vdots \\ \boldsymbol{b}_m \end{pmatrix}.$$

Then

$$\mathbf{A}' = \begin{pmatrix} \boldsymbol{a}_1' \\ \vdots \\ \boldsymbol{a}_n' \end{pmatrix} = \begin{pmatrix} \boldsymbol{b}_1' & \cdots & \boldsymbol{b}_m' \end{pmatrix}.$$

It is easy to see that

$$\operatorname{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n) = \mathcal{R}_{\mathrm{T}}, \quad \text{and} \quad \operatorname{span}(\boldsymbol{a}_1', \ldots, \boldsymbol{a}_n') = \mathcal{R}_{\mathrm{T}^*}.$$

It follows from Exercise 6.31 that $\dim \mathcal{R}_{\mathrm{T}} = \dim \mathcal{R}_{\mathrm{T}^*}$. □

7

OPERATORS ON INNER-PRODUCT SPACES

"As You Should Verify"

Remark 7.1 (p.131). If T is normal, then $\mathrm{T} - \lambda \mathrm{Id}$ is normal, too.

Proof. Note that $(\mathrm{T} - \lambda \mathrm{Id})^* = \mathrm{T}^* - \overline{\lambda}\mathrm{Id}$. For any $\boldsymbol{v} \in V$,

$$\begin{aligned}(\mathrm{T} - \lambda \mathrm{Id})(\mathrm{T}^* - \overline{\lambda}\mathrm{Id})\boldsymbol{v} &= (\mathrm{T} - \lambda \mathrm{Id})(\mathrm{T}^*\boldsymbol{v} - \overline{\lambda}\boldsymbol{v}) \\ &= \mathrm{T}(\mathrm{T}^*\boldsymbol{v} - \overline{\lambda}\boldsymbol{v}) - \lambda \cdot (\mathrm{T}^*\boldsymbol{v} - \overline{\lambda}\boldsymbol{v}) \\ &= \mathrm{T}\mathrm{T}^*\boldsymbol{v} - \overline{\lambda}\mathrm{T}\boldsymbol{v} - \lambda\mathrm{T}^*\boldsymbol{v} + |\lambda|^2\,\boldsymbol{v},\end{aligned}$$

and

$$\begin{aligned}(\mathrm{T}^* - \overline{\lambda}\mathrm{Id})(\mathrm{T} - \lambda \mathrm{Id})\boldsymbol{v} &= (\mathrm{T}^* - \overline{\lambda}\mathrm{Id})\,(\mathrm{T}\boldsymbol{v} - \lambda\boldsymbol{v}) \\ &= \mathrm{T}^*\,(\mathrm{T}\boldsymbol{v} - \lambda\boldsymbol{v}) - \overline{\lambda} \cdot (\mathrm{T}\boldsymbol{v} - \lambda\boldsymbol{v}) \\ &= \mathrm{T}^*\mathrm{T}\boldsymbol{v} - \lambda\mathrm{T}^*\boldsymbol{v} - \overline{\lambda}\mathrm{T}\boldsymbol{v} + |\lambda|^2\,\boldsymbol{v}.\end{aligned}$$

Hence, $(\mathrm{T} - \lambda \mathrm{Id})(\mathrm{T}^* - \overline{\lambda}\mathrm{Id}) = (\mathrm{T}^* - \overline{\lambda}\mathrm{Id})(\mathrm{T} - \lambda \mathrm{Id})$ since $\mathrm{T}\mathrm{T}^* = \mathrm{T}^*\mathrm{T}$. □

Exercises

► Exercise 7.2 (7.1). *Make $\mathfrak{P}_2(\mathbb{R})$ into an inner-product space by defining $\langle p, q\rangle = \int_0^1 p\,(x)\,q\,(x)\,\mathrm{d}x$. Define $\mathrm{T} \in \mathfrak{L}(\mathfrak{P}_2(\mathbb{R}))$ by $\mathrm{T}\left(a_0 + a_1 x + a_2 x^2\right) = a_1 x$.*

a. *Show that* T *is not self-adjoint.*

b. *The matrix of* T *with respect to the basis $\left(1, x, x^2\right)$ is*

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix equals its conjugate transpose, even though T *is not self-adjoint. Explain why this is not a contradiction.*

PROOF. (a) Suppose T is self-adjoint, that is, $\mathrm{T} = \mathrm{T}^*$. Take any $p, q \in \mathfrak{P}_2(\mathbb{R})$ with $p(x) = a_0 + a_1 x + a_2 x^2$ and $q(x) = b_0 + b_1 x + b_2 x^2$. Then $\langle \mathrm{T}p, q\rangle = \langle p, \mathrm{T}^* q\rangle = \langle p, \mathrm{T}q\rangle$ implies that

$$\int_0^1 (a_1 x)\left(b_0 + b_1 x + b_2 x^2\right) \mathrm{d}x = \int_0^1 \left(a_0 + a_1 x + a_2 x^2\right)(b_1 x)\ \mathrm{d}x,$$

that is,

$$\frac{a_1 b_0}{2} + \frac{a_1 b_1}{3} + \frac{a_1 b_2}{4} = \frac{a_0 b_1}{2} + \frac{a_1 b_1}{3} + \frac{a_2 b_1}{4}. \tag{7.1}$$

Let $a_1 = 0$, then (7.1) becomes $0 = a_0 b_1/2$, which fails to hold for any $a_0 b_1 \neq 0$. Therefore, $\mathrm{T} \neq \mathrm{T}^*$.

(b) $\left(1, x, x^2\right)$ is not an orthonormal basis. See Proposition 6.47. □

► EXERCISE 7.3 (7.2). *Prove or give a counterexample: the product of any two self-adjoint operators on a finite-dimensional inner-product space is self-adjoint.*

PROOF. The claim is incorrect. Let $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V)$ be two self-adjoint operators. Then $(\mathrm{ST})^* = \mathrm{T}^*\mathrm{S}^* = \mathrm{TS}$. It is not necessarily that $\mathrm{ST} = \mathrm{TS}$ since multiplication is not commutable.

For example, let $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(\mathbb{R}^2)$ be defined by the following matrices (with respect to the stand basis of $\mathbb{R}^2$):

$$\mathcal{M}(\mathrm{S}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \mathcal{M}(\mathrm{T}) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then both S and T are self-adjoint, but ST is not since $\mathcal{M}(\mathrm{S})\mathcal{M}(\mathrm{T}) \neq \mathcal{M}(\mathrm{T})\mathcal{M}(\mathrm{S})$. □

► EXERCISE 7.4 (7.3). a. *Show that if V is a real inner-product space, then the set of self-adjoint operators on V is a subspace of $\mathfrak{L}(V)$.*

b. *Show that if V is a complex inner-product space, then the set of self-adjoint operators on V is not a subspace of $\mathfrak{L}(V)$.*

PROOF. (a) Let $\mathfrak{L}^{\mathrm{sa}}(V)$ be the set of self-adjoint operators. Obviously, $0 = 0^*$ since for any $\boldsymbol{v}, \boldsymbol{w}$ we have $0 = \langle 0\boldsymbol{v}, \boldsymbol{w}\rangle = \langle \boldsymbol{v}, 0\boldsymbol{w}\rangle = \langle \boldsymbol{v}, 0^*\boldsymbol{w}\rangle$. To see $\mathfrak{L}^{\mathrm{sa}}(V)$ is closed under addition, let $\mathrm{S}, \mathrm{T} \in \mathfrak{L}^{\mathrm{sa}}(V)$. Then $(\mathrm{S}+\mathrm{T})^* = \mathrm{S}^* + \mathrm{T}^* = \mathrm{S} + \mathrm{T}$ implies that $\mathrm{S} + \mathrm{T} \in \mathfrak{L}^{\mathrm{sa}}(V)$. Finally, for any $a \in \mathbb{F}$ and $\mathrm{T} \in \mathfrak{L}^{\mathrm{sa}}(V)$, we have $(a\mathrm{T})^* = a\mathrm{T}^* = a\mathrm{T} \in \mathfrak{L}^{\mathrm{sa}}(V)$.

(b) If V is a complex inner-product, then $(a\mathrm{T})^* = \overline{a}\mathrm{T}^* = \overline{a}\mathrm{T}$, so $\mathfrak{L}^{\mathrm{sa}}(V)$ is not a subspace of $\mathfrak{L}(V)$. □

► EXERCISE 7.5 (7.4). *Suppose $\mathbf{P} \in \mathfrak{L}(V)$ is such that $\mathbf{P}^2 = \mathbf{P}$. Prove that $\mathbf{P}$ is an orthogonal projection if and only if $\mathbf{P}$ is self-adjoint.*

PROOF. If $\mathbf{P}^2 = \mathbf{P}$, then $V = \mathcal{N}_{\mathbf{P}} \oplus \mathcal{R}_{\mathbf{P}}$ (by Exercise 5.22), and $\mathbf{P}\boldsymbol{w} = \boldsymbol{w}$ for every $\boldsymbol{w} \in \mathcal{R}_{\mathbf{P}}$ (by Exercise 6.17).

Suppose first that $\mathbf{P} = \mathbf{P}^*$. Take arbitrary $\boldsymbol{u} \in \mathcal{N}_{\mathbf{P}}$ and $\boldsymbol{w} \in \mathcal{R}_{\mathbf{P}}$. Then

$$\langle \boldsymbol{u}, \boldsymbol{w} \rangle = \langle \boldsymbol{u}, \mathbf{P}\boldsymbol{w} \rangle = \left\langle \boldsymbol{u}, \mathbf{P}^*\boldsymbol{w} \right\rangle = \langle \mathbf{P}\boldsymbol{u}, \boldsymbol{w} \rangle = \langle \mathbf{0}, \boldsymbol{w} \rangle = 0.$$

Hence, $\mathcal{N}_{\mathbf{P}} \perp \mathcal{R}_{\mathbf{P}}$ and so $\mathbf{P} = \mathbf{P}_{\mathcal{R}_{\mathbf{P}}}$.

Now suppose that $\mathbf{P}$ is an orthogonal projection. Then there exists a subspace U of V such that $V = U \oplus U^{\perp}$ and $\mathbf{P}\boldsymbol{v} = \boldsymbol{u}$ if $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$ with $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in U^{\perp}$. Take arbitrary $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$ with $\boldsymbol{v}_1 = \boldsymbol{u}_1 + \boldsymbol{w}_1$ and $\boldsymbol{v}_2 = \boldsymbol{u}_2 + \boldsymbol{w}_2$. Then $\langle \mathbf{P}\boldsymbol{v}_1, \boldsymbol{v}_2 \rangle = \langle \boldsymbol{u}_1, \boldsymbol{u}_2 + \boldsymbol{w}_2 \rangle = \langle \boldsymbol{u}_1, \boldsymbol{u}_2 \rangle$. Similarly, $\langle \boldsymbol{v}_1, \mathbf{P}\boldsymbol{v}_2 \rangle = \langle \boldsymbol{u}_1 + \boldsymbol{w}_1, \boldsymbol{u}_2 \rangle = \langle \boldsymbol{u}_1, \boldsymbol{u}_2 \rangle$. Thus, $\mathbf{P} = \mathbf{P}^*$. □

► EXERCISE 7.6 (7.5). *Show that if* $\dim V \geqslant 2$*, then the set of normal operators on* V *is not a subspace of* $\mathfrak{L}(V)$*.*

PROOF. Let $\mathfrak{L}^{\mathrm{n}}(V)$ denote the set of normal operators on V and $\dim V \geqslant 2$. Let $\mathrm{S}, \mathrm{T} \in \mathfrak{L}^{\mathrm{n}}(V)$. It is easy to see that

$$\begin{aligned}(\mathrm{S}+\mathrm{T})(\mathrm{S}+\mathrm{T})^* &= (\mathrm{S}+\mathrm{T})\left(\mathrm{S}^*+\mathrm{T}^*\right)\\ &\neq \left(\mathrm{S}^*+\mathrm{T}^*\right)(\mathrm{S}+\mathrm{T})\end{aligned}$$

generally since matrix multiplication is not commutable. □

► EXERCISE 7.7 (7.6). *Prove that if* $\mathrm{T} \in \mathfrak{L}(V)$ *is normal, then* $\mathcal{R}_{\mathrm{T}} = \mathcal{R}_{\mathrm{T}^*}$*.*

PROOF. $\mathrm{T} \in \mathfrak{L}(V)$ is normal if and only if $\|\mathrm{T}\boldsymbol{v}\| = \|\mathrm{T}^*\boldsymbol{v}\|$ for all $\boldsymbol{v} \in V$ (by Proposition 7.6). Then $\boldsymbol{v} \in \mathcal{N}_{\mathrm{T}} \iff \|\mathrm{T}\boldsymbol{v}\| = 0 \iff \|\mathrm{T}^*\boldsymbol{v}\| = 0 \iff \boldsymbol{v} \in \mathcal{N}_{\mathrm{T}^*}$, i.e., $\mathcal{N}_{\mathrm{T}} = \mathcal{N}_{\mathrm{T}^*}$. It follows from Proposition 6.46 that

$$\mathcal{R}_{\mathrm{T}^*} = \mathcal{N}_{\mathrm{T}}^{\perp} = \mathcal{N}_{\mathrm{T}^*}^{\perp} = \mathcal{R}_{\mathrm{T}}.$$ □

► EXERCISE 7.8 (7.7). *Prove that if* $\mathrm{T} \in \mathfrak{L}(V)$ *is normal, then* $\mathcal{N}_{\mathrm{T}^k} = \mathcal{N}_{\mathrm{T}}$ *and* $\mathcal{R}_{\mathrm{T}^k} = \mathcal{R}_{\mathrm{T}}$ *for every positive integer* k*.*

PROOF. It is evident that $\mathcal{N}_{\mathrm{T}} \subseteq \mathcal{N}_{\mathrm{T}^k}$. So we take any $\boldsymbol{v} \in \mathcal{N}_{\mathrm{T}^k}$ with $\boldsymbol{v} \neq \mathbf{0}$ (if $\mathcal{N}_{\mathrm{T}^k} = \{\mathbf{0}\}$, there is nothing to prove). Then

$$\begin{aligned}\left\langle \mathrm{T}^*\mathrm{T}^{k-1}\boldsymbol{v}, \mathrm{T}^*\mathrm{T}^{k-1}\boldsymbol{v} \right\rangle = \left\langle \mathrm{T}\mathrm{T}^*\mathrm{T}^{k-1}\boldsymbol{v}, \mathrm{T}^{k-1}\boldsymbol{v} \right\rangle &= \left\langle \mathrm{T}^*\mathrm{T}\mathrm{T}^{k-1}\boldsymbol{v}, \mathrm{T}^{k-1}\boldsymbol{v} \right\rangle\\ &= \left\langle \mathrm{T}^*\mathrm{T}^{k}\boldsymbol{v}, \mathrm{T}^{k-1}\boldsymbol{v} \right\rangle\\ &= 0,\end{aligned}$$

and so $\left(\mathrm{T}^*\mathrm{T}^{k-1}\right)\boldsymbol{v} = 0$. Now

$$\left\langle \mathrm{T}^{k-1}\boldsymbol{v}, \mathrm{T}^{k-1}\boldsymbol{v} \right\rangle = \left\langle \mathrm{T}^{k-2}\boldsymbol{v}, \mathrm{T}^*\mathrm{T}^{k-1}\boldsymbol{v} \right\rangle = 0$$

implies that $\mathrm{T}^{k-1}\boldsymbol{v} = 0$, that is, $\boldsymbol{v} \in \mathcal{N}_{\mathrm{T}^{k-1}}$. With the same logic, we can show that $\boldsymbol{v} \in \mathcal{N}_{\mathrm{T}^{k-2}}, \ldots, \boldsymbol{v} \in \mathcal{N}_{\mathrm{T}}$. □

► EXERCISE 7.9 (7.8). *Prove that there does not exist a self-adjoint operator* $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^3)$ *such that* $\mathrm{T}(1,2,3) = (0,0,0)$ *and* $\mathrm{T}(2,5,7) = (2,5,7)$.

PROOF. Suppose there exists such a operator $\mathrm{T} \in \mathfrak{L}(\mathbb{R}^3)$. Then

$$\langle \mathrm{T}(1,2,3), (2,5,7) \rangle = \langle (0,0,0), (2,5,7) \rangle = 0,$$

but

$$\langle (1,2,3), \mathrm{T}(2,5,7) \rangle = \langle (1,2,3), (2,5,7) \rangle \neq 0. \qquad \square$$

► EXERCISE 7.10 (7.9). *Prove that a normal operator on a complex inner-product space is self-adjoint if and only if all its eigenvalues are real.*

PROOF. It follows from Proposition 7.1 that every eigenvalue of a self-adjoint operator is real, so the "only if" part is clear.

To see the "if" part, let $\mathrm{T} \in \mathfrak{L}(V)$ be a normal operator, and all its eigenvalues be real. Then by the Complex Spectral Theorem, V has an orthonormal basis consisting of eigenvectors of T. Hence, $\mathcal{M}(\mathrm{T})$ is diagonal with respect this basis, and so the conjugate transpose of $\mathcal{M}(\mathrm{T})$ equals to $\mathcal{M}(\mathrm{T})$ since all eigenvalues are real. $\square$

► EXERCISE 7.11 (7.10). *Suppose V is a complex inner-product space and* $\mathrm{T} \in \mathfrak{L}(V)$ *is a normal operator such that* $\mathrm{T}^9 = \mathrm{T}^8$. *Prove that* T *is self-adjoint and* $\mathrm{T}^2 = \mathrm{T}$.

PROOF. Let $\mathrm{T} \in \mathfrak{L}(V)$ be normal and $\boldsymbol{v} \in V$. Then by Exercise 7.8,

$$\mathrm{T}^8(\mathrm{T}\boldsymbol{v} - \boldsymbol{v}) = 0 \Longrightarrow \mathrm{T}\boldsymbol{v} - \boldsymbol{v} \in \mathcal{N}_{\mathrm{T}^8} = \mathcal{N}_{\mathrm{T}} \Longrightarrow \mathrm{T}(\mathrm{T}\boldsymbol{v} - \boldsymbol{v}) = 0 \Longrightarrow \mathrm{T}^2 = \mathrm{T}.$$

By the Complex Spectral Theorem, there exists an orthonormal basis of V such that $\mathcal{M}(\mathrm{T})$ is diagonal, and the entries on the diagonal line consists of the eigenvalues $(\lambda_1, \ldots, \lambda_n)$ of T. Now $\mathrm{T}^2 = \mathrm{T}$ implies that $\mathcal{M}(\mathrm{T})\mathcal{M}(\mathrm{T}) = \mathcal{M}(\mathrm{T})$; that is,

$$\lambda_i^2 = \lambda_i, \quad i = 1, \ldots, n.$$

Then each $\lambda_i \in \mathbb{R}$. It follows from Exercise 7.10 that T is self-adjoint. $\square$

► EXERCISE 7.12 (7.11). *Suppose V is a complex inner-product space. Prove that every normal operator on V has a square root.*

PROOF. By the Complex Spectral Theorem, there exists an orthonormal basis of V such that $\mathcal{M}(\mathrm{T})$ is diagonal, and the entries on the diagonal line consists of the eigenvalues $(\lambda_1, \ldots, \lambda_n)$ of T. Let $\mathrm{S} \in \mathfrak{L}(V)$ be an operator whose matrix is

$$\mathcal{M}(\mathrm{S}) = \begin{pmatrix} \sqrt{\lambda_1} & 0 & \cdots & 0 \\ 0 & \sqrt{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\lambda_n} \end{pmatrix}.$$

Then $\mathrm{S}^2 = \mathrm{T}$; that is, S is a square root of T. □

► EXERCISE 7.13 (7.12). *Give an example of a real inner-product space V and $\mathrm{T} \in \mathfrak{L}(V)$ and real numbers α, β with $\alpha^2 < 4\beta$ such that $\mathrm{T}^2 + \alpha\mathrm{T} + \beta\mathrm{Id}$ is not invertible.*

PROOF. We use a normal, but not self-adjoint operator on V (See Lemma 7.15). Let

$$\mathcal{M}(\mathrm{T}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\mathcal{M}(\mathrm{T}^2) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

If we let $\alpha = 0$ and $\beta = 1$, then

$$\left(\mathrm{T}^2 + \mathrm{Id}\right)(x, y) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

for all $(x, y) \in \mathbb{R}^2$. Thus, $\mathrm{T}^2 + \mathrm{Id}$ is not injective, and so is not invertible. □

► EXERCISE 7.14 (7.13). *Prove or give a counterexample: every self-adjoint operator on V has a cube root.*

PROOF. By the Spectral Theorem, for any self-adjoint operator on V there is a orthonormal basis $(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n)$ such that

$$\mathcal{M}(\mathrm{T}) = \begin{pmatrix} \lambda_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \lambda_n \end{pmatrix},$$

where there may some i with $\lambda_i = 0$. Then it is clear that there exists a matrix $\mathcal{M}(\mathrm{S})$ with

$$\mathcal{M}(\mathrm{S}) = \begin{pmatrix} \sqrt[3]{\lambda_1} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \sqrt[3]{\lambda_n} \end{pmatrix}$$

such that $[\mathcal{M}(\mathrm{S})]^3 = \mathcal{M}(\mathrm{T})$. Let S be the operator with the matrix $\mathcal{M}(\mathrm{S})$ and so S is the cube root of T. □

► EXERCISE 7.15 (7.14). *Suppose* $\mathrm{T} \in \mathfrak{L}(V)$ *is self-adjoint,* $\lambda \in \mathbb{F}$*, and* $\varepsilon > 0$*. Prove that if there exists* $\boldsymbol{v} \in V$ *such that* $\|\boldsymbol{v}\| = 1$ *and* $\|\mathrm{T}\boldsymbol{v} - \lambda\boldsymbol{v}\| < \varepsilon$*, then* T *has an eigenvalue* λ' *such that* $\left|\lambda - \lambda'\right| < \varepsilon$*.*

PROOF. By the Spectral Theorem, there exists an orthonormal basis $(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n)$ consisting of eigenvectors of T. Write $\boldsymbol{v} = \sum_{i=1}^n a_i \boldsymbol{e}_i$, where $a_i \in \mathbb{F}$. Since $\|\boldsymbol{v}\| = 1$, we have

$$1 = \left\| \sum_{i=1}^n a_i \boldsymbol{e}_i \right\|^2 = |a_1|^2 + \cdots + |a_n|^2 .$$

Suppose that $|\lambda - \lambda_i| \geqslant \varepsilon$ for all eigenvalues $\lambda_i \in \mathbb{F}$. Then

$$\begin{aligned}
\|\mathrm{T}\boldsymbol{v} - \lambda\boldsymbol{v}\|^2 = \left\| \mathrm{T}\left(\sum_{i=1}^n a_i \boldsymbol{e}_i\right) - \lambda \sum_{i=1}^n a_i \boldsymbol{e}_i \right\|^2 &= \left\| \sum_{i=1}^n a_i \lambda_i \boldsymbol{e}_i - \sum_{i=1}^n a_i \lambda \boldsymbol{e}_i \right\|^2 \\
&= \left\| \sum_{i=1}^n a_i \left(\lambda_i - \lambda\right) \boldsymbol{e}_i \right\|^2 \\
&= \sum_{i=1}^n |a_i|^2 \cdot |\lambda_i - \lambda|^2 \\
&\geqslant \sum_{i=1}^n |a_i|^2 \cdot \varepsilon^2 \\
&= \varepsilon^2,
\end{aligned}$$

that is, $\|\mathrm{T}\boldsymbol{v} - \lambda\boldsymbol{v}\| \geqslant \varepsilon$. A contradiction. Thus, there exists some eigenvalue λ' so that $\left|\lambda - \lambda'\right| < \varepsilon$. □

► EXERCISE 7.16 (7.15). *Suppose* U *is a finite-dimensional real vector space and* $\mathrm{T} \in \mathfrak{L}(U)$*. Prove that* U *has a basis consisting of eigenvectors of* T *if and only if there is an inner product on* U *that makes* T *into a self-adjoint operator.*

PROOF. Suppose first that U has a basis consisting of eigenvectors $(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n)$ of T. Let the corresponding eigenvalues be $(\lambda_1, \ldots, \lambda_n)$. Then

$$\mathcal{M}(\mathrm{T}) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Define $\langle \cdot, \cdot \rangle : U \times U \to \mathbb{R}$ by letting

$$\left\langle \boldsymbol{e}_i, \boldsymbol{e}_j \right\rangle = \delta_{ij} .$$

Then, for arbitrary $\boldsymbol{u}, \boldsymbol{w} \in U$,

$$\begin{aligned}\langle \mathrm{T}\boldsymbol{u}, \boldsymbol{w}\rangle = \left\langle \sum_{i=1}^{n} a_i \mathrm{T}\boldsymbol{e}_i, \sum_{i=1}^{n} b_i \boldsymbol{e}_i \right\rangle &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j \left\langle \mathrm{T}\boldsymbol{e}_i, \boldsymbol{e}_j \right\rangle \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i \lambda_i b_j \left\langle \boldsymbol{e}_i, \boldsymbol{e}_j \right\rangle \\ &= \sum_{i=1}^{n} a_i \lambda_i b_i.\end{aligned}$$

Similarly, $\langle \boldsymbol{u}, \mathrm{T}\boldsymbol{w}\rangle = \sum_{i=1}^{n} a_i \lambda_i b_i$. Hence $\mathrm{T} = \mathrm{T}^*$.

The other direction follows from the Real Spectral Theorem directly. □

► EXERCISE 7.17 (7.16). *Give an example of an operator* T *on an inner-product space such that* T *has an invariant subspace whose orthogonal complement is not invariant under* T.

SOLUTION. Let $(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m)$ be an orthonormal basis of U. Extend to an orthonormal basis $(\boldsymbol{e}_1, \dots, \boldsymbol{e}_m, \boldsymbol{f}_1, \dots, \boldsymbol{f}_n)$ of V. Let U is invariant under T, but $U^\perp$ is not invariant under T. Then $\mathcal{M}(\mathrm{T})$ takes the following form

$$\mathcal{M}(\mathrm{T}) = \begin{array}{c} \\ \boldsymbol{e}_1 \\ \vdots \\ \boldsymbol{e}_m \\ \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_n \end{array} \begin{array}{c} \begin{array}{cccccc} \boldsymbol{e}_1 & \cdots & \boldsymbol{e}_m & \boldsymbol{f}_1 & \cdots & \boldsymbol{f}_n \end{array} \\ \begin{pmatrix} & & & & & \\ & \mathbf{A} & & & \mathbf{B} & \\ & & & & & \\ & & & & & \\ & \mathbf{0} & & & \mathbf{C} & \\ & & & & & \end{pmatrix} \end{array}.$$

Since $(\boldsymbol{f}_1, \dots, \boldsymbol{f}_n)$ is a orthonormal basis of $U^\perp$, we know that $U^\perp$ is not invariant if $\mathbf{C} \neq \mathbf{0}$.

For example, let $V = \mathbb{R}^2$, U be the x-axis, and $U^\perp$ be the y-axis. Let $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ be the standard basis of $\mathbb{R}^2$. Let

$$\mathcal{M}(\mathrm{T}) = \begin{array}{c} \\ \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \end{array} \begin{array}{c} \begin{array}{cc} \boldsymbol{e}_1 & \boldsymbol{e}_2 \end{array} \\ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \end{array}$$

Notice that T is not normal:

$$\mathcal{M}(\mathrm{T})\mathcal{M}(\mathrm{T}^*) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \text{but} \quad \mathcal{M}(\mathrm{T}^*)\mathcal{M}(\mathrm{T}) = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$ □

► EXERCISE 7.18 (7.17). *Prove that the sum of any two positive operators on V is positive.*

PROOF. Let $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(V)$ be positive. Then

$$(\mathrm{S} + \mathrm{T})^* = \mathrm{S}^* + \mathrm{T}^* = \mathrm{S} + \mathrm{T};$$

that is, $S + T$ is self-adjoint. Also, for an arbitrary $\boldsymbol{v} \in V$,

$$\langle (S + T)\boldsymbol{v}, \boldsymbol{v}\rangle = \langle S\boldsymbol{v}, \boldsymbol{v}\rangle + \langle T\boldsymbol{v}, \boldsymbol{v}\rangle \geqslant 0.$$

Hence, $S + T$ is positive. □

► EXERCISE 7.19 (7.18). *Prove that if* $T \in \mathfrak{L}(V)$ *is positive, then so is* T^k *for every positive integer* k.

PROOF. It is evident that T^k is self-adjoint. Pick an arbitrary $\boldsymbol{v} \in V$. If $k = 2$, then $\left\langle T^2\boldsymbol{v}, \boldsymbol{v}\right\rangle = \langle T\boldsymbol{v}, T\boldsymbol{v}\rangle = \|T\boldsymbol{v}\| \geqslant 0$. Now suppose that $\left\langle T^\ell\boldsymbol{v}, \boldsymbol{v}\right\rangle \geqslant 0$ for all integer $\ell < k$. Then

$$\left\langle T^k\boldsymbol{v}, \boldsymbol{v}\right\rangle = \left\langle T^{k-1}\boldsymbol{v}, T\boldsymbol{v}\right\rangle = \left\langle T^{k-2}(T\boldsymbol{v}), T\boldsymbol{v}\right\rangle \geqslant 0$$

by the induction hypothesis. □

► EXERCISE 7.20 (7.19). *Suppose that* T *is a positive operator on* V. *Prove that* T *is invertible if and only if* $\langle T\boldsymbol{v}, \boldsymbol{v}\rangle > 0$ *for every* $\boldsymbol{v} \in V \smallsetminus \{\boldsymbol{0}\}$.

PROOF. First assume that $\langle T\boldsymbol{v}, \boldsymbol{v}\rangle > 0$ for every $\boldsymbol{v} \in V \smallsetminus \{\boldsymbol{0}\}$. Then $T\boldsymbol{v} \neq \boldsymbol{0}$; that is, T is injective, which means that T is invertible.

Now suppose that T is invertible. Since T is self-adjoint, there exists an orthonormal basis $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ consisting of eigenvectors of T by the Real Spectral Theorem. Let $(\lambda_1, \ldots, \lambda_n)$ be the corresponding eigenvalues. Since T injective, we know that $T\boldsymbol{v}_i \neq \boldsymbol{0}$ for all $i = 1, \ldots, n$; hence, $\lambda_i \neq 0$ for all $i = 1, \ldots, n$.

For every $\boldsymbol{v} \in V \smallsetminus \{\boldsymbol{0}\}$, there exists a list $(a_1, \ldots, a_n)$, not all zero, such that $\boldsymbol{v} = \sum_{i=1}^n a_i\boldsymbol{v}_i$. Then

$$\langle T\boldsymbol{v}, \boldsymbol{v}\rangle = \left\langle \sum_{i=1}^n a_i T\boldsymbol{v}_i, \sum_{i=1}^n a_i\boldsymbol{v}_i \right\rangle = \left\langle \sum_{i=1}^n a_i\lambda_i\boldsymbol{v}_i, \sum_{i=1}^n a_i\boldsymbol{v}_i \right\rangle = \sum_{i=1}^n \lambda_i |a_i|^2 > 0. \qquad \square$$

► EXERCISE 7.21 (7.20). *Prove or disprove: the identity operator on* $\mathbb{F}^2$ *has infinitely many self-adjoint square roots.*

PROOF. Let

$$\mathcal{M}(S) = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}.$$

Then $S^2 = \mathrm{Id}$. Hence, there are infinitely many self-adjoint square roots. □

Part II

Linear Algebra and Its Application (Lax, 2007)

8

FUNDAMENTALS

► EXERCISE 8.1 (1.1). *Show that the zero of vector addition is unique.*

PROOF. Suppose that $\mathbf{0}$ and $\mathbf{0}'$ are both additive identities for some vector. Then $\mathbf{0}' = \mathbf{0}' + \mathbf{0} = \mathbf{0}$. □

► EXERCISE 8.2 (1.2). *Show that the vector with all components zero serves as the zero element of classical vector addition.*

PROOF. Let $\mathbf{0} = (0, \ldots, 0)$. Then $\boldsymbol{x} + \mathbf{0} = (x_1, \ldots, x_n) + (0, \ldots, 0) = (x_1, \ldots, x_n) = \boldsymbol{x}$. □

EXAMPLE 8.3 (Examples of Linear Spaces).

(i) Set of all row vectors: $(a_1, \ldots, a_n)$, $a_j \in K$; addition, multiplication defined componentwise. This space is denoted as K^n.

(ii) Set of all real-valued functions $f(x)$ defined on the real line, $K = \mathbb{R}$.

(iii) Set of all functions with values in K, defined on an arbitrary set S.

(iv) Set of all polynomials of degree less than n with coefficients in K.

► EXERCISE 8.4 (1.3). *Show that (i) and (iv) are isomorphic.*

PROOF. Let $\mathfrak{P}_{n-1}(K)$ denote the set of all polynomials of degree less than n with coefficients in K, that is,

$$\mathfrak{P}_{n-1}(K) = \left\{ a_1 + a_2 x + \cdots + a_n x^{n-1} \,\middle|\, a_1, \ldots, a_n \in K \right\}.$$

Then, $(a_1, \ldots, a_n) \mapsto a_1 + a_2 x + \cdots + a_n x^{n-1}$ is an isomorphism. □

► EXERCISE 8.5 (1.4). *Show that if S has n elements, (i) and (iii) are isomorphic.*

PROOF. Let $|S| = n$. Then any function $f \in K^S$ can be written as

$$\big(f(s_1), \ldots, f(s_n)\big) = (a_1, \ldots, a_n),$$

where $s_1, \ldots, s_n \in S$. □

► EXERCISE 8.6 (1.5). *Show that when $K = \mathbb{R}$, (iv) is isomorphic with (iii) when S consists of n distinct points of $\mathbb{R}$.*

PROOF. We need to show that $\mathbb{R}^S$ is isomorphic to $\mathfrak{P}_{n-1}(\mathbb{R})$. We can write each $f \in \mathbb{R}^S$ as $(a_1, \ldots, a_n)$, and consider the map $(a_1, \ldots, a_n) \mapsto a_1 + a_2 x + \cdots + a_n x^{n-1}$. □

► EXERCISE 8.7 (1.6). *Prove that $Y + Z$ is a linear subspace of X if Y and Z are.*

PROOF. If $\boldsymbol{y}_1 + \boldsymbol{z}_1, \boldsymbol{y}_2 + \boldsymbol{z}_2 \in Y + Z$, then $(\boldsymbol{y}_1 + \boldsymbol{z}_1) + (\boldsymbol{y}_2 + \boldsymbol{z}_2) = (\boldsymbol{y}_1 + \boldsymbol{y}_2) + (\boldsymbol{z}_1 + \boldsymbol{z}_2) \in Y + Z$; if $\boldsymbol{y} + \boldsymbol{z} \in Y + Z$ and $k \in K$, then $k(\boldsymbol{y} + \boldsymbol{z}) = k\boldsymbol{y} + k\boldsymbol{z} \in Y + Z$. □

► EXERCISE 8.8 (1.7). *Prove that if Y and Z are linear subspaces of X, so is $Y \cap Z$.*

PROOF. If $\boldsymbol{x}, \boldsymbol{y} \in Y \cap Z$, then $\boldsymbol{x} + \boldsymbol{y} \in Y$ and $\boldsymbol{x} + \boldsymbol{y} \in Z$, which imply that $\boldsymbol{x} + \boldsymbol{y} \in Y \cap Z$; if $\boldsymbol{x} \in Y \cap Z$, then $\boldsymbol{x} \in Y$ and $\boldsymbol{x} \in Z$; since both Y and X are subspaces of X, we have $k\boldsymbol{x} \in Y$ and $k\boldsymbol{y} \in Z$ for all $k \in K$, that is $k\boldsymbol{x} \in Y \cap Z$. □

► EXERCISE 8.9 (1.8). *Show that the set $\{\mathbf{0}\}$ consisting of the zero element of a linear space X is a subspace of X. It is called the trivial subspace.*

PROOF. Trial. □

► EXERCISE 8.10 (1.9). *Show that the set of all linear combinations of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j$ is a subspace of X, and that is the smallest subspace of X containing $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j$. This is called the* subspace spanned *by $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j$.*

PROOF. Let $\operatorname{span}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j) \equiv \left\{\boldsymbol{x} : \boldsymbol{x} = \sum_{i=1}^{j} k_i \boldsymbol{x}_i\right\}$. Let $\boldsymbol{x} = \sum_{i=1}^{j} k_i \boldsymbol{x}_i$ and $\boldsymbol{x}' = \sum_{i=1}^{j} k_i' \boldsymbol{x}_i$. Then

$$\boldsymbol{x} + \boldsymbol{x}' = \sum_{i=1}^{j} \left(k_i + k_i'\right) \boldsymbol{x}_i,$$

and

$$k\boldsymbol{x} = \sum_{i=1}^{j} (k k_i) \boldsymbol{x}_i.$$

Hence, the set of all linear combinations of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j$ is a subspace of X.

Since $\boldsymbol{x}_i = 1 \cdot \boldsymbol{x}_i + \sum_{\ell \neq i} 0 \cdot \boldsymbol{x}_\ell$, each $\boldsymbol{x}_i$ is a linear combination of $\left(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j\right)$. Thus, $\operatorname{span}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j)$ contains each $\boldsymbol{x}_i$. Conversely, because subspaces are closed under scalar multiplication and addition, every subspace of V containing each $\boldsymbol{x}_i$ must contain $\operatorname{span}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j)$. □

► EXERCISE 8.11 (1.10). *Show that if the vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j$ are linearly independent, then none of the $\boldsymbol{x}_i$ is the zero vector.*

PROOF. Suppose that there is a vector $\boldsymbol{x}_i = \mathbf{0}$. Then

$$k_i \cdot \mathbf{0} + \sum_{\ell \neq i} 0 \cdot \boldsymbol{x}_\ell = 0, \quad \forall\, k \neq 0,$$

that is, the list $(\boldsymbol{x}_1, \dots, \boldsymbol{x}_j)$ is linearly dependent. □

► EXERCISE 8.12 (1.11). *Prove that if X is finite dimensional and the direct sum of $Y_1, \dots, Y_m$, then* $\dim X = \sum_{j=1}^m \dim Y_j$.

PROOF. Let $\left(\boldsymbol{y}_1^1, \dots, \boldsymbol{y}_{n_1}^1\right)$ be a basis of Y_1, …, $\left(\boldsymbol{y}_1^m, \dots, \boldsymbol{y}_{n_m}^m\right)$ be a basis of Y_m. We show that the list $B = \left(\boldsymbol{y}_1^1, \dots, \boldsymbol{y}_{n_1}^1, \dots, \boldsymbol{y}_1^m, \dots, \boldsymbol{y}_{n_m}^m\right)$ is a basis of $X = Y_1 \oplus \cdots \oplus Y_m$. To see $X = \operatorname{span}(B)$, note that for any $\boldsymbol{x} \in X$, there exists a unique list $(\boldsymbol{y}_1, \dots, \boldsymbol{y}_m)$ with $\boldsymbol{y}_i \in Y_i$ such that $\boldsymbol{x} = \sum_{i=1}^m \boldsymbol{y}_i$. But each $\boldsymbol{y}_i$ can be uniquely represented as $\boldsymbol{y}_i = \sum_{j=1}^{n_i} = a_j^i \boldsymbol{y}_j^i$; thus,

$$\boldsymbol{x} = \sum_{i=1}^m \sum_{j=1}^{n_i} a_j^i \boldsymbol{y}_j^i.$$

To see that the list B is linearly independent, suppose that there exists a list of scalars $\boldsymbol{b} = \left(b_1^1, \dots, b_{n_1}^1, \dots, b_1^m, \dots, b_{m_n}^m\right)$, such that

$$b_1^1 \boldsymbol{y}_1^1 + \cdots + b_{n_1}^1 \boldsymbol{y}_{n_1}^1 + \cdots + b_1^m \boldsymbol{y}_1^m + \cdots + b_{m_n}^m \boldsymbol{y}_{m_n}^m = \mathbf{0}_X.$$

But $\mathbf{0}_X = 0\boldsymbol{y}_1^1 + \cdots + 0\boldsymbol{y}_{n_1}^1 + \cdots + 0\boldsymbol{y}_1^m + \cdots + 0\boldsymbol{y}_{m_n}^m$ and $X = Y_1 \oplus \cdots \oplus Y_m$ implies that all the scalars are zero, that is, B is linearly independent. Therefore, $\dim X = \sum_{j=1}^m \dim Y_j$. □

► EXERCISE 8.13 (1.12). *Show that every finite-dimensional space X over K is isomorphic to K^n, $n = \dim X$. Show that this isomorphism is not unique when n is > 1.*

PROOF. Let $(\boldsymbol{x}_1, \dots, \boldsymbol{x}_n)$ be a basis of X, and $(\boldsymbol{e}_1, \dots, \boldsymbol{e}_n)$ be a basis of K^n. Define a linear map $\mathrm{T} \in \mathfrak{L}(X, K^n)$ by letting $\mathrm{T}\boldsymbol{x}_i = \boldsymbol{e}_i$. Then for any $\boldsymbol{x} = \sum_{i=1}^n a_i \boldsymbol{x}_i \in X$, we have

$$\mathrm{T}\boldsymbol{x} = \mathrm{T}\left(\sum_{i=1}^n a_i \boldsymbol{x}_i\right) = \sum_{i=1}^n a_i \mathrm{T}\boldsymbol{x}_i = \sum_{i=1}^n a_i \boldsymbol{e}_i.$$

We first show that T is surjective. For any $\boldsymbol{k} \in K^n$, there exists $(k_1, \dots, k_n)$ such that $\boldsymbol{k} = \sum_{i=1}^n k_i \boldsymbol{e}_i$, and so there exists $\boldsymbol{x}_{\boldsymbol{k}} = \sum_{i=1}^n k_i \boldsymbol{x}_i \in X$ such that $\mathrm{T}\boldsymbol{x}_{\boldsymbol{k}} = \sum_{i=1}^n k_i \boldsymbol{e}_i = \boldsymbol{k}$. To see T is injective, let

$$\mathrm{T}\left(\sum_{i=1}^n a_i \boldsymbol{x}_i\right) = \mathrm{T}\left(\sum_{i=1}^n b_i \boldsymbol{x}_i\right),$$

that is,

$$\sum_{i=1}^{n} a_i \boldsymbol{e}_i = \sum_{i=1}^{n} b_i \boldsymbol{e}_i \iff \sum_{i=1}^{n} (a_i - b_i)\, \boldsymbol{e}_i = \boldsymbol{0} \iff a_i = b_i \quad \forall\, i = 1, \ldots, n$$

since $(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n)$ is linearly independent. Thus, $\sum_{i=1}^{n} a_i \boldsymbol{x}_i = \sum_{i=1}^{n} b_i \boldsymbol{x}_i$.

The isomorphism is not unique when $n > 1$ since there are many, many basis. □

► EXERCISE 8.14 (1.13). *Congruence mod Y is an equivalence relation. Show further that if $\boldsymbol{x}_1 \equiv \boldsymbol{x}_2$, then $k\boldsymbol{x}_1 \equiv k\boldsymbol{x}_2$ for every scalar k.*

PROOF. (i) If $\boldsymbol{x}_1 \equiv \boldsymbol{x}_2$, then $\boldsymbol{x}_1 - \boldsymbol{x}_2 \in Y$, which means that $\boldsymbol{x}_2 - \boldsymbol{x}_1 = -(\boldsymbol{x}_1 - \boldsymbol{x}_2) \in Y$ since Y is a subspace; (ii) $\boldsymbol{x} - \boldsymbol{x} = \boldsymbol{0} \in Y$; (iii) if $\boldsymbol{x}_1 - \boldsymbol{x}_2 \in Y$ and $\boldsymbol{x}_2 - \boldsymbol{x}_3 \in Y$, then $\boldsymbol{x}_1 - \boldsymbol{x}_3 = (\boldsymbol{x}_1 - \boldsymbol{x}_2) + (\boldsymbol{x}_2 - \boldsymbol{x}_3) \in Y$, that is, $\boldsymbol{x}_1 \equiv \boldsymbol{x}_3$.

If $\boldsymbol{x}_1 \equiv \boldsymbol{x}_2 \mod Y$, then $\boldsymbol{x}_1 - \boldsymbol{x}_2 \in Y$ and so $k(\boldsymbol{x}_1 - \boldsymbol{x}_2) \in Y$ since Y is a subspace of X. But then $k\boldsymbol{x}_1 - k\boldsymbol{x}_2 \in Y$, i.e., $k\boldsymbol{x}_1 \equiv k\boldsymbol{x}_2 \mod Y$. □

► EXERCISE 8.15 (1.14). *Show that two congruence classes are either identical or disjoint.*

PROOF. Let $\boldsymbol{x}_3 \in [\boldsymbol{x}_1] \cap [\boldsymbol{x}_2]$. Then $\boldsymbol{x}_1 \equiv \boldsymbol{x}_3$ and $\boldsymbol{x}_3 \equiv \boldsymbol{x}_2$. By transitivity of $\equiv$ we have $\boldsymbol{x}_1 \equiv \boldsymbol{x}_2$, that is, $[\boldsymbol{x}_1] = [\boldsymbol{x}_2]$. □

► EXERCISE 8.16 (1.15). *Show that the above definition of addition and multiplication by scalars is independent of the choice of representatives in the congruence class.*[1]

PROOF. By definition, $[\boldsymbol{x}] + [\boldsymbol{z}] = [\boldsymbol{x} + \boldsymbol{z}] = (\boldsymbol{x} + \boldsymbol{z}) + Y$, and $k[\boldsymbol{x}] = [k\boldsymbol{x}] = k\boldsymbol{x} + Y$. Note that $[\boldsymbol{x}'] = [\boldsymbol{x}]$ if $\boldsymbol{x}' \in [\boldsymbol{x}]$. □

► EXERCISE 8.17 (1.16). *Denote by X the linear space of all polynomials $p(t)$ of degree $< n$, and denote by Y the set of polynomials that are zero at $t_1, \ldots, t_j$, $j < n$.*

a. *Show that Y is a subspace of X.*

b. *Determine* $\dim Y$.

c. *Determine* $\dim X/Y$.

PROOF.

a. Any $p \in \mathfrak{P}_{n-1}(K)$ with roots $t_1, \ldots, t_j$ can be written in the form

$$q(t) \prod_{i=1}^{j} (t - t_j),$$

where $q(t) \in \mathfrak{P}_{n-1-j}(K)$. These clearly form a vector space.

[1] We have $[\boldsymbol{x}] = \boldsymbol{x} + Y$. **Proof:** If $\boldsymbol{z} \in [\boldsymbol{x}]$, then there exists $\boldsymbol{y} \in Y$ such that $\boldsymbol{z} - \boldsymbol{x} = \boldsymbol{y}$; then $\boldsymbol{z} = \boldsymbol{x} + \boldsymbol{y} \in \boldsymbol{x} + Y$. Conversely, if $\boldsymbol{z} \in \boldsymbol{x} + Y$, then $\boldsymbol{z} = \boldsymbol{x} + \boldsymbol{y}$ for some $\boldsymbol{y} \in Y$; hence, $\boldsymbol{z} - \boldsymbol{x} = \boldsymbol{y} \in Y$, i.e., $\boldsymbol{z} \in [\boldsymbol{x}]$.

b. $\dim Y = n - j$.

c. $\dim X/Y = \dim X - \dim Y = n - (n - j) = j$.

□

► EXERCISE 8.18 (1.17). *A subspace Y of a finite-dimensional linear space X whose dimension is the same as the dimension of X is all of X.*

PROOF. Suppose that $Y \subsetneq X$, then there exists $\boldsymbol{x} \in X \smallsetminus Y$ and $\boldsymbol{x} \neq \mathbf{0}_X$ since $\mathbf{0}_X \in Y$. Let $[\boldsymbol{x}] = \boldsymbol{x} + Y$. Thus, $[\boldsymbol{x}] \in X/Y$ and $[\boldsymbol{x}] \neq Y = \mathbf{0}_{X/Y}$ and so $\dim X/Y \geqslant 1$, which implies that $\dim Y = \dim X - \dim X/Y < \dim X$ by Theorem 1.6. A contradiction. □

► EXERCISE 8.19 (1.18). *Show that* $\dim X_1 \oplus X_2 = \dim X_1 + \dim X_2$.

PROOF. $X_1 \oplus X_2$ implies that $X_1 \cap X_2 = \{\mathbf{0}\}$, that is, $\dim X_1 \cap X_2 = 0$. Therefore, $\dim X_1 \oplus X_2 = \dim X_1 + \dim X_2$. See Exercise 8.12. □

► EXERCISE 8.20 (1.19). *X is a linear space, Y a subspace. Show that $Y \oplus X/Y$ is isomorphic to X.*

PROOF. According to Exercise 8.13, we only need to show that $\dim Y \oplus X/Y = \dim X$. This holds since

$$\dim Y \oplus X/Y = \dim Y + \dim X/Y = \dim X.$$

□

► EXERCISE 8.21 (1.20). *Which of the following sets of vectors $\boldsymbol{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ are a subspace of $\mathbb{R}^n$? Explain your answer.*

a. *All $\boldsymbol{x}$ such that $x_1 \geqslant 0$.*

b. *All $\boldsymbol{x}$ such that $x_1 + x_2 = 0$.*

c. *All $\boldsymbol{x}$ such that $x_1 + x_2 + 1 = 0$.*

d. *All $\boldsymbol{x}$ such that $x_1 = 0$.*

e. *All $\boldsymbol{x}$ such that x_1 is an integer.*

PROOF.

a. No. $-\boldsymbol{x}$ is not in that set if $x_1 > 0$.

b. Yes.

c. No. $k\boldsymbol{x}$ is not in that set if $k \neq 1$.

d. Yes.

e. No. $k\boldsymbol{x}$ is not in that set if $k \notin \mathbb{Z}$.

□

► EXERCISE 8.22 (1.21). *Let U, V, and W be subspaces of some finite-dimensional vector space X. Is the statement*

$$\begin{aligned}\dim U + V + W = &\dim U + \dim V + \dim W - \dim U \cap V \\ &- \dim U \cap W - \dim V \cap W + \dim U \cap V \cap W,\end{aligned}$$

true or false? If true, prove it. If false, provide a counterexample.

PROOF. It is false. See Exercise 2.17. □

9

DUALITY

REMARK 9.1 (Theorem 3). The bilinear function $\langle \boldsymbol{x}, \ell \rangle$ gives a natural identification of X with X''.

PROOF. For $\langle \boldsymbol{x}, \ell \rangle$, fix $\boldsymbol{x} = \boldsymbol{x}_0$, then we observe that the function of the vectors in X', whose value at ℓ is $\langle \boldsymbol{x}_0, \ell \rangle = \ell(\boldsymbol{x}_0)$, is a scalar-valued function that happens to be linear [**Proof:** Let $z_0 \in X''$ be so defined. For any $\ell, \ell' \in X'$, we have $z_0(\ell + \ell') = \langle \boldsymbol{x}_0, \ell + \ell' \rangle = (\ell + \ell')(\boldsymbol{x}_0) = \ell(\boldsymbol{x}_0) + \ell'(\boldsymbol{x}_0) = z_0(\ell) + z_0(\ell')$. For any $k \in K$ and $\ell \in X'$, we have $z_0(k\ell) = \langle \boldsymbol{x}_0, k\ell \rangle = (k\ell)(\boldsymbol{x}_0) = k\ell(\boldsymbol{x}_0) = k \cdot z_0(\ell)$.] Thus,

> $\langle \boldsymbol{x}_0, \ell \rangle$ defines a linear functional on X', and consequently, and element of X''.

By this method we have exhibited *some* linear functionals on X'; have we exhibited them all? For the finite-dimensional case the following theorem furnishes the affirmative answer.

> If X is a finite-dimensional vector space, then corresponding to every linear functional z_0 on X' there is a vector $\boldsymbol{x}_0 \in X$ such that $z_0(\ell) = \langle \boldsymbol{x}_0, \ell \rangle = \ell(\boldsymbol{x}_0)$ for every $\ell \in X'$; the correspondence $z_0 \leftrightarrow \boldsymbol{x}_0$ between X'' and X is an isomorphism.

Proof: To every $\boldsymbol{x}_0 \in X$, we make correspond a vector $z_{\boldsymbol{x}_0} \in X''$ defined by $z_{\boldsymbol{x}_0}(\ell) = \ell(\boldsymbol{x}_0)$ for every $\ell \in X'$. We first show that the transformation $\boldsymbol{x}_0 \mapsto z_{\boldsymbol{x}_0}$ is linear. For any $\boldsymbol{x}_0, \boldsymbol{x}_1 \in X$, we have $\boldsymbol{x}_0 + \boldsymbol{x}_1 \mapsto z_{\boldsymbol{x}_0 + \boldsymbol{x}_1}$; by definition, $z_{\boldsymbol{x}_0 + \boldsymbol{x}_1}(\ell) = \ell(\boldsymbol{x}_0 + \boldsymbol{x}_1) = \ell(\boldsymbol{x}_0) + \ell(\boldsymbol{x}_1) = z\boldsymbol{x}_0(\ell) + z_{\boldsymbol{x}_1}(\ell)$ for any $\ell \in X'$. For any $k \in K$ and $\boldsymbol{x}_0 \in X$, we have $k\boldsymbol{x}_0 \mapsto z_{k\boldsymbol{x}_0}$ and so $z_{k\boldsymbol{x}_0}(\ell) = \ell(k\boldsymbol{x}_0) = k \cdot \ell(\boldsymbol{x}_0) = k \cdot z_{\boldsymbol{x}_0}$ for any $\ell \in X'$.

We shall show that this transformation is injective. Take any $z_{\boldsymbol{x}_1}, z_{\boldsymbol{x}_2} \in X''$ with $z_{\boldsymbol{x}_1} = z_{\boldsymbol{x}_2}$. To say that $z_{\boldsymbol{x}_1} = z_{\boldsymbol{x}_2}$ means that $\langle \boldsymbol{x}_1, \ell \rangle = \langle \boldsymbol{x}_2, \ell \rangle$ for every $\ell \in X'$. But then $\boldsymbol{x}_1 = \boldsymbol{x}_2$ by Exercise 9.3 (iii).

Therefore, the set $Z \equiv \{z_{\boldsymbol{x}} : \boldsymbol{x} \in X\}$ is a subspace of X'' since Z is the range under a linear map, and Z is isomorphic to X'', and so $\dim Z = \dim X$. Since $\dim X = \dim X' = \dim X''$, we have $\dim Z = \dim X''$. It follows that $X'' = Z$ by Exercise 8.18. □

REMARK 9.2 (p. 16). $Y^{\perp}$ is isomorphic to $(X/Y)'$.

PROOF. Given $\ell \in Y^{\perp}$, we make correspond a linear functional $L_\ell \in (X/Y)'$ defined by

$$L_\ell[\boldsymbol{x}] = \ell(\boldsymbol{x}).$$

We show first that $L_\ell[\boldsymbol{x}] = \ell(\boldsymbol{x})$ is well defined.[1] Let $[\boldsymbol{x}_1] = [\boldsymbol{x}]$; then there exists $\boldsymbol{y}_1 \in Y$ such that $\boldsymbol{x}_1 = \boldsymbol{x} + \boldsymbol{y}_1$, as depicted in Figure 9.1. Thus,

$$\ell(\boldsymbol{x}_1) = \ell(\boldsymbol{x} + \boldsymbol{y}_1) = \ell(\boldsymbol{x}) + \ell(\boldsymbol{y}_1) = \ell(\boldsymbol{y}),$$

that is, $L_\ell[\boldsymbol{x}_1] = L_\ell[\boldsymbol{x}]$ if $[\boldsymbol{x}_1] = [\boldsymbol{x}]$. We then show L_ℓ such defined is linear. For any $[\boldsymbol{x}], [\boldsymbol{y}] \in X/Y$, we have $L_\ell([\boldsymbol{x}] + [\boldsymbol{y}]) \overset{\langle 1 \rangle}{=} L_\ell[\boldsymbol{x} + \boldsymbol{y}] = \ell(\boldsymbol{x} + \boldsymbol{y}) = \ell(\boldsymbol{x}) + \ell(\boldsymbol{y}) = L_\ell[\boldsymbol{x}] + L_\ell[\boldsymbol{y}]$, where $\langle 1 \rangle$ holds since $[\boldsymbol{x}] + [\boldsymbol{y}] = (\boldsymbol{x} + Y) + (\boldsymbol{y} + Y) = (\boldsymbol{x} + \boldsymbol{y}) + Y = [\boldsymbol{x} + \boldsymbol{y}]$. To see L_ℓ is homogenous, take any $k \in K$ and $[\boldsymbol{x}] \in X/Y$. Then $L_\ell(k \cdot [\boldsymbol{x}]) = L_\ell[k\boldsymbol{x}] = \ell(k\boldsymbol{x}) = k\ell(\boldsymbol{x}) = k \cdot L_\ell[\boldsymbol{x}]$.

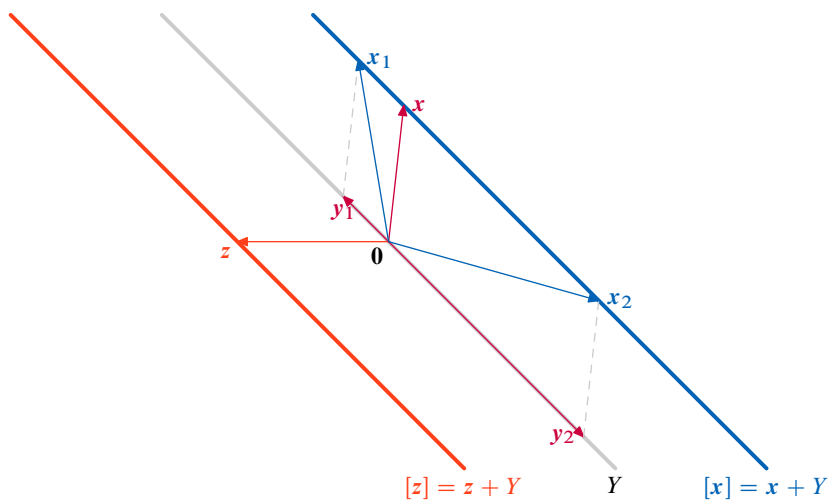

FIGURE 9.1. $Y^{\perp} \cong (X/Y)'$

Conversely, given any $L \in (X/Y)'$, define a linear functional ℓ_L on X as

$$\ell_L(\boldsymbol{x}) = L[\boldsymbol{x}].$$

It follows from the above definition that $\ell_L \in Y^{\perp}$: for any $\boldsymbol{y} \in Y$, we have $[\boldsymbol{y}] = \boldsymbol{y} + Y = Y$ and so $\ell_L(\boldsymbol{y}) = L[Y] = 0$. This also proves that the correspondence between $Y^{\perp}$ and $(X/Y)'$ is surjective. We finally to show that the mapping $\ell \mapsto L_\ell$ is injective. Take two $\ell_1, \ell_2 \in Y^{\perp}$ such that $L_{\ell_1} = L_{\ell_2}$, where $L_{\ell_1}, L_{\ell_2} \in (X/Y)'$. To say $L_{\ell_1} = L_{\ell_2}$ means that $L_{\ell_1}[\boldsymbol{x}] = L_{\ell_2}[\boldsymbol{x}]$ for all $[\boldsymbol{x}] \in X/Y$, but which means that $\ell_1(\boldsymbol{x}) = \ell_2(\boldsymbol{x})$ for all $\boldsymbol{x} \in X$, i.e., $\ell_1 = \ell_2$. Thus, $\ell \mapsto L_\ell$ is injective. □

[1] To see the relation between congruence classes and affine sets (linear manifolds), refer Rockafellar (1970).

► EXERCISE 9.3 (2.1). *Given a nonzero vector $\boldsymbol{x}_1 \in X$, show that there is a linear function ℓ such that $\ell(\boldsymbol{x}_1) \neq 0$.*

PROOF. See Halmos (1974, Sec. 15).

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i) If X is an n-dimensional vector space, if $(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ is a basis of X, and if $(a_1, \ldots, a_n)$ is any list of n scalars, then there is one and only one linear functional ℓ on X such that $\langle \boldsymbol{x}_i, \ell \rangle = a_i$ for $i = 1, \ldots, n$.

Proof: Every $\boldsymbol{x} \in X$ can be represented uniquely as $\boldsymbol{x} = \sum_{i=1}^n k_i \boldsymbol{x}_i$, where $k_i \in K$. If ℓ is any linear functional, then

$$\langle \boldsymbol{x}, \ell \rangle = \left\langle \sum_{i=1}^n k_i \boldsymbol{x}_i, \ell \right\rangle = k_1 \langle \boldsymbol{x}_1, \ell \rangle + \cdots + k_n \langle \boldsymbol{x}_i, \ell \rangle .$$

From this relation the uniqueness of ℓ is clear: if $\langle \boldsymbol{x}_i, \ell \rangle = a_i$, then the value of $\langle \boldsymbol{x}, \ell \rangle$ is determined, for every $\boldsymbol{x}$, by $\langle \boldsymbol{x}, \ell \rangle = \sum_{i=1}^n k_i a_i$. The argument can also be turned around; if we define ℓ by

$$\langle \boldsymbol{x}, \ell \rangle = k_1 a_1 + \cdots k_n a_n,$$

then ℓ is indeed a linear functional, and $\langle \boldsymbol{x}_i, \ell \rangle = a_i$.

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ii) If X is an n-dimensional vector space and if $B = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ is a basis of X, then there is a uniquely determined basis B' in X', $B' = (\ell_1, \ldots, \ell_n)$, with the property that $\langle \boldsymbol{x}_i, \ell_j \rangle = \delta_{ij}$. Consequently the dual space of an n-dimensional space is n-dimensional.

Proof: It follows from (i) that, for each $j = 1, \ldots, n$, a unique $\ell_i \in X'$ can be found so that $\langle \boldsymbol{x}_i, \ell_j \rangle = \delta_{ij}$; we have only to prove that the list $B' = (\ell_1, \ldots, \ell_n)$ is a basis in X'. In the first place, B' is linearly independent, for if we had $a_1 \ell_1 + \cdots + a_n \ell_n = 0$, in other words, if

$$\langle \boldsymbol{x}, a_1 \ell_1 + \cdots + a_n \ell_n \rangle = a_1 \langle \boldsymbol{x}, \ell_1 \rangle + \cdots + a_n \langle \boldsymbol{x}, \ell_n \rangle = 0$$

for all $\boldsymbol{x} \in X$, then we should have, for $\boldsymbol{x} = \boldsymbol{x}_i$,

$$0 = \sum_{j=1}^n a_j \langle \boldsymbol{x}_i, \ell_j \rangle = \sum_{j=1}^n a_j \delta_{ij} = a_i .$$

In the second place, $X' = \operatorname{span}(\ell_1, \ldots, \ell_n)$. To prove this, write $\langle \boldsymbol{x}_i, \ell \rangle = a_i$; then, for $\boldsymbol{x} = \sum_{i=1}^n k_i \boldsymbol{x}_i$, we have

$$\langle \boldsymbol{x}, \ell \rangle = \left\langle \sum_{i=1}^{n} k_i \boldsymbol{x}_i, \ell \right\rangle = \sum_{i=1}^{n} k_i \langle \boldsymbol{x}_i, \ell \rangle = \sum_{i=1}^{n} k_i a_i.$$

On the other hand,

$$\left\langle \boldsymbol{x}, \ell_j \right\rangle = \sum_{i=1}^{n} k_i \left\langle \boldsymbol{x}_i, \ell_j \right\rangle = k_j,$$

so that, substituting in the preceding equation, we get

$$\langle \boldsymbol{x}, \ell \rangle = \sum_{i=1}^{n} k_i a_i = \sum_{i=1}^{n} a_i \cdot \langle \boldsymbol{x}, \ell_i \rangle = \left\langle \boldsymbol{x}, \sum_{i=1}^{n} a_i \ell_i \right\rangle.$$

Consequently $\ell = \sum_{i=1}^{n} a_i \ell_i$, and the proof of (ii) is complete.

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iii) For any non-zero vector $\boldsymbol{x} \in X$ there corresponds a $\ell \in X'$ such that $\langle \boldsymbol{x}, \ell \rangle \neq 0$.

Proof: Let $(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ be a basis of X, and let $(\ell_1, \ldots, \ell_n)$ be the dual basis in X'. If $\boldsymbol{x} = \sum_{i=1}^{n} k_i \boldsymbol{x}_i$, then $\left\langle \boldsymbol{x}, \ell_j \right\rangle = k_j$. Hence if $\langle \boldsymbol{x}, \ell \rangle = 0$ for all ℓ, in particular, if $\left\langle \boldsymbol{x}, \ell_j \right\rangle = 0$ for $j = 1, \ldots, n$, then $k_j = 0$ and so $\boldsymbol{x} = \mathbf{0}_X$. □

► EXERCISE 9.4 (2.2). *Verify that $Y^{\perp}$ is a subspace of X'.*

PROOF. (i) Obviously that $0 \in Y^{\perp}$ since $\langle \boldsymbol{x}, 0 \rangle = 0$ for any $\boldsymbol{x} \in X$, including $\boldsymbol{y} \in Y \subseteq X$. (ii) Let $\ell, m \in Y^{\perp}$. Then $\langle \boldsymbol{y}, \ell \rangle = 0 = \langle \boldsymbol{y}, m \rangle$ for all $\boldsymbol{y} \in Y$ and so $\langle \boldsymbol{y}, \ell + m \rangle = \langle \boldsymbol{y}, \ell \rangle + \langle \boldsymbol{y}, m \rangle = 0$, i.e., $\ell + m \in Y^{\perp}$. (iii) If $\ell \in Y^{\perp}$, then $k \langle \boldsymbol{y}, \ell \rangle = 0$ for any $\boldsymbol{y} \in Y$, and so $k\ell \in Y^{\perp}$. Thus $Y^{\perp}$ is a subspace of X'. □

► EXERCISE 9.5 (2.3). *Denote by Y the smallest subspace containing S. Then $S^{\perp} = Y^{\perp}$.*

PROOF. It is clear that $Y^{\perp} \subseteq S^{\perp}$. If $S = \varnothing$, then $Y = \{\mathbf{0}\}$ and the conclusion is obvious. Similarly, the proof is trivial if $S = \{\mathbf{0}\}$. So we suppose that $S \neq \varnothing$ and $S \neq \{\mathbf{0}\}$. Take any $\boldsymbol{y}_1 \in S$ with $\boldsymbol{y}_1 \neq \mathbf{0}$. If $S \subseteq \operatorname{span}(\boldsymbol{y}_1)$, let $Y = \operatorname{span}(\boldsymbol{y}_1)$; if there is $\boldsymbol{y}_2 \in S \setminus \operatorname{span}(\boldsymbol{y}_1)$, let $Y = \operatorname{span}(\boldsymbol{y}_1, \boldsymbol{y}_2)$; .... Since the embedding vector space is finite-dimensional, the process will be ended with a list $(\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n)$ with $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n \in S$, and this list is a basis of Y. Then for any $\ell \in S^{\perp}$ and any $\boldsymbol{y} \in Y$, we have

$$\langle \boldsymbol{y}, \ell \rangle = \left\langle \sum_{i=1}^{n} k_i \boldsymbol{y}_i, \ell \right\rangle = \sum_{i=1}^{n} k_i \langle \boldsymbol{y}_i, \ell \rangle = 0$$

since $\ell(\boldsymbol{y}_i) = 0$. Thus, $\ell \in Y^{\perp}$. □

► EXERCISE 9.6 (2.4). *In Theorem 7 take the interval I to be $[-1, 1]$, and take $n = 3$. Choose the three points to be $t_1 = -a$, $t_2 = 0$, and $t_3 = a$.*

a. *Determine the weights* m_1, m_2, m_3 *so that* $\int_I p(t)\,\mathrm{d}t = m_1 p(t_1) + m_2 p(t_2) + m_3 p(t_3)$ *holds for all polynomials* $p \in \mathfrak{P}_2(K)$.

b. *Show that for* $a > \sqrt{1/3}$, *all three weights are positive.*

c. *Show that for* $a = \sqrt{3/5}$, *(9) holds for all* $p \in \mathfrak{P}_5(K)$.

PROOF.

a. If $p(t) = t$, then $\int_{-1}^{1} t\,\mathrm{d}t = 0$ and so $0 = m_1(-a) + m_3 a$, i.e., $m_1 = m_3$. Then (9) can be rewritten as

$$\int_{-1}^{1} p(t)\,\mathrm{d}t = m_1\left[p(-a) + p(a)\right] + m_2 p(0). \tag{9.1}$$

Take $p(t) = 1$ now. Then $2 = \int_{-1}^{1} \mathrm{d}t = 2m_1 + m_2$, i.e., $m_2 = 2(1 - m_1)$. So we rewrite (9.1) as

$$\int_{-1}^{1} p(t)\,\mathrm{d}t = m_1\left[p(-a) + p(a)\right] + 2(1 - m_1)\,p(0). \tag{9.2}$$

Now let $p(t) = t^2$ and hence $p(0) = 0$. We then have $\frac{2}{3} = \int_{-1}^{1} t^2\,\mathrm{d}t = m_1 2a^2$ implies that

$$m_1 = m_3 = \frac{1}{3a^2}, \quad \text{and} \quad m_2 = 2 - \frac{2}{3a^2}.$$

□

10

LINEAR MAPPINGS

► EXERCISE 10.1 (3.1). *The image of a subspace of X under a linear map* T *is a subspace of U. The inverse image of a subspace of U, that is the set of all vectors in X mapped by* T *into the subspace, is a subspace of X.*

PROOF. Let Y be a subspace of X; then $\mathbf{0}_X \in Y$ and so $\mathbf{0}_U = \mathrm{T}\mathbf{0}_X \in \mathrm{T}[Y]$. To see $\mathrm{T}[Y]$ is closed under addition, take any $\mathrm{T}\boldsymbol{x}, \mathrm{T}\boldsymbol{y} \in \mathrm{T}[Y]$. Then $\boldsymbol{x} + \boldsymbol{y} \in Y$ and $\mathrm{T}\boldsymbol{x} + \mathrm{T}\boldsymbol{y} = \mathrm{T}(\boldsymbol{x} + \boldsymbol{y}) \in \mathrm{T}[Y]$; to see $\mathrm{T}[Y]$ is closed under scalar multiplication, take any $k \in K$ and $\mathrm{T}\boldsymbol{x} \in \mathrm{T}[Y]$; then $k \ln \in Y$ and $k\mathrm{T}\boldsymbol{x} = \mathrm{T}(k\boldsymbol{x}) \in \mathrm{T}[Y]$. Thus $\mathrm{T}[Y]$ is a subspace of U.

We then show that $\mathrm{T}^{-1}[V]$ is a subspace of X is V is a subspace of U. (i) $\mathbf{0}_X \in \mathrm{T}^{-1}[V]$ since $\mathbf{0}_U \in V$ and $\mathrm{T}\mathbf{0}_X = \mathbf{0}_U$. (ii) For any $\boldsymbol{x}, \boldsymbol{y} \in \mathrm{T}^{-1}[V]$, we have $\mathrm{T}\boldsymbol{x}, \mathrm{T}\boldsymbol{y} \in V$ and so $\mathrm{T}(\boldsymbol{x} + \boldsymbol{y}) = \mathrm{T}\boldsymbol{x} + \mathrm{T}\boldsymbol{y} \in V$, i.e., $\boldsymbol{x} + \boldsymbol{y} \in \mathrm{T}^{-1}[V]$. (iii) For any $k \in K$ and $\boldsymbol{x} \in \mathrm{T}^{-1}[V]$, we have $\mathrm{T}\boldsymbol{x} \in V$ and $k\mathrm{T}\boldsymbol{x} \in V$; then $\mathrm{T}(k\boldsymbol{x})\, k\mathrm{T}\boldsymbol{x} \in V$, that is, $k\boldsymbol{x} \in k\mathrm{T}\boldsymbol{x}$. □

► EXERCISE 10.2 (3.3).

a. *The composite of linear mappings is also a linear mapping.*

b. *Composition is distributive with respect to the addition of linear maps, that is,* $(\mathrm{R} + \mathrm{S}) \circ \mathrm{T} = \mathrm{R} \circ \mathrm{T} + \mathrm{S} \circ \mathrm{T}$ *and* $\mathrm{S} \circ (\mathrm{T} + \mathrm{P}) = \mathrm{S} \circ \mathrm{T} + \mathrm{S} \circ \mathrm{P}$, *where* R *and* S *map* $U \to V$ *and* P *and* T *map* $X \to U$.

PROOF.

a. Let $\mathrm{S}, \mathrm{T} \in \mathfrak{L}(X, U)$ and consider $\mathrm{S} \circ \mathrm{T}$. To see $\mathrm{S} \circ \mathrm{T}$ is additive, take any $\boldsymbol{x}, \boldsymbol{y} \in X$; then $(\mathrm{S} \circ \mathrm{T})(\boldsymbol{x} + \boldsymbol{y}) = \mathrm{S}\big[\mathrm{T}(\boldsymbol{x} + \boldsymbol{y})\big] = \mathrm{S}[\mathrm{T}\boldsymbol{x} + \mathrm{T}\boldsymbol{y}] = (\mathrm{ST})\boldsymbol{x} + (\mathrm{ST})\boldsymbol{y} = (\mathrm{S} \circ \mathrm{T})\boldsymbol{x} + (\mathrm{S} \circ \mathrm{T})\boldsymbol{y}$. To see $\mathrm{S} \circ \mathrm{T}$ is homogenous, take any $k \in K$ and $\boldsymbol{x} \in X$. Then $(\mathrm{S} \circ \mathrm{T})(k\boldsymbol{x}) = \mathrm{S}\big[\mathrm{T}(k\boldsymbol{x})\big] = \mathrm{S}(k\mathrm{T}\boldsymbol{x}) = k\mathrm{ST}\boldsymbol{x} = k(\mathrm{S} \circ \mathrm{T})\boldsymbol{x}$.

b. Let
$$V \underset{\mathrm{S}}{\overset{\mathrm{R}}{\leftarrow}} U \underset{\mathrm{T}}{\overset{\mathrm{P}}{\leftarrow}} X.$$
For any $\boldsymbol{x} \in X$, we have $\big[(\mathrm{R} + \mathrm{S}) \circ \mathrm{T}\big](\boldsymbol{x}) = (\mathrm{R} + \mathrm{S})(\mathrm{T}\boldsymbol{x}) = \mathrm{R}(\mathrm{T}\boldsymbol{x}) + \mathrm{S}(\mathrm{T}\boldsymbol{x}) = (\mathrm{R} \circ \mathrm{T})\boldsymbol{x} + (\mathrm{S} \circ \mathrm{T})\boldsymbol{x}$. The other claim is proved similarly.

□

► EXERCISE 10.3 (3.7). *Show that whenever meaningful,*

$$(\mathrm{ST})' = \mathrm{T}'\mathrm{S}', \quad (\mathrm{T}+\mathrm{R})' = \mathrm{T}' + \mathrm{R}', \quad \textit{and} \quad \left(\mathrm{T}^{-1}\right)' = \left(\mathrm{T}'\right)^{-1}.$$

PROOF. For a generic linear mapping $\mathrm{T} \in \mathfrak{L}(X, U)$, we have the following diagram:

$$\begin{aligned} X &\xrightarrow{\mathrm{T}} U, \\ X' &\xleftarrow{\mathrm{T}'} U'. \end{aligned}$$

For the first equality[1], let $\mathbb{F} \xleftarrow{\ell} V \xleftarrow{\mathrm{S}} U \xleftarrow{\mathrm{T}} X$, i.e., $\mathbb{F} \xleftarrow{\ell} V \xleftarrow{\mathrm{ST}} X$, and so $(\mathrm{ST})' : V' \to X'$. We have

$$\left\langle \left(\mathrm{T}'\mathrm{S}'\right)\ell, \boldsymbol{x}\right\rangle = \left\langle \mathrm{T}'\left(\mathrm{S}'\ell\right), \boldsymbol{x}\right\rangle = \left\langle \mathrm{S}'\ell, \mathrm{T}\boldsymbol{x}\right\rangle = \langle \ell, (\mathrm{ST})\,\boldsymbol{x}\rangle = \left\langle (\mathrm{ST})'\,\ell, \boldsymbol{x}\right\rangle,$$

and this establish the first equality. As for the second equality,

$$\begin{aligned} \left\langle \left(\mathrm{T}' + \mathrm{S}'\right)\ell, \boldsymbol{x}\right\rangle &= \left\langle \mathrm{T}'\ell + \mathrm{S}'\ell, \boldsymbol{x}\right\rangle \\ &= \left\langle \mathrm{T}'\ell, \boldsymbol{x}\right\rangle + \left\langle \mathrm{S}'\ell, \boldsymbol{x}\right\rangle \\ &= \langle \ell, \mathrm{T}\boldsymbol{x}\rangle + \langle \ell, \mathrm{S}\boldsymbol{x}\rangle \\ &= \langle \ell, (\mathrm{T}+\mathrm{S})\,\boldsymbol{x}\rangle \\ &= \left\langle (\mathrm{T}+\mathrm{S})'\,\ell, \boldsymbol{x}\right\rangle. \end{aligned}$$

Finally, let $\mathbb{F} \xleftarrow{\ell} U \xleftarrow{\mathrm{T}} X$, then $\mathrm{T}' : U' \to X'$ and $\left(\mathrm{T}'\right)^{-1} : \mathcal{R}_{\mathrm{T}'} \to U'$. Take any $m \in \mathcal{R}_{\mathrm{T}'}$; then there exists $\ell \in U'$ such that $\left(\mathrm{T}'\right)^{-1}(m) = \ell$, or equivalently, $\mathrm{T}'\ell = m$. Now consider $\left(\mathrm{T}^{-1}\right)'(m)$. Then

$$\left(\mathrm{T}^{-1}\right)'(m) = \left(\mathrm{T}^{-1}\right)'\left(\mathrm{T}'\ell\right) = \left(\mathrm{T}\mathrm{T}^{-1}\right)'\ell = \mathrm{Id}'\ell = \ell$$

since $\mathrm{Id}' = \mathrm{Id}$. □

► EXERCISE 10.4 (3.8). *Show that if X'' is identified with X and U'' with U, then* $\mathrm{T}'' = \mathrm{T}$.

PROOF. We have

$$\langle \mathrm{T}\ell, \boldsymbol{x}\rangle = \left\langle \ell, \mathrm{T}'\boldsymbol{x}\right\rangle = \left\langle \mathrm{T}''\ell, \boldsymbol{x}\right\rangle.$$

□

► EXERCISE 10.5 (3.9). *Show that if* $\mathrm{A} \in \mathfrak{L}(X)$ *is a left inverse of* $\mathrm{B} \in \mathfrak{L}(X)$*, that is,* $\mathrm{A}\,\mathrm{B} = \mathrm{Id}$*, then it is also a right inverse:* $\mathrm{B}\,\mathrm{A} = \mathrm{Id}$.

[1] Notation Warning: We occasionally use $\mathbb{F}$, instead of $\boldsymbol{K}$, to denote the field. From now on we also be back to Lax's notation of linear mapping, that is, $\ell\,(\boldsymbol{x}) = \langle \ell, \boldsymbol{x}\rangle$.

PROOF. $\mathrm{A\,B} = \mathrm{Id} \Longrightarrow \mathrm{A\,B\,A} = \mathrm{A} \Longrightarrow \mathrm{A}\,(\mathrm{B\,A}) = \mathrm{A} \Longrightarrow \mathrm{B\,A} = \mathrm{Id}$. □

► EXERCISE 10.6 (3.10). *Show that if* M *is invertible, and similar to* K, *then* K *also is invertible, and* K^{-1} *is similar to* M^{-1}.

PROOF. M similar to K means that $\mathrm{K} = \mathrm{S\,M\,S}^{-1}$; then

$$\mathrm{K}^{-1} = \left[(\mathrm{S\,M})\,\mathrm{S}^{-1}\right]^{-1} = \mathrm{S}\,(\mathrm{S\,M})^{-1} = \mathrm{S\,M}^{-1}\,\mathrm{S}^{-1},$$

and

$$\mathrm{M} = \mathrm{S}^{-1}\,\mathrm{K\,S} \Longrightarrow \mathrm{M}^{-1} = \mathrm{S}^{-1}\,\mathrm{K}^{-1}\,\mathrm{S}.$$

□

► EXERCISE 10.7 (3.11). *If either* A *or* B *in* $\mathfrak{L}(X)$ *is invertible, then* A B *and* B A *are similar.*

PROOF. Suppose B is invertible. Then

$$\mathrm{B}\,(\mathrm{A\,B})\,\mathrm{B}^{-1} = \mathrm{B\,A},$$

i.e., A B similar to B A. □

► EXERCISE 10.8 (3.14). *Suppose* T *is a linear map of rank 1 of a finite dimensional vector space into itself.*

a. *Show there exists a unique number* c *such that* $\mathrm{T}^2 = c\mathrm{T}$.

b. *Show that if* $c \neq 1$ *then* $\mathrm{Id} - \mathrm{T}$ *has an inverse.*

PROOF. Let $\mathrm{T} \in \mathfrak{L}(X)$. By definition $\operatorname{rank}(\mathrm{T}) = 1$ means that $\dim \mathcal{R}_{\mathrm{T}} = 1$. Let $\dim X = n$. Then

$$\dim X = \dim \mathcal{N}_{\mathrm{T}} + \dim \mathcal{R}_{\mathrm{T}}$$

implies that

$$\dim \mathcal{N}_{\mathrm{T}} = n - 1.$$

Let $(\boldsymbol{v})$, where $\boldsymbol{v} \neq \mathbf{0}_X$, be a basis of $\mathcal{R}_T$, and extend it to a basis $(\boldsymbol{v}, \boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-1})$ of X. Since $\boldsymbol{u}_i \notin \operatorname{span}(\boldsymbol{v})$ for all $\boldsymbol{u}_i$, we have $\mathrm{T}\boldsymbol{u}_i = \mathbf{0}_X$; since $\mathcal{R}_{\mathrm{T}} = \operatorname{span}(\boldsymbol{v})$, there exists $c \in K$ such that $\mathrm{T}\boldsymbol{v} = a\boldsymbol{v}$. For any $\boldsymbol{x} \in X$, there exists a list of scalars $(b, k_1, \ldots, k_{n-1})$ such that $\boldsymbol{x} = b\boldsymbol{v} + \sum_{i=1}^{n-1} k_i \boldsymbol{u}_i$. Then

$$\mathrm{T}\boldsymbol{x} = b\mathrm{T}\boldsymbol{v} = b\,(c\boldsymbol{v}) = cb\boldsymbol{v},$$

and

$$\mathrm{T}^2\boldsymbol{x} = \mathrm{T}\,(\mathrm{T}\boldsymbol{x}) = \mathrm{T}\,(cb\boldsymbol{v}) = cb\mathrm{T}\boldsymbol{v} = c^2 b\boldsymbol{v} = c\,(cb\boldsymbol{v}) = c\mathrm{T}\boldsymbol{x}.$$

Since the above display holds for any $\boldsymbol{x} \in X$, we have $\mathrm{T}^2 = c\mathrm{T}$. □

► EXERCISE 10.9 (3.15[2][3]). *Suppose* T *and* S *are linear maps of a finite dimensional vector space into itself. Show that* $\operatorname{rank}(\mathrm{ST}) \leqslant \operatorname{rank}(\mathrm{S})$. *Show that* $\dim \mathcal{N}_{\mathrm{ST}} \leqslant \dim \mathcal{N}_{\mathrm{S}} + \dim \mathcal{N}_{\mathrm{T}}$.

PROOF. Let $X \overset{\mathrm{S}}{\leftarrow} X \overset{\mathrm{T}}{\leftarrow} X$. By definition, $\operatorname{rank}(\mathrm{ST}) \leqslant \operatorname{rank}(\mathrm{S})$ if and only if $\dim \mathcal{R}_{\mathrm{ST}} \leqslant \dim \mathcal{R}_{\mathrm{S}}$. But this is obvious. As for the second claim, we have

$$\mathcal{R}_{\mathrm{ST}} = (\mathrm{ST})[X] = \mathrm{S}\big[\mathrm{T}[X]\big] = \mathrm{S}[\mathcal{R}_{\mathrm{T}}],$$

so that

$$\operatorname{rank}(\mathrm{ST}) = \dim \mathcal{R}_{\mathrm{ST}} = \dim \mathrm{S}[\mathcal{R}_{\mathrm{T}}].$$

If M is a subspace of dimension m, say, and if N is any complement of M so that $X = M + N$, then[4]

$$\mathcal{R}_{\mathrm{S}} = \mathrm{S}[X] = \mathrm{S}[M] + \mathrm{S}[N].$$

It follows that

$$\operatorname{rank}(\mathrm{S}) = \dim \mathcal{R}_{\mathrm{S}} \leqslant \dim \mathrm{S}[M] + \dim \mathrm{S}[N] \leqslant \dim \mathrm{S}[M] + \dim N,$$

and hence that

$$\dim X - \mathcal{N}_{\mathrm{S}} \leqslant \dim \mathrm{S}[M] + \dim X - m.$$

If in particular

$$M = \mathcal{R}_{\mathrm{T}} = \mathrm{T}[X],$$

then the last inequality implies that

$$\operatorname{rank}(\mathrm{T}) - \mathcal{N}_{\mathrm{S}} \leqslant \operatorname{rank}(\mathrm{ST}),$$

or, equivalently, that

$$\dim X - \dim \mathcal{N}_{\mathrm{S}} - \dim \mathcal{N}_{\mathrm{T}} \leqslant \dim X - \dim \mathcal{N}_{\mathrm{ST}},$$

that is,

$$\dim \mathcal{N}_{\mathrm{ST}} \leqslant \dim \mathcal{N}_{\mathrm{S}} + \dim \mathcal{N}_{\mathrm{T}}.$$

□

[2] See Exercise 3.18.
[3] See Halmos (1995, Problem 95, p. 270).
[4] Theorem 1.5 (b).

11

MATRICES

12

DETERMINANT AND TRACE

► EXERCISE 12.1 (5.1). *Prove the properties of signature:*[1]

$$\operatorname{sign}(\pi) = \pm 1. \tag{5-a}$$

$$\operatorname{sign}(\pi \circ \sigma) = \operatorname{sign}(\pi) \operatorname{sign}(\sigma). \tag{5-b}$$

PROOF. The discriminant of $(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ is $P(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = \prod_{i<j} (\boldsymbol{x}_i - \boldsymbol{x}_j)$. Thus

$$(P\pi)(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = \prod_{i<j} \left(\boldsymbol{x}_{\pi_i} - \boldsymbol{x}_{\pi_j}\right).$$

A typical factor in $P\pi$ is $\boldsymbol{x}_{\pi_i} - \boldsymbol{x}_{\pi_j}$. Now if $\pi_i < \pi_j$, this is also a factor of P, while if $\pi_i > \pi_j$, then $-\left(\boldsymbol{x}_{\pi_i} - \boldsymbol{x}_{\pi_j}\right)$ is a factor of P. Consequently, $P\pi = +P$ if the number of inversions of the natural order in π is even and $P\pi = -P$ if it is odd. Then (5-a) holds since

$$\operatorname{sign}(\pi) = \frac{P\pi}{P} = \pm 1.$$

We not prove (5-b). Let $P = \prod_{i<j} (\boldsymbol{x}_i - \boldsymbol{x}_j)$. Then, since $P\pi = \operatorname{sign}(\pi) P$, we have

$$\begin{aligned}
(P\pi\sigma)(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) &= \left[(P\pi)\sigma\right](\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) \\
&= \operatorname{sign}(\sigma)(P\pi)(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) \\
&= \operatorname{sign}(\sigma) \operatorname{sign}(\pi) P(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n).
\end{aligned}$$

But $P\pi\sigma = \sigma\pi\sigma P$. Hence, $\operatorname{sign}(\pi\sigma) = \operatorname{sign}(\pi) \operatorname{sign}(\sigma)$. □

► EXERCISE 12.2 (5.2). *Prove that transposition has the following properties:*

a. *The signature of a transposition t is minus one:*

$$\operatorname{sign}(t) = -1. \tag{5-c}$$

[1] See Robinson (2003, Sec. 3.1) for a detailed discussion of permutation, signature function, and so on.

b. *Every permutation π can be written as a composition of transpositions:*

$$\pi = t_k \circ \cdots \circ t_1. \tag{5-d}$$

PROOF. (5-c) is clear. For (5-d), see Robinson (2003, 3.1.3 & 3.1.4, p. 34-35). □

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